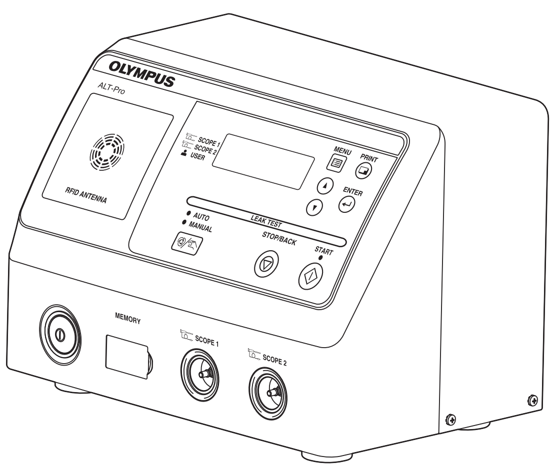
INSTRUCTIONS

AUTOMATED ENDOSCOPE LEAK TESTER

ALT-Pro



Contents

Labels and Symbols

The meaning(s) of the symbol(s) shown on the component packaging, the back cover of this instruction manual and/or this equipment are as follows:

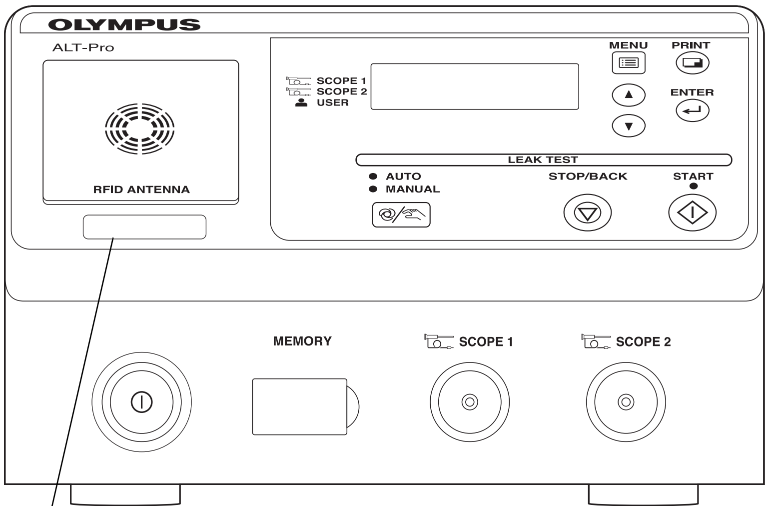
Symbol	Description
	Refer to instructions.
SN	Serial number
	Manufacturer
EC REP	Authorized representative in the European Community
	Date of manufacture
	Importer (into European Union)

For US Customers only

For a Symbols Glossary, visit us: http://www.olympus-global.com/en/common/pdf/symbolsglossary.pdf

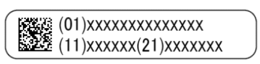
Safety-related labels and symbols are attached to this equipment at the locations shown below. If labels or symbols are missing or illegible, contact Olympus.

❍ Front panel

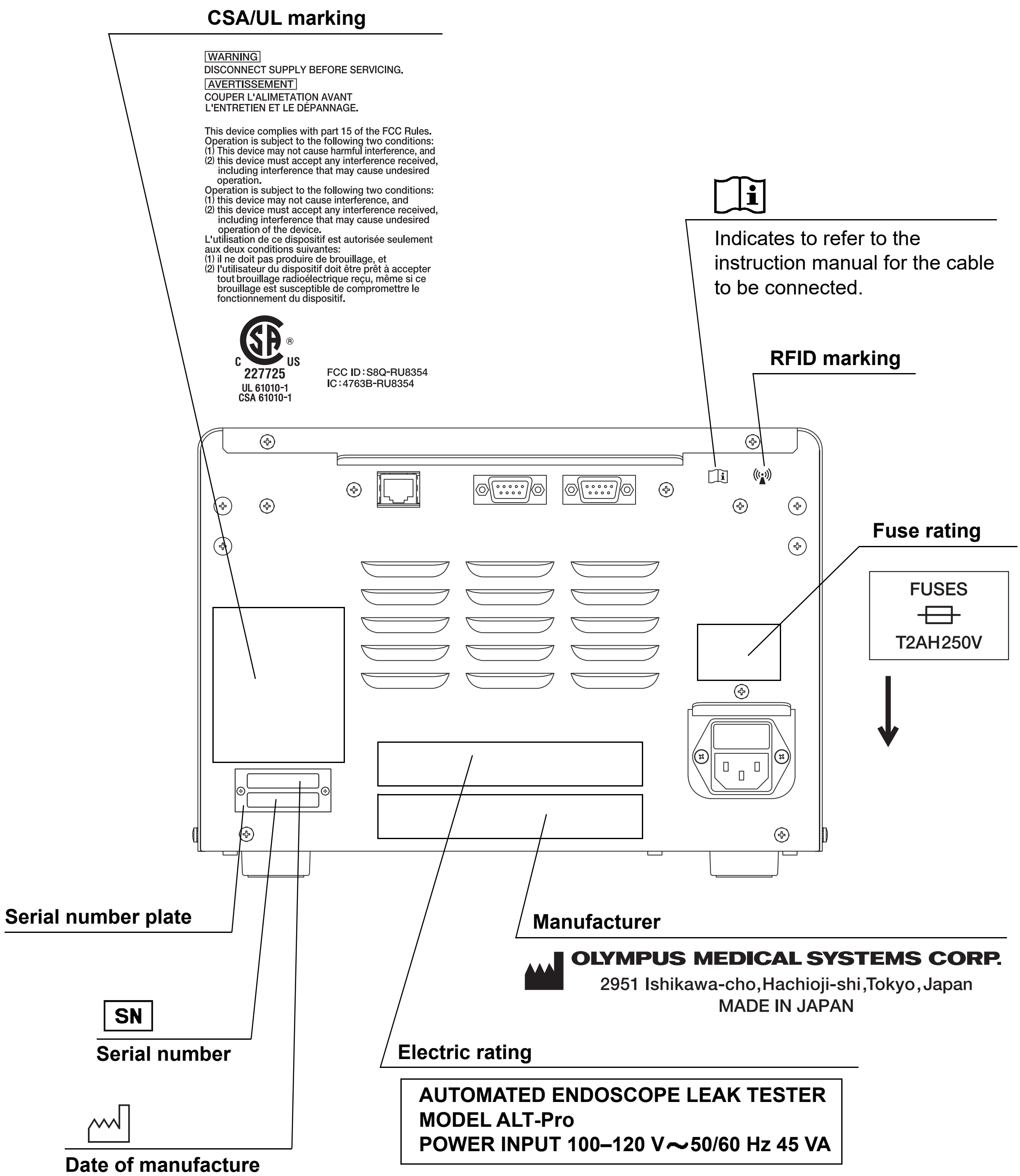


UDI label

The UDI label is required by some countries' regulations regarding the identification of a medical device also known as Unique Device Identification (UDI).



❍ Rear panel


CSA/UL marking
WARNING
DISCONNECT SUPPLY BEFORE SERVICING.
AVERTISSEMENT
COUPER L'ALIMETATION AVANT
L'ENTRETIEN ET LE DÉPANNAGE.
This device complies with part 15 of the FCC Rules.
Operation is subject to the following two conditions:
(1) This device may not cause harmful interference, and
(2) this device must accept any interference received, including interference that may cause undesired operation.
Operation is subject to the following two conditions:
(1) this device may not cause interference, and
(2) this device must accept any interference received, including interference that may cause undesired operation of the device.
L'utilisation de ce dispositif est autorisée seulement aux deux conditions suivantes:
(1) il ne doit pas produire de brouillage, et
(2) l'utilisateur du dispositif doit être prêt à accepter tout brouillage radioélectrique reçu, même si ce brouillage est susceptible de compromettre le fonctionnement du dispositif.
C US
227725
UL 61010-1
CSA 61010-1
FCC ID：S8Q-RU8354
IC：4763B-RU8354
Indicates to refer to the instruction manual for the cable to be connected.
RFID marking
Fuse rating
FUSES
T2AH250V
Serial number plate
SN
Serial number
Date of manufacture
Manufacturer
OLYMPUS MEDICAL SYSTEMS CORP.
2951 Ishikawa-cho, Hachioji-shi, Tokyo, Japan
MADE IN JAPAN
Electric rating
AUTOMATED ENDOSCOPE LEAK TESTER
MODEL ALT-Pro
POWER INPUT 100–120 V～50/60 Hz 45 VA

Important Information — Please Read Before Use

Indications for use

This equipment is intended to be used to perform and record leakage testing on Olympus flexible endoscopes.

Contraindications

None known.

Instruction manual

This instruction manual contains essential information on using this AUTOMATED ENDOSCOPE LEAK TESTER (hereinafter referred to as “this equipment”) safely and effectively. Before use, thoroughly review this manual and the manuals of all equipment that will be used during the procedure, and use all equipment as instructed. If the equipment is used in a manner not specified by the manufacturer, the protection provided by the equipment may be impaired.
Keep this and all related instruction manuals in a safe, accessible location.
If you have any questions or comments about any information in this manual, contact Olympus.

Terms used in this manual

Leakage test:

A leakage test can detect a pin hole in the external surface of the endoscope and/or internal working channel wall, and any other leak-related damage.
When the endoscope is immersed in a liquid for cleaning and disinfection, the liquid may go into the endoscope through a pin hole or other damaged area, resulting in malfunction of electrical components inside the endoscope. Detecting such a pin hole and damaged area before liquid immersion of the endoscope can prevent potential damage of the endoscope caused by the entry of the liquid into the endoscope.

Process:

A general term for the leakage testing, self-check, and other operations this Equipment performs.

Portable memory:

Olympus-designated USB memory: this recording medium is used in combination with this equipment to save leakage test data stored in the internal memory of this equipment.

Master data:

These are unedited leak test log data and error log data downloaded to personal computers.

CSV file format:

This is one data format most frequently used by statistical processing software. Each item of data (record) is accommodated in a single line, in which each item is delimited by "," (a comma). This data is useful when executing statistical processing using commercially available statistical processing software.

KE server

Computer where KE-BASE SYSTEM SOFTWARE (Ver.2.1.0 or later) is installed.

Wall mains outlet:

A three-pin outlet having an exclusive terminal for grounding.

Error code:

A code consisting of [E] and a three-digit number is displayed on the LCD, if there is a problem with this equipment. When an error code is displayed, check the error code list to find out what corrective measures to take.

■ *User qualifications*

The operator of this equipment must be sufficiently trained in reprocessing of endoscopes. The medical literature reports cases of infections due to inappropriate cleaning, disinfection, and /or sterilization. Thoroughly review and understand the following items before use:

- Cleaning, disinfection, and sterilization procedures described in the instruction manuals for the endoscope and ancillary equipment.
- Professional health and safety standards.
- Applicable guidelines on cleaning, disinfection, and sterilization of endoscopy equipment.
- Structure and handling of endoscopic equipment.
- Personal protective equipment requirements to minimize exposure to chemicals and infectious materials.

This manual does not explain or discuss details of cleaning, disinfection, and sterilization.

Instrument compatibility

Use this equipment in combination with ancillary equipment listed in System Chart in Appendix. Using incompatible equipment can result in patient or operator injury and equipment damage and/or malfunction. Refer to the “List of Compatible Endoscopes <ALT-Pro>” to find what models of endoscope can be automatically leak-tested.
Do not use this equipment in combination with non-Olympus endoscopes; Olympus has not tested the efficacy of detecting leaks with this equipment other than those combinations designated.

Care and storage

After use, reprocess and store this equipment referring to the instructions in Chapter 6, “Routine Maintenance” in this manual. Inappropriate care and storage could present an infection control risk and/or cause equipment damage or malfunction.

Maintenance and inspection

The ALT-Pro requires routine maintenance and inspection. In addition to checks before use, the person in charge of maintenance and administration of the medical equipment at the hospital should periodically check all of the items described in this manual. If any irregularity is observed, do not use this equipment and follow the procedure as described in Section 8.1, “Troubleshooting guide”. If the irregularity still presents, this equipment needs repairing. Contact Olympus.

Repair and modification

Do not disassemble, modify, or attempt to repair this equipment and its accessories. Doing so could result in operator or patient injury and/or equipment damage or malfunction. Some problems that appear to be malfunctions may be corrected by referring to Section 8.1, “Troubleshooting guide”. If the problem persists, do not use this equipment and contact Olympus.

Disposal of this equipment

Follow all applicable national and local guidelines in disposing of this equipment.

Signal words

The following signal words are used throughout this manual:

WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
CAUTION	Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices or potential equipment damage.
NOTE	Indicates additional helpful information.

Precautions

Follow the warnings and cautions given below when handling this equipment. This information is supplemented by the warnings given in each chapter.
A very small pin hole may not be detected in automated leakage testing. The ALT-Pro is capable of detecting leaks that cause fluid invasion to the inside of the endoscope; however, if more precise leak detection is required, perform manual leakage testing.

WARNING

- Strictly observe the following cautions. Failure to do so may place the patient and medical personnel in danger of an electric shock.
  - Keep fluids away from all electrical equipment. If fluids are spilled on or into this equipment, stop operation immediately and contact Olympus.
  - Do not prepare, inspect, or use this equipment with wet hand.
- Do not install this equipment in any place where any of the following are present.
  - High oxygen concentration
  - Oxidizing substance such as Nitrous Oxide (N_2O)
  - Flammable anesthetic gas
  - Flammable liquid

  This equipment is not explosion-proof and may explode or cause a fire under these conditions.
- Do not insert an EndoTherapy accessory or other object through an opening including the air vents of this equipment. Also, do not allow any liquid including water or disinfectant solution to flow into the opening. Contact with an electrical part inside this equipment could cause an electric shock and/or malfunction of this equipment.

WARNING

- This equipment may interfere with other medical electronic equipment used in combination with it. Before use, refer to Appendix to confirm the compatibility of this equipment with all equipment to be used.
- Always use the power cord provided with this equipment. Otherwise, equipment failure or power cord burnout could result. Also, remember that the provided power cord is for exclusive use with this equipment and should not be used with other equipment.
- Do not bend, pull, or twist the power cord. An electric shock, equipment damage, or a fire can result.
- This equipment can be set up to use the RFID (Radio Frequency Identification) function. Be aware that the radio waves emitted from the RFID antenna of this equipment may cause medical equipment such as pacemakers to malfunction. If any interference with this equipment is observed, immediately move away from the RFID antenna or set the power switch to OFF. Call your doctor if you do not begin to feel better.

CAUTION

- Certain endoscopes cannot be leak-tested automatically in this equipment. Refer to the “List of Compatible Endoscopes <ALT-Pro>” provided with this equipment.
- Do not use this equipment in any place where it may be subject to strong electromagnetic radiation (for example, in the vicinity of a microwave therapeutic equipment, short wave medical treatment equipment, MRI, and cellular/portable phone, etc). Doing so may result in malfunction of this equipment.
- Do not press the buttons on the front panel of this equipment with a sharp or hard object. Doing so may damage the buttons.
- Avoid applying excessive force to the connectors, as this may damage this equipment.
- Do not use this equipment in a dusty environment. Otherwise, damage and/or malfunction can occur.
- Be sure that this equipment is not used adjacent to other equipment (other than the components of this equipment or system) to avoid electromagnetic interference.
- Be sure that this equipment is not stacked with other equipment and do not put anything on top of this equipment. An external force applied to this equipment may result in damage and deformation.

CAUTION

- This equipment emits RF (Radio Frequency) energy to perform intended functions. Therefore, it may cause electromagnetic interference in nearby electronic equipment, and is labeled with the following symbol. If electromagnetic interference occurs, mitigation measures may be necessary, such as moving the electronic equipment away, reorienting or relocating this instrument, or shielding the location.

Summary of Functions of this Equipment

❍ Automated leakage testing (ALT)

This equipment feeds air into the endoscope and computes changed air pressure to check automatically for a hole in the endoscope or damage to the endoscope due to a puncture or hole. The automated leakage testing is performed in air, not in water.
→Refer to Section 5.8, “Automated leakage testing”.

❍ Manual leakage testing (MLT)

This equipment feeds air into the endoscope to visually check for any bubbles emerging from the endoscope when placed in water.
→Refer to Section 5.9, “Manual leakage testing”.

❍ Self-check

This equipment has a self-check function to verify that it is working properly.
→Refer to Section 4.2, “Self-check”.

❍ Printing

The results of leakage tests and self-checks and the records of errors can be printed from a connected printer.
→Refer to Section 7.10, “Manual printing”.

❍ Data download

The results of leakage tests and self-checks and the records of errors can be downloaded to the portable memory.
→Refer to Section 7.3, “Download to the portable memory”.

❍ Data transmission

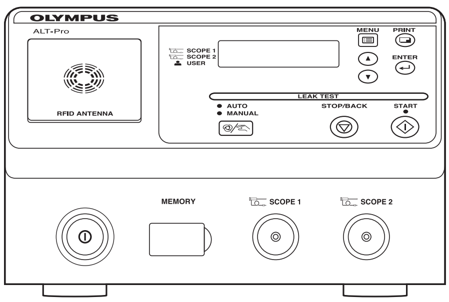
The results of leakage tests and the records of errors can be transmitted to the server.
→Refer to Section 7.12, “Setting the network”.

Chapter 1 Checking the Package Contents

1.1 Checking the package contents

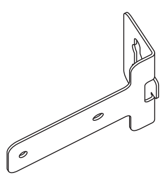
Match all items in the package with the components shown below. Inspect each item for damage. If this equipment is damaged, a component is missing, or you have any questions, do not use this equipment; immediately contact Olympus.

❍ AUTOMATED ENDOSCOPE LEAK TESTER

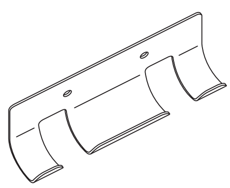


AUTOMATED ENDOSCOPE LEAK TESTER (ALT-Pro)

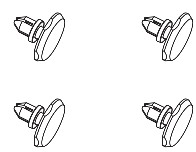
❍ Tube hanger



Bracket (×1)

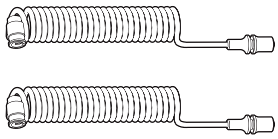


Hanger parts (×1)

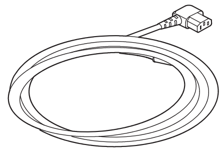


One-touch rivets (×4)

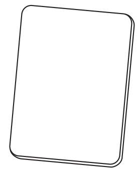
❍ Accessories



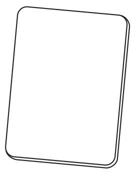
ALT-Pro leak test air tubes (×2) (MAJ-2009)



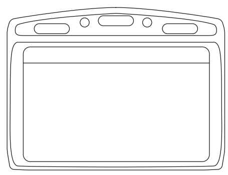
Power cord*1



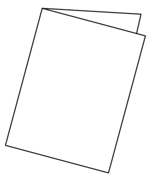
User ID master card



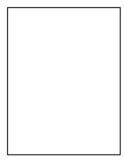
Scope ID master card



Card holders (×2)



Instruction manual

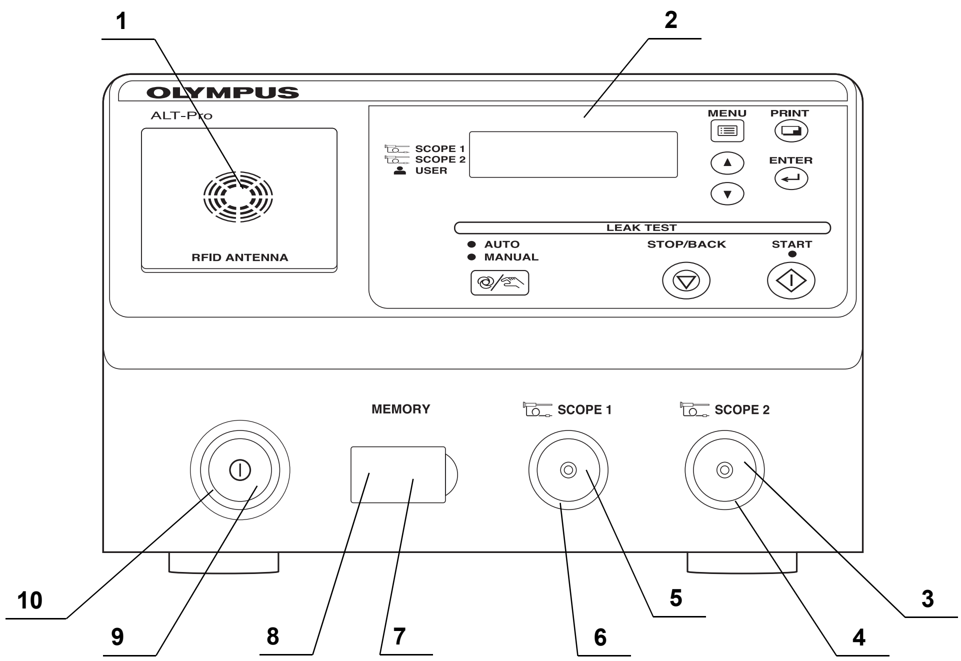


List of Compatible Endoscopes <for ALT-Pro>

*1 Power cord MAJ-2270 is compatible.

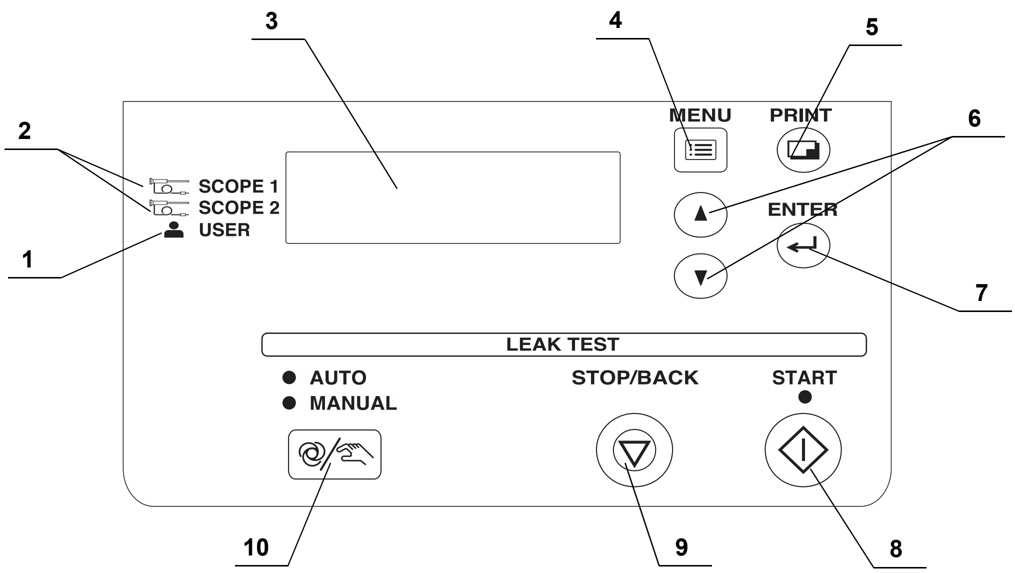
Chapter 2 Nomenclature and Functions

2.1 Front panel



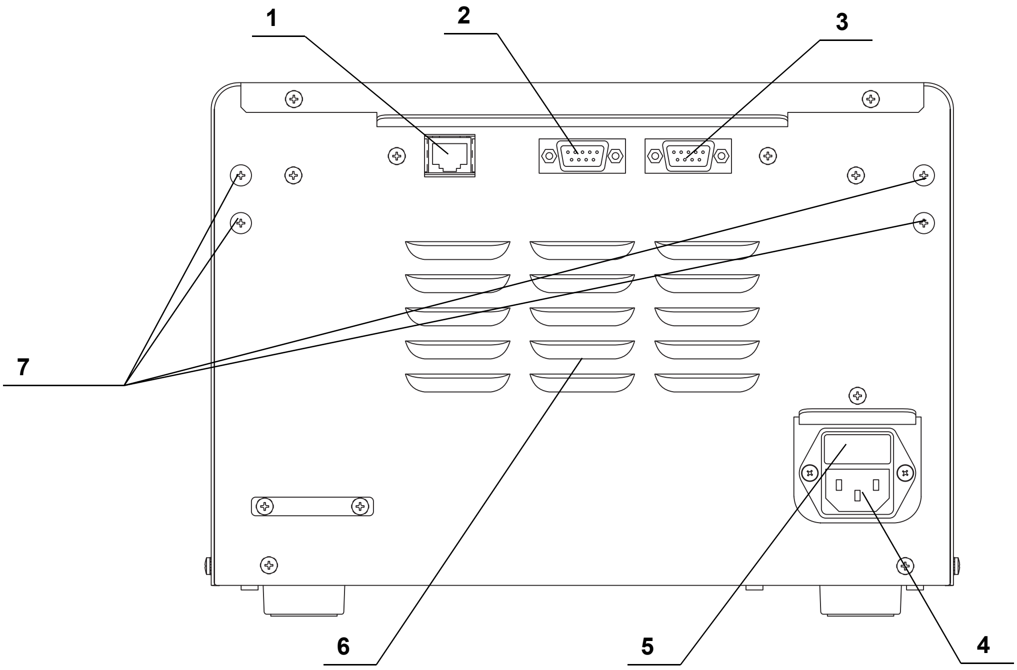
No.	Name	Description
1	RFID reader	Reads user IDs and scope IDs into this equipment
2	CONTROL PANEL	Used to control and set up this equipment.
3	Tube connector 2	Accepts the ALT-Pro leak test air tube
4	Tube connector 2 indicator	Blinks when the scope ID has been recognized and then lights when the endoscope has been connected to the ALT-Pro leak test air tube.
5	Tube connector 1	Accepts the ALT-Pro leak test air tube
6	Tube connector 1 indicator	Blinks when the scope ID has been recognized and then lights when the endoscope has been connected to the ALT-Pro leak test air tube.
7	Portable memory port	Accepts an (optional) portable memory. To insert the portable memory into this port, open the memory port cap.
8	Memory port cap	Prevent water from entering and damaging the portable memory port.
9	Power switch	Press to switch the power ON/OFF. The green light is lit when this equipment is turned ON.
10	Power indicator	Lights up when electric power is supplied.

2.2 Control panel



No.	Name	Description
1	User ID indicator	Lights up when the user ID is recognized.
2	Endoscope ID indicator	Lights up when the scope ID is recognized.
3	LCD (Liquid Crystal Display)	Displays the test process and results.
4	MENU button	In the standby state, the function menus can be displayed on the LCD by pressing the button.
5	PRINT button	In the standby state, the printing menus can be displayed on the LCD by pressing the button.
6	▼, ▲ button	Pressed to set the program, time, and date.
7	ENTER button	Pressed to confirm the end of the process and settings.
8	START button	Pressed to start the selected program.
9	STOP/BACK button	Pressed to interrupt the test procedure or cancel settings.
10	SELECT button	Pressed to select ALT or MLT.

2.3 Rear panel

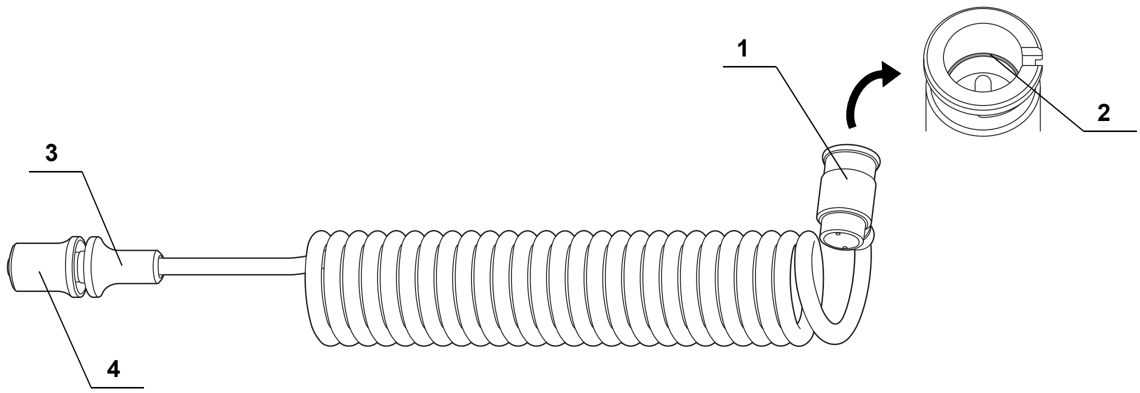


No.	Name	Description
1	100 BASE-TX terminal	Connects the Olympus-designed server equipment. Establishes the communication with the Olympus-designed server equipment. For other uses, do not connect the network cable.
2	OPTION terminal	Not used. (A terminal only for service persons.)
3	PRINTER OUT terminal	Use the ALT-Pro interface cable to connect the printer to this equipment.
4	AC power inlet	Accepts AC power supply.
5	Fuse box	Contains a fuse for this equipment.
6	Air vent	Draws in fresh air from the room, not into the room, and prevents the temperature from rising in this equipment.
7	Projections	Fixture for mounting the tubing hanger.

CAUTION

Never touch the pin inside the PRINTER OUT terminal and OPTION terminal. Static electricity may cause a malfunction or damage of this equipment.

2.4 ALT-Pro leak test air tube MAJ-2009

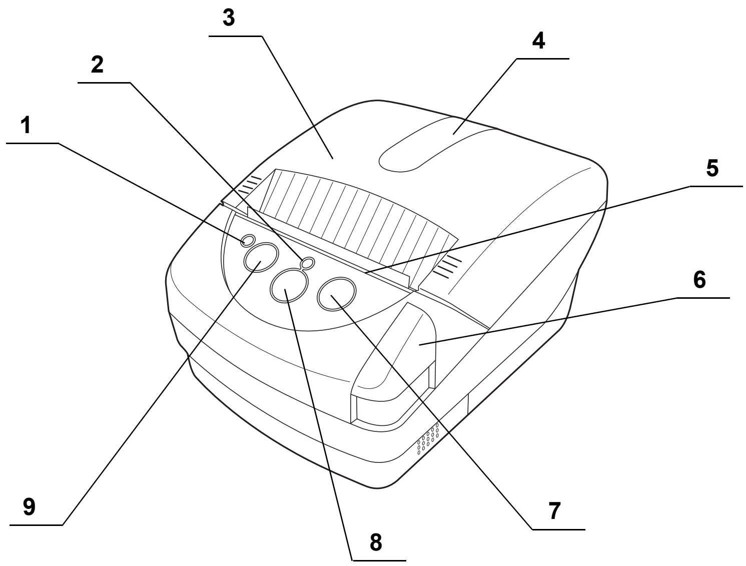


Ch.2

No.	Name	Description
1	Endoscope side connector	Connected to the venting connector of Olympus endoscope or water-resistant cap.
2	O-ring	(Black) Rubber to keep the ALT-Pro leak test air tube airtight.
3	Connecting grip	Held to connect the ALT-Pro leak test air tube to this equipment.
4	Removing grip	Held to remove the ALT-Pro leak test air tube from this equipment.

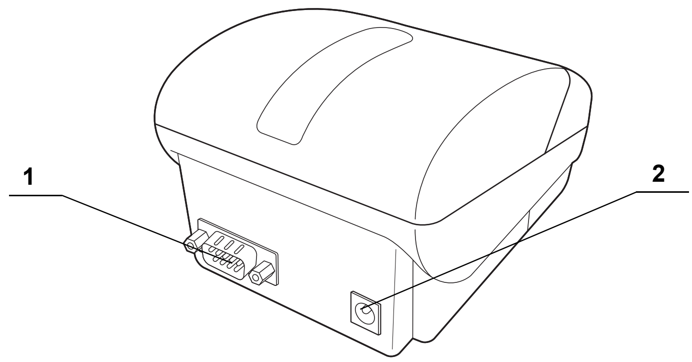
2.5 Printer MAJ-1937 (optional)

❍ Front panel



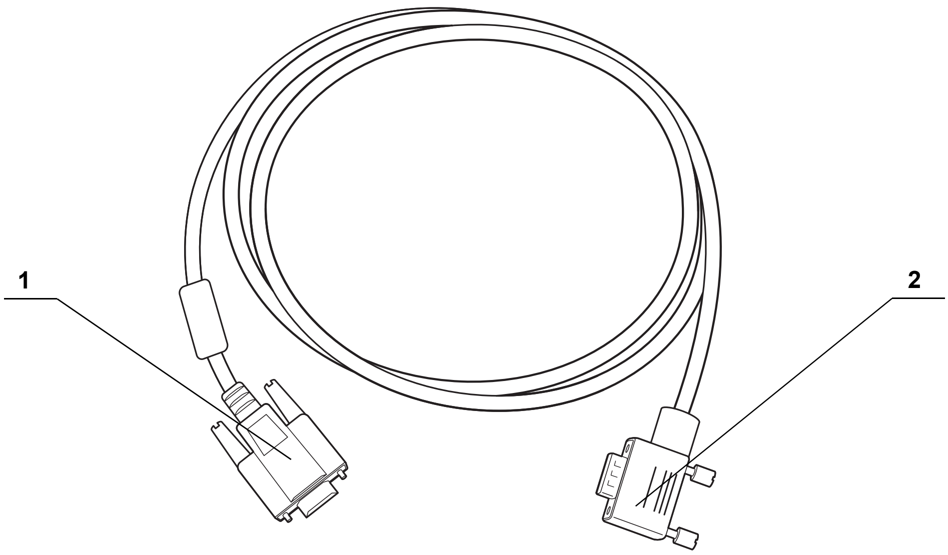
No.	Name	Description
1	ERROR LED	Indicates printer errors.
2	POWER LED	Indicates Power ON and OFF.
3	Paper cover	Stores paper under this cover.
4	Transparent window to check the printer paper roll	Check through this transparent window whether the paper is set properly or how much it remains.
5	Paper cutter	Pull the paper toward you to cut paper.
6	Cover open button	Pressed to open the paper cover when the paper roll is set.
7	FEED button	Pressed to feed the printer paper roll.
8	POWER button	Pressed to turn this equipment ON and OFF. To turn ON/OFF this equipment, hold down the power button (for approximately one second) until the POWER LED lights ON/OFF.
9	SELECT button	No used.

❍ Rear panel



No.	Name	Description
1	IF Connector	Accepts the ALT-Pro interface cable
2	Power Connector	Accepts the exclusive power cord.

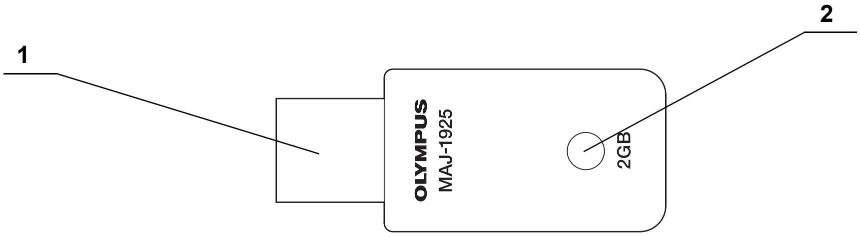
2.6 ALT-Pro interface cable MAJ-2052 (Optional)



No.	Name	Description
1	Printer side connector	Accepts the IF connector of the printer.
2	Equipment side connector	Accepts the PRINTER OUT terminal of this equipment.

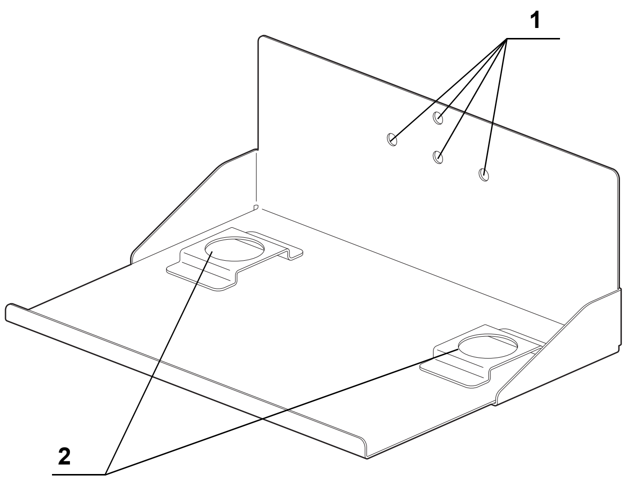
* The ALT-Pro interface cable is 1.5 m long.

2.7 Portable memory MAJ-1925 (Optional)



No.	Name	Description
1	Connector	Insert the connector into the Portable memory port.
2	LED	The LED is blinking while the endoscopic images, patient data, and setting information are saved or read.

2.8 ALT-Pro wall-mount holder MAJ-2020 (Optional)



No.	Name	Description
1	Mounting holes	Holes for screws to fix the ALT-Pro wall mount holder to a wall.
2	Foot base	Accepts the feet of ALT-Pro.

Chapter 3 Installation and Connection

3.1 Installation of this equipment

This equipment is not water-proof. Install this equipment referring to the following installation conditions. Use of the ALT-Pro wall-mount holder (MAJ-2020) (Optional) enables you to install this equipment on the wall. To mount this equipment on the wall, refer to Section 3.2, “Installation of the ALT-Pro wall-mount holder on the wall” in this instruction manual.

WARNING

- Do not spray water on this equipment. Otherwise, a short can occur in the internal circuit, resulting in an electric shock or a fire.
- Check the location of this equipment to see if it is within 2 m away from the location of the endoscope where an air leak test is performed (the sink or basin). If the ALT-Pro leak test air tube is pulled 2 m or over by applying an excessive force, this equipment will be toppled to fall, resulting in injury.
- Do not use a humidifier near this instrument. Dew condensation may occur inside this equipment, causing a malfunction.

CAUTION

- Do not place this equipment on its side or upside down.
- Keep the air vents of this equipment clear. The air vents are on the rear panel. Blockage can cause overheating and equipment damage.
- In installing this equipment, taking care not to step on the power cord or cables. The power cord or cables may be damaged.

❍ Installation location conditions

To ensure safe use of this equipment, be sure to observe the following conditions that are requisite for installation of this equipment.

- Place this equipment on a flat location.
- The power cord (3 m/9.8 ft) should be able to reach the power outlet.
- This equipment should not be exposed to direct sunlight.
- The location should be clean and free of dirt or dust.
- The ambient temperature should not exceed 40°C (104°F).
- The relative humidity should be between 30% and 85%.
- The elevation above sea level should not exceed 3000 meters.
- This equipment should be used indoors.

3.2 Installation of the ALT-Pro wall-mount holder on the wall

For installation of the ALT-Pro wall-mount holder on the wall, ask a qualified installation specialist to carry out the installation. Olympus is not liable for any injury or damage that occurs as a result of improper installation or mishandling.

WARNING

- Do not tighten the screws and/or bolts for installation into any locations on the wall where the screws and/or bolts may contact with the wires and/or ducts in the wall. The wires and/or ducts may be damaged, inflicting an electric shock or causing water leakage.
- Only a qualified installation specialist must install/remove the ALT-Pro wall-mount holder on/from the wall. Improper installation by any unqualified person may cause the ALT-Pro wall-mount holder to drop, which may result in an injury.
- Never install the ALT-Pro wall-mount holder on a location that cannot support the weight of the housing. Insufficient strength of the installation location of the ALT-Pro wall-mount holder may cause the housing to drop, resulting in an injury.
- Make sure the wall of the installation location is strong enough, taking a long term use of this equipment into consideration. If the strength at the installation location becomes insufficient due to a long term use, the ALT-Pro wall-mount holder may drop, resulting in an injury.

WARNING

- Install the ALT-Pro wall-mount holder in a method suited to the wall structure or material.
- Install the ALT-Pro wall-mount holder only on a vertical wall. Installation to other types of wall may cause the ALT-Pro wall-mount holder to drop, resulting in an injury.
- Ensure safety around the installation location in installing the ALT-Pro wall-mount holder on the wall. Installation work in a small place may cause an injury.
- Do not place equipment other than the ALT-Pro leak test air tubes on the ALT-Pro wall mount holder. Equipment falling off the holder may cause an injury.
- Do not pull the power cord and cables once this equipment has been installed on the ALT-Pro wall-mount holder. This equipment may fall, causing injury.

CAUTION

- Do not install the ALT-Pro wall-mount holder upside down. This equipment may fall off.
- After installing the ALT-Pro wall-mount holder on the wall, install this equipment, taking care not to step on the power cord or cables. The power cord or cables may be damaged.

❍ Installation location conditions

To ensure safe use of the ALT-Pro wall-mount holder, be sure to observe the following conditions that are requisite for installation of the ALT-Pro wall-mount holder and the conditions described in Section 3.1, “Installation of this equipment”.

- The wall for installation should be vertical.
- The wall should be strong enough to hold the weight of the ALT-Pro wall-mount holder and this equipment. (See Figure 3.1, the dimensional drawing)

Secure ALT-Pro wall-mount holder to a wall using anchors or bolts being able to withstand 26 kg load.

❍ Installing the ALT-Pro wall-mount holder

Check	Required items
	ALT-Pro wall-mount holder

Table 3.1

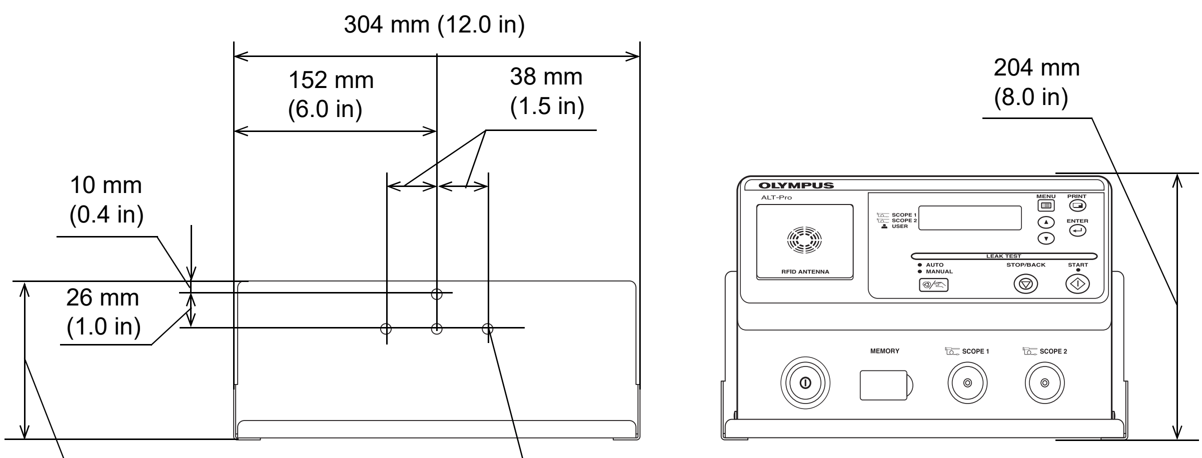
1 Locate the best position on the wall to install the ALT-Pro wall-mount holder. Make sure that the position meets the conditions of the installation location as described below.

Make sure that this equipment is placed in a location where its power cord can reach the wall mains outlet with a margin and within two meters from the installation location of the endoscope where leak air testing is performed.

WARNING

Referring to the drawings of Figure 3.1 showing the dimensions of the installation location of the ALT-Pro wall-mount holder, make sure the strength of the wall surface of the four installation positions. If the strength of the installation is insufficient, add some reinforcement. Otherwise, the ALT-Pro wall-mount holder may drop, breaking the wall or injuring the operator.

2 Referring to the drawings of Figure 3.1 showing the dimensions and weight of the ALT-Pro wall-mount holder, make sure the strength of the wall surface of the four installation positions. If the strength is insufficient, add reinforcement.



Weight of the ALT-Pro:
Approx.6.5 kg (When accessories are connected)
Weight of the ALT-Pro wall-mount holder:
Approx.1.3 kg

Figure 3.1

3 If screws or nuts need screwing into a concrete wall, determine the hole positions based on the dimension drawing above and embed anchor bolts or nuts equivalent to "6 mm nominal diameter of bolt". When embedding the bolts into the wall, protrusion of the bolt from the wall surface should be 10 mm to 15 mm.

CAUTION

Securely hold the ALT-Pro wall-mount holder and install it so that it will not fall off. If the ALT-Pro wall-mount holder falls off, it may be damaged.

4 Hold the ALT-Pro wall-mount holder in the direction as shown in Figure 3.1, align the holder with the mounting holes, and secure it with screws or nuts. The screw or nuts to be used for installation should be commercially available and equivalent to "6 mm nominal diameter of nut or bolt," which is suited to the material of the wall.

5 Make sure that all nuts or bolts are securely tightened. If any of the nuts or bolts loosen, tighten them again.

3.3 Installation of the tube hanger

Installing the tube hanger on this equipment enables you to hook the endoscope side connector of the ALT-Pro leak test air tube(s) when they are not in use.

CAUTION

Avoid placing anything other than the endoscope side connector of the ALT-Pro leak test air tubes on the tube hanger. Otherwise the tube hunger may get damaged.

1 To install the tube hanger on the right side viewed from the front, align the holes of the bracket and the hanger part as shown in Figure 3.2 (A).
(To install the tube hanger on the left side viewed from the front, align the holes of the bracket and the hanger part as shown in Figure 3.2 (B).)

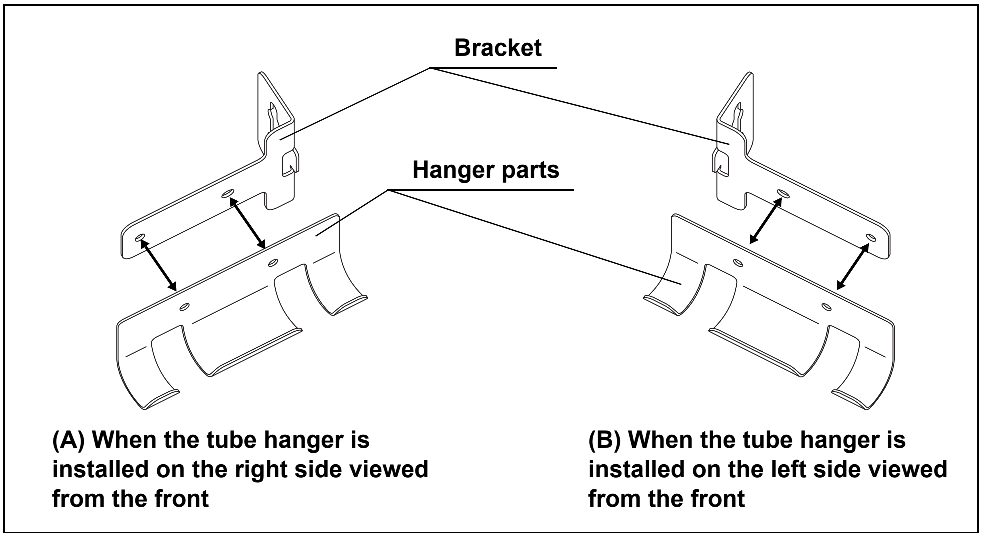


Figure 3.2

2 Align the holes of the bracket and the hanger part and insert the one-touch rivets into the holes from the hanger part side (see Figure 3.3). Insert the new one-touch rivets until they hit the surface of hanger part, and secure the bracket to the hanger part. (See Figure 3.4)

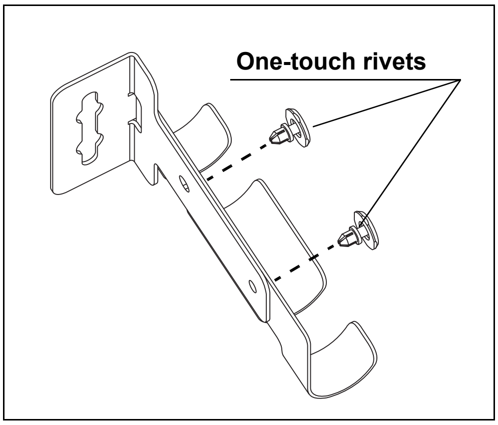


Figure 3.3

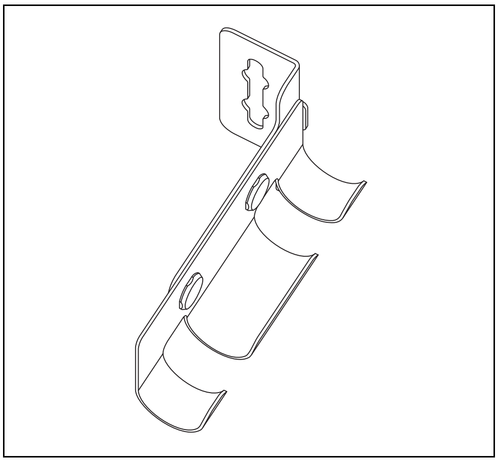


Figure 3.4

3 Align the pins on the rear panel of this equipment with the grooves of the bracket. Push the bracket onto this equipment and slide the bracket downward into position. (To install the tube hanger in the left side viewed from the front, match the groove of the bracket with the pins on the right.)

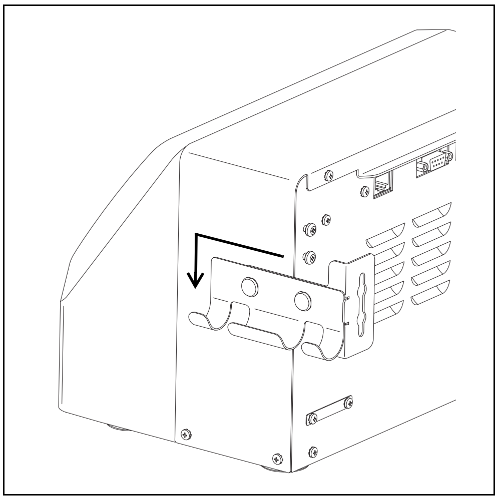


Figure 3.5

Ch.3

3.4 Connecting the power supply

WARNING

- Be sure to connect the power plug of the power cord directly to a grounded, hospital grade, wall mains outlet. If this equipment is not grounded properly, it can cause an electric shock and/or a fire.
- Do not connect the power plug to the 2-pole power circuit with a 3-pole to 2-pole adapter. It can prevent proper grounding and cause an electric shock.
- Do not allow the power plug to get wet. A wet power plug may cause an electric shock.
- Confirm that the grounded wall mains outlet offer adequate electric capacity. In sufficient capacity may result in a fire or activation of the circuit breaker of the facility, which turn OFF the power to all this equipment connected to the same line.
- Do not bend, pull, or twist the power cord. Visually inspect the entire length of the power cord for cracks or cuts. Equipment damage including separation of the power plug and disconnection of the cord wire as well as a fire or an electric shock can result.
- Be sure to connect the power plug securely to prevent its accidental disconnection during operation. Otherwise, this equipment may not function.
- Do not push the rear of this equipment against the wall after connecting the power cord. The power supply cord may break, resulting in a fire or an electric shock.
- Always use the power cord provided with this equipment. Otherwise, equipment failure or power cord burnout may result. Also, remember that the provided power cord is for use only with this equipment and should not be used with any other equipment.

1 Make sure that this equipment is turned OFF.

2 Insert the power cord all the way into the connector on the rear panel (bottom right).

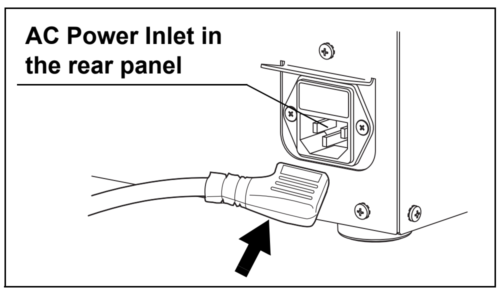


Figure 3.6

3 Connect the power cord plug into a hospital-grade, wall mains outlet that meets the input power conditions indicated on the rating plate.

3.5 Connecting the printer

Connecting a printer (optional) to this equipment enables you to print the results of leakage testing of the endoscope.

WARNING

- Use the ALT-Pro interface cable (optional) for this equipment. Using a different interface cable may result in malfunction of this equipment or damage to the interface cable.
- Do not wet the ALT-Pro interface cable. A wet ALT-Pro interface cable may cause an electric shock.
- Do not bend, pull, or twist the ALT-Pro interface cable. This may cause fall-off of the connector or disconnection of the ALT-Pro interface cable; a fire or an electric shock may result.
- Be sure to connect the ALT-Pro interface cable to prevent its accidental disconnection during operation. Otherwise, this equipment may not function.
- Make sure that each connector or terminal is free from water on it before connecting the ALT-Pro interface cable. Any water may damage this equipment or printer.
- Do not push the rear of this equipment against the wall after connecting the interface cable. The interface cable may bend, resulting in a fire or an electric shock.

CAUTION

- Read through the instruction manual for the printer before using it.
- Install the printer in an area where it will not get sprayed. Otherwise, the printer may be damaged.

1 Make sure that this equipment and printer are turned OFF.

2 Make sure that the power cord of the printer is connected to the printer.

3 Connect the ALT-Pro interface cable to the IF connector of the printer and rotate the screws to fix the cable.

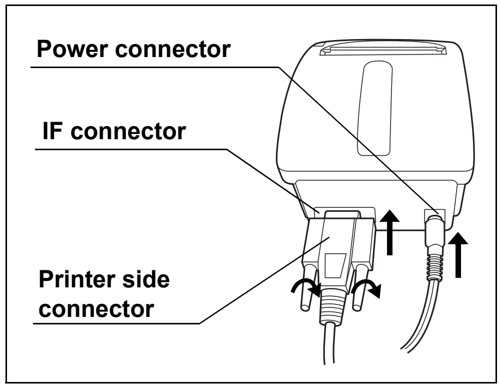


Figure 3.7

4 Connect the ALT-Pro interface cable to the PRINTER OUT terminal of this equipment and rotate the screws to fix the cable.

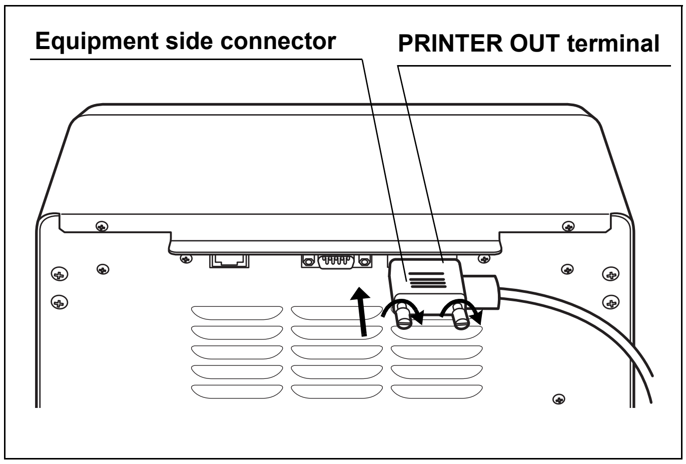


Figure 3.8

3.6 Connection of the server

Refer to Section 7.12, “Setting the network” for the connection of the server.

NOTE

When this equipment is connected to the server after completion of the network setting, characters “Sv” appear on the top right of the LCD screen. For the information displayed on the LCD in each Communication status, See Table 3.2.

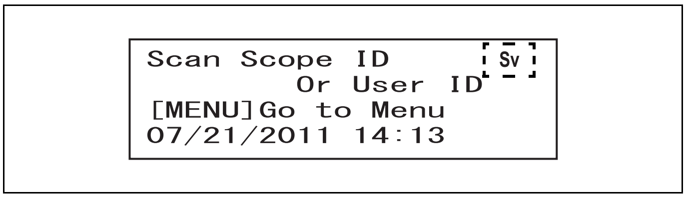


Figure 3.9

Ch.3

Communication status	LCD display
Server communication enabled in Connected status	Characters “Sv” are displayed.
Server communication enabled in Disconnected status	Characters “Sv” flash.
Server communication disabled	Characters “Sv” are not displayed.

Table 3.2

Chapter 4 Inspection Before Use

To ensure that this equipment operates safely and reliably, inspect all parts before use.

Check	Inspection items before use
	4.1 Installing the ALT-Pro leak test air tubes and turning ON the power
	4.2 Self-check
	4.3 Checking the printer paper roll

Table 4.1

WARNING

Before each use, inspect this equipment as instructed below. Otherwise, this equipment may not work properly.
Should any irregularity be observed, do not use this equipment and contact Olympus.
Damage or irregularity may compromise patient or user safety and may result in more-severe equipment damage.

4.1 Installing the ALT-Pro leak test air tubes and turning ON the power

CAUTION

- Over time, the ALT-Pro leak test air tube will deteriorate because it is subject to wear with each use. If any of the following are found with the ALT-Pro leak test air tube, do not use it and replace it with a new one.
  Otherwise, it may cause malfunction of the endoscope.
- Check that the outer surface of the ALT-Pro leak test air tubes is free from scratches and flaws. Any scratch or flaw may prevent proper leakage testing.
- Make sure that the surfaces of the removing grip and tube connector are completely dry before holding the connecting grip of the ALT-Pro leak test air tube to connect the ALT-Pro leak test air tube to the tube connector. If they are wet, thoroughly wipe them with a clean cloth. Otherwise, fluid may enter the endoscope and the endoscope may fail.
- Wipe any debris, dust, and dirt away from the surfaces of the removing grip and tube connector using a clean cloth before holding the connecting grip of the ALT-Pro leak test air tube to connect the ALT-Pro leak test air tube to the tube connector. Otherwise, proper leakage testing cannot be performed.

CAUTION

- Be sure the ALT-Pro leak test air tubes are not tangled. Otherwise, an unintentional force may act on the tubes. As a result, the tubes may get damaged or flattened, preventing proper leakage testing.
- Always keep the air vents clear. Blockage can cause overheating and equipment damage.
- Securely insert the both ALT-Pro leak test air tubes into the tube connectors. Otherwise, an error code is displayed on the LCD in self-check performed when this equipment is turned ON. Also, the inside of endoscope may not be pressurized properly in leakage testing, resulting in a failure of leakage testing.
- Do not turn ON this equipment within 5 seconds after switching it OFF. This equipment may not perform properly or fail.

NOTE

- To inform users of the time to replace the internal pump, “Contact Olympus for Pump Replacement” will be lit up on the LCD upon power activation. When this indication is lit up, it is time to replace the pump. Contact Olympus for the pump replacement.
  - If the facility wants to continue to use this equipment before pump replacement, this equipment can be used continuously if there is no error in self-check. In that case, press the ENTER button to proceed to self-check. However, note that this indication appears every time turning on the power until the pump is replaced.
  - This indication is lit up when the total operation time of the pump reaches 700 hours. Refer to Section 7.8, “Checking equipment status” to check the total operation time of the pump.
- Disconnect the water-resistant cap from the ALT-Pro leak test air tube if connected. Otherwise, an error code is displayed on the LCD when self-check is performed.

1 Check that the outer surface of the ALT-Pro leak test air tubes is free from scratches and flaws.

2 Wipe any debris, dust, and dirt away from the surfaces of the removing grip and tube connector using a clean cloth before holding the connecting grip of the ALT-Pro leak test air tube to connect the ALT-Pro leak test air tube to the tube connector.

3 Visually check that the inside of the removing grip of the ALT-Pro leak test air tubes and the outside of the tube connector are free from water. If any water is observed, wipe it with clean cloth.

4 Using one hand to hold the equipment in place, hold the connecting grips of the ALT-Pro leak test air tube with the other hand and insert them into the tube connector 1 and tube connector 2 of this equipment until they stop. Check that they do not come out by pulling them slightly.

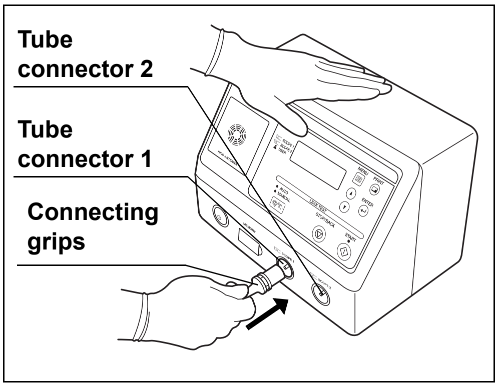


Figure 4.1

5 Set each ALT-Pro leak test air tubes in the tube hanger.

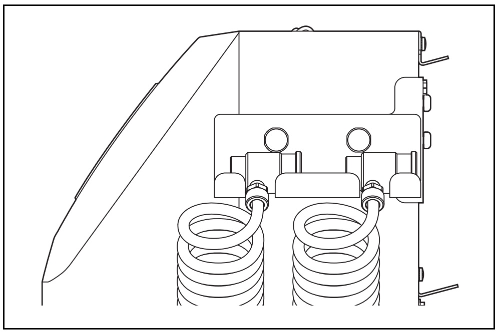


Figure 4.2

6 When using the printer, turn ON the power switch of the printer.

7 Turn ON the power switch of this equipment.

8 The power indicator will be turned ON. The startup screen and the connection status of the printer will be displayed on the LCD. (See Table 4.2)

Indicator	Description
OLYMPUS ALT–Pro Auto Print Disabled	Displayed when Auto Print is not set.
OLYMPUS ALT–Pro Auto Print Activated Printer ON	Displayed when Auto Print is set and the power of printer is turned ON.
OLYMPUS ALT–Pro Auto Print Activated Turn the Printer ON [↵]Begin Self Check	Displayed when Auto Print is set and the power of printer is not turned ON • When a printer is used, a self-check (Section 4.2) automatically starts by turning ON the printer. • When a printer is not used, a self-check (Section 4.2) automatically starts by pressing the ENTER button.

Table 4.2

4.2 Self-check

The self-check is performed to inspect for a leak from the channel and the ALT-Pro leak test air tubes and examine whether the pump or valves works properly. This self-check is automatically performed when the power is turned ON, but it can be also manually performed when necessary. For the manual self-check, refer to Section 7.9, “Performing only self-check”.

CAUTION

- Do not apply excessive force on the ALT-Pro leak test air tubes by forcibly bending them or placing anything on it. Doing so may prevent a proper self-check and damage the ALT-Pro leak test air tubes.
- Do not touch the ALT-Pro leak test air tubes during self-check. Otherwise, the self-check cannot be conducted properly.
- If the error code [112] is displayed on the LCD as a result of the self-check, follow the instruction in Section 8.1, “Troubleshooting guide”.

Ch.4

NOTE

When self-check is complete, the results of self-check are automatically recorded in this equipment.

1 Turn ON this equipment. A self-check will automatically start. The “Self Check In Process” indicator will be displayed on the LCD. The number of mark “■” of the progress bar will increase to show the progress of the self-check until all the process is completed.

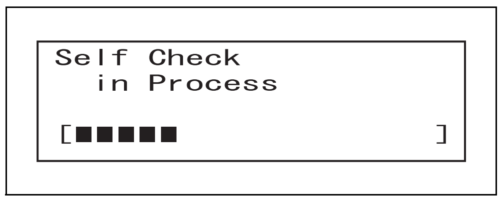


Figure 4.3

2 A short beep sounds when the self-check is completed properly. The “endoscope ID indicator (SCOPE1)” and “User ID indicator (USER)” blinks as shown in Figure 4.4, and the standby screen appears on the LCD.

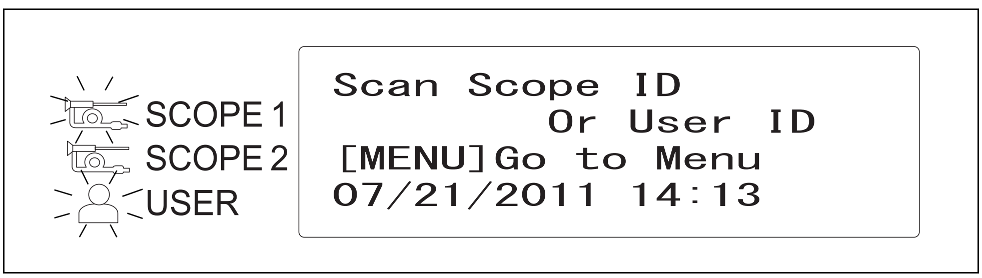


Figure 4.4

3 When using this equipment for the first time, set the date and time according to Section 7.6, “Setting date” and Section 7.7, “Setting time”.

4 Turn OFF this equipment.

If the error code [E112] is displayed during self-check

If the error code [E112] is displayed in the self-check, make sure of the following items.

- Check that the two ALT-Pro leak test air tubes are connected properly.
- Check that the water-resistant cap or endoscope is not connected to the ALT-Pro leak test air tubes.

After that, press the Enter button to display the Standby screen and restart a self-check referring to Section 7.9.

If the error code [E112] is still displayed, an irregularity may have occurred in this equipment or ALT-Pro leak test air tubes.
Follow the conditions below.

In case of having extra ALT-Pro leak test air tubes:

To clarify whether the cause of this error results from this equipment or ALT-Pro leak test air tubes, connect extra ALT-Pro leak test air tubes and perform a self-check.

- If the error code [E112] is displayed, that means an irregularity may have occurred in this equipment. Turn OFF the power switch and contact Olympus. When requesting repair, make sure to send not only this equipment but also ALT-Pro leak test air tubes to OLYMPUS.
- If the self-check is completed properly, that means an irregularity may have occurred in the ALT-Pro leak test air tubes. Replace the ALT-Pro leak test air tubes.

In case of not having extra ALT-Pro leak test air tubes:

Turn OFF the power switch and contact Olympus. When requesting repair, make sure to send not only this equipment but also ALT-Pro leak test air tubes to OLYMPUS.

4.3 Checking the printer paper roll

CAUTION

Do not start printing if the printer paper roll is incorrectly installed. Otherwise, the printer paper roll may jam and/or printing may fail.

Make sure that the printer paper roll has been installed in this equipment. If not, replace the printer paper roll according to Section 6.4, "Installing the printer paper roll".

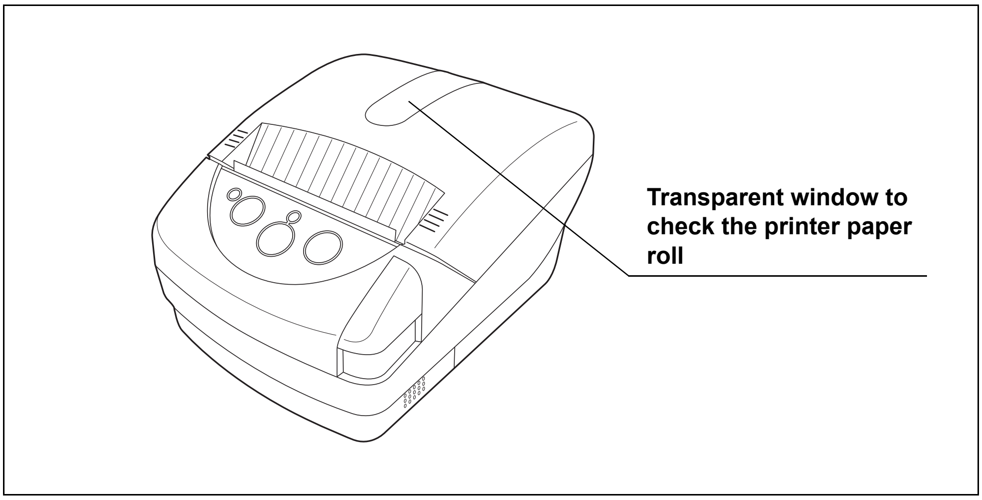


Figure 4.5

Chapter 5 Operation

This chapter explains how to perform a leakage test of the endoscope with this equipment.

5.1 Precautions for operation

WARNING

- Be sure to wear protective equipment such as protective eye wear, face mask, moisture-resistant gloves that fit properly and are long enough so that your skin is not exposed. Otherwise, dangerous chemicals and/or potentially infectious material such as blood and/or mucus of the patient may cause an infection.
- Wear gloves long enough to protect your skin and exchange them regularly so that they do not get torn.
- Should any irregularity be observed, do not use this equipment. Damage or irregularity may cause an electric shock, burns, and/or a fire.
- Do not allow the power cord to become wet. A wet power cord may cause an electric shock.
- After leakage testing with this equipment, reprocess the endoscope following all applicable national and local guideline. Endoscopy or endoscopic therapy with an endoscope not reprocessed may result in infections of a patient or operator.

CAUTION

- Always wait for more than three minutes after withdrawing the endoscope from the body, perform an automated leakage test. The endoscope upon removal from the body may be exposed to a sudden temperature change, resulting in an inaccurate test.
- Select the Immediate Mode as the automated leakage test mode (ALT mode) if less than 15 minutes have passed since the removal of the endoscope from the body. If 15 minutes or more have passed, select the Standard Mode. The endoscope upon removal from the body may have a temperature greatly different from ambient temperature and undergo a sudden temperature change. Thus selecting inappropriate ALT mode may result in an inaccurate test. (Refer to Section 7.5, “Setting automated leakage test mode”.)

CAUTION

- In some endoscope models, however, the Immediate Mode needs to be selected even if 15 minutes or more has passed since their removal from the body. To find these models, refer to “List of Compatible Endoscopes <ALT-Pro>”. Selecting the Standard Mode to perform an automated leakage test of these models may result in an inaccurate test.
- With some endoscope models (Additional Wait Time Scopes = AWT-Scopes), short beeps are generated and the monitor displays the screen as shown in Figure 5.1 when the scope ID recognition is performed. When an AWT scope is used, do not proceed to automated leakage test until 9 minutes or more have elapsed after withdrawal of the endoscope from the patient body. Otherwise, the test results may become inaccurate due to the temperature change. For the mode of automated leakage test of the AWT-Scope, select the Immediate Mode if the time elapsed from the endoscope withdrawal is less than 27 minutes or the Standard Mode when it is 27 minutes or more.

```
Caution
Additional Wait Time
Required Endoscope
See Instructions
```

Figure 5.1

Scope ID recognition	Time elapsed after withdrawal from body	Test Mode
Figure 5.1 is displayed.	≥9 min., <27 min.	Immediate Mode
	≥27 min.	Standard Mode

Table 5.1

- Install endoscopes to be automatically leak-tested in a location not exposed to direct sunlight and air conditioning. A rapid temperature change in the endoscopes may cause improper automated leakage testing.
- To avoid loss of master data, it is recommended that data in the ALT-Pro be regularly downloaded and managed according to Section 7.3, “Download to the portable memory” and Section 7.4, “Management of portable memory data in personal computer”.

Ch.5

NOTE

- To perform a leakage test in this equipment, an endoscope must be of a type with a built-in scope ID or have a scope ID tag. In addition, the user ID card is required.
- This equipment can be stopped at any time during operation by pressing the STOP/BACK button. When a process is interrupted, be sure to execute it again from the beginning.
- “Immediate Mode” has been set as the factory default setting for automated leakage testing. Refer to Section 7.5, “Setting automated leakage test mode” for the setting of the automated leakage test mode.

5.2 Operation workflow

The chart below shows the leakage test workflow. Review and understand each step before use to obtain an accurate leakage test result.

1	**Prepare equipment before use** **→Section 5.3 on page 43**
2	**Turn the power ON** **→Section 5.4 on page 44**
3	**Have the scope ID recognized** **→Section 5.5 on page 45**
4	**Connect the ALT-Pro leak test air tube with the endoscope** **→Section 5.6 on page 51**
5	**Have the user ID recognized** **→Section 5.7 on page 53**
6	**Perform leakage testing** • Automated leakage testing→Section 5.8 on page 54 • Manual leakage testing→Section 5.9 on page 64
7	**Care for the ALT-Pro leak test air tubes** **→Section 6.2 on page 70**
8	**Proceed to the next step** • To continue leakage testing→Section 5.5 on page 45 • To finish leakage testing→Section 6.3 on page 73

5.3 Preparation before use

CAUTION

Use a basin deep enough to allow complete immersion of the endoscope and large enough for the insertion section of the endoscope and the universal cord to be coiled into a diameter of 17 cm or over. If the insertion section or universal cord is coiled into a smaller diameter, the leakage testing may not be conducted properly.

Prepare equipment as shown below.

Check	Required items
	This Automated Endoscope Leak Tester
	ALT-Pro leak test air tubes (MAJ-2009) (× 2)
	Personal protective equipment
	A clean and large basin (Prepare two basins if two endoscopes are tested.)
	Endoscope with a built-in scope ID memory chip or a scope ID tag (MAJ-1545)
	Water-resistant cap (only for EVIS series endoscope)
	ID chip (MAJ-1546) (optional)
	Printer (MAJ-1937) (optional)
	ALT-Pro interface cable (MAJ-2052) (optional)

5.4 Power activation

NOTE

- When the printer is used, turn ON the printer before turning ON this equipment.
- For indicators on the printer after power activation, see Table 4.2 in Section 4.1, “Installing the ALT-Pro leak test air tubes and turning ON the power”.

1 Confirm that ALT-Pro leak test air tubes are surely connected to tube connector 1 and 2 of this equipment. Also confirm that there is no endoscope or water-resistant cap connected to the endoscope connectors of the ALT-Pro leak test air tubes at this point.

NOTE

An error code [E112] is displayed on turning power ON if the ALT-Pro leak test air tubes are not securely connected to this equipment, or if an endoscope or water-resistant cap is connected to the ALT-Pro leak test air tubes.

2 Turn the power switch ON.

3 A self-check starts automatically.

4 A short beep sounds when the self-check is completed properly. As seen in Figure 5.2, the endoscope ID indicator (SCOPE1) and User ID indicator (USER) indicators blink. The Standby screen is displayed on the LCD.

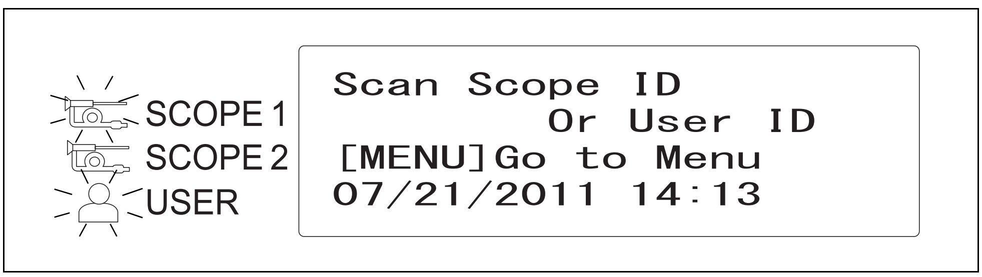


Figure 5.2

5.5 Recognition of scope ID

To keep an endoscopic leak test log, have this equipment recognize the individual scope ID that identifies the endoscope to be tested.

CAUTION

- Be sure to have this equipment recognize the scope ID of the endoscopes to be leak-tested. Otherwise, the leakage testing cannot be properly recorded, and automated leakage testing cannot be accurately performed.
- Do not bang the scope ID tag or endoscope connector onto the RFID reader during scope ID recognition. This may damage the scope ID tag, endoscope connector, and RFID reader.
- Some endoscope models need to be tested alone in automated leakage testing. To find these models, refer to “List of Compatible Endoscopes <ALT-Pro>”. If an endoscope of this model is tested simultaneously with another endoscope, the result of the automated leakage test may be inaccurate.
- With some endoscope models (Additional Wait Time Scopes = AWT-Scopes), short beeps are generated and the monitor displays the screen as shown in Figure 5.3 when the scope ID recognition is performed. When an AWT scope is used, do not proceed to automated leakage test until 9 minutes or more have elapsed after withdrawal of the endoscope from the patient body. Otherwise, the test results may become inaccurate due to the temperature change. For the mode of automated leakage test of the AWT-Scope, select the Immediate Mode if the time elapsed from the endoscope withdrawal is less than 27 minutes or the Standard Mode when it is 27 minutes or more.

```
Caution
Additional Wait Time
Required Endoscope
See Instructions
```

Figure 5.3

Scope ID recognition	Time elapsed after withdrawal from body	Test Mode
Figure 5.3 is displayed.	≥9 min., <27 min.	Immediate Mode
	≥27 min.	Standard Mode

Table 5.2

Ch.5

NOTE

- The START button on the main control panel does not function unless the User ID has been recognized.
- Two endoscopes may not always be tested simultaneously depending on the compatibility of the endoscopes. Refer to the “List of Compatible Endoscopes <ALT-Pro>” for the endoscopes that cannot be tested simultaneously. If you attempt to have this equipment recognize the second ID, the panel will display the message as shown in Figure 5.4, and the scope ID cannot be detected. Press the ENTER button to restart the user ID detection procedure.

```
Caution
Incorrect Endoscope
Combination
[↵]Enter
```

Figure 5.4

- Endoscope without the scope ID card can be manually leak-tested by using the scope ID master card. (Such endoscopes cannot be automatically leak-tested.) However, two of such endoscopes cannot be tested simultaneously. Perform manual leak tests of such endoscopes individually. If the scope ID master card is properly recognized, the screen will display the indicator as shown in Figure 5.5.


```
Connect Scope1
ScopeID Master
```

Figure 5.5

- If the scope ID master card is used, the history logs of the endoscope cannot be managed.
- If any irregularity is observed on the scope ID recognition of this equipment, or if any irregularity is observed on the scope ID, contact Olympus.
- For addition or reissue of a scope ID, contact Olympus.

1 Confirm that ALT-Pro leak test air tubes are surely connected to tube connector 1 and 2 of this equipment. Also confirm that any endoscope nor no water-resistant caps are disconnected to the endoscope connectors of ALT-Pro leak test air tubes.

NOTE

The error code [E123] is displayed on the monitor when you try to read the scope ID with the following conditions:

- The ALT-Pro leak test air tubes are disconnected
- An endoscope or water-resistant cap is connected to the ALT-Pro leak test air tubes

2 Hold the internal ID-type endoscope connector or the external ID tag to the RFID reader of this equipment, and scan the tag with the reader until a short beep sounds.

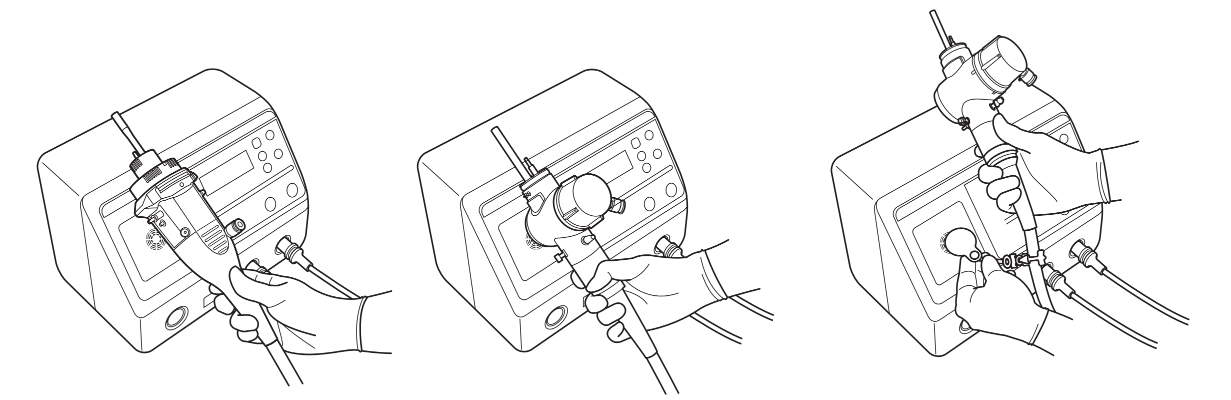


Figure 5.6

Ch.5

NOTE

With some endoscope models (Additional Wait Time Scopes = AWT-Scopes), short beeps are generated and the monitor displays the screen as shown in Figure 5.7 when the scope ID recognition is performed. Proceed to the “next operation” in Table 5.3 according to the test mode being selected for the AWT-Scope and the time elapsed after withdrawal of the endoscope from the patient body.

```
Caution
Additional Wait Time
Required Endoscope
See Instructions
```

Figure 5.7

Test mode being selected	Time elapsed after withdrawal from body	Next operation
Immediate Mode	<9 min.	After pressing the STOP/BACK button, press the ENTER button to put the screen on standby. After waiting for over 9 minutes, start the operation from Step 2.
	≥9 min., <27 min.	Go directly to Step 3.
Standard Mode	≥27 min.	After pressing the STOP/BACK button, press the ENTER button to put the screen on standby. After waiting for over 27 minutes, start the operation from Step 2. (Alternatively, if the elapsed time is 9 minutes or more, change the test mode to the Immediate Mode and start operation from Step 2.)
	<27 min.	Go directly to Step 3.

Table 5.3

3 Make sure that the endoscope ID indicator "SCOPE 1" lights up on the control panel.

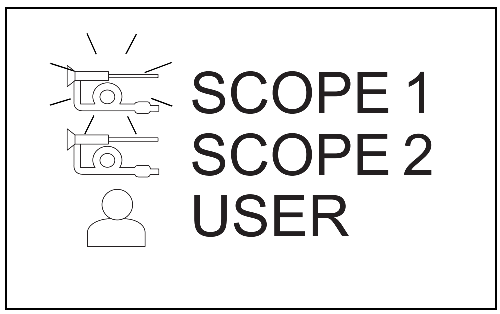


Figure 5.8

4 Check that the tube connector 1 indicator is blinking. Connect an endoscope to the ALT-Pro leak test air tube connected to tube connector 1 according to the instruction described in Section 5.6, "Connecting ALT-Pro leak test air tubes and endoscope".

NOTE

- An error code [E123] is displayed if the endoscope is connected to the ALT-Pro leak test air tube before the tube connector 1 indicator is lit. If this error code is displayed, press the Enter button and detach the endoscope from the ALT-Pro leak test air tube. Go back to beginning of Section 5.5 and repeat the steps.
- The error code [E122] is displayed when, in Step 4 above, an endoscope is connected to the ALT-Pro leak test air tube that has been connected to tube connector 2. Press the Enter button and detach the endoscope from the ALT-Pro leak test air tube. Go back to beginning of Section 5.5 and repeat the steps.

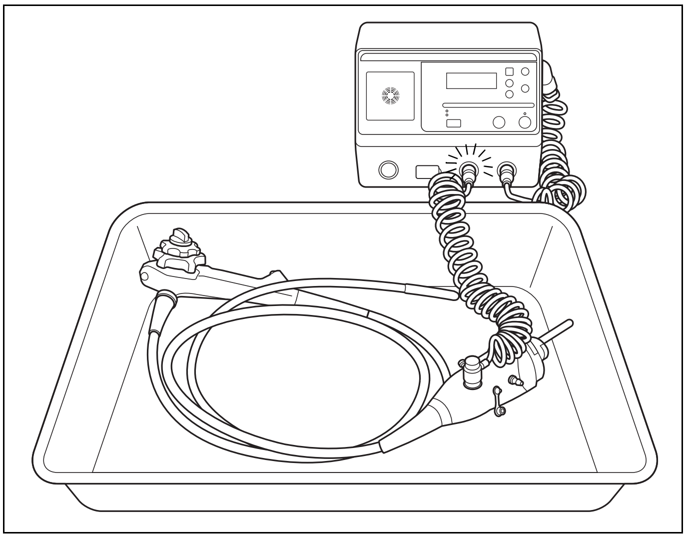


Figure 5.9

5 To leak test a second endoscope, repeat Step 1 through 4 above, and connect the second endoscope to the ALT-Pro leak test air tube connected to “tube connector 2".

NOTE

- When the first endoscope is detected, the endoscope ID indicator “SCOPE 1” lights up.
- When the second endoscope is detected, the endoscope ID indicator “SCOPE 2” lights up.
- To replace the endoscope to be tested, press the STOP/BACK button and disconnect the endoscope from the ALT-Pro leak test air tube. Go back to the beginning of Section 5.5 and repeat the steps.

Ch.5

The LCD indicators change as shown in Figure 5.10 below when the ID is entered or when the endoscope is connected.

```
Scan Scope ID
        Or User ID
[MENU]Go to Menu
07/21/2011 14:13
```

(a) Standby screen (Waiting for ID entry)

Recognition of scope ID (of a first endoscope)

```
Connect Scope1
GIF-XXX     1234567
```

Endoscope Code No./Serial No.

(b) Waiting for the connection of an endoscope

Connect the endoscope to Scope Connector 1

```
Scan User/Scope2 ID
GIF-XXX     1234567
```

(c) Waiting for ID entry

Recognition of scope ID (of a second endoscope)

```
Connect Scope2
GIF-XXX     1234567
GIF-YYY     7654321
```

(d) Waiting for the connection of another endoscope

Connect the endoscope to Scope Connector 2

```
Scan User ID
GIF-XXX     1234567
GIF-YYY     7654321
```

(e) Standby screen (Waiting for ID entry

Figure 5.10

NOTE

When the STOP/BACK button is pressed in any screen from (b) to (e) in Figure 5.10 above, the screen returns to “(a) Standby screen (Waiting for the input of ID)”. In this case, all the scope IDs and user IDs need recognizing again.

5.6 Connecting ALT-Pro leak test air tubes and endoscope

Set the endoscope for automated leakage testing and connect it to the ALT-Pro leak test air tube.

CAUTION

- For automated leakage test confirm that the basin is dry. It should not contain any water or disinfectant. If there is any pin hole in the endoscope, the fluid may enter the endoscope, resulting in malfunction of the endoscope.
- For an endoscope that requires a water-resistant cap, be sure to inspect and attach the cap as described in the instruction manual for the endoscope. Otherwise, the inside of endoscope may not be pressurized, resulting in inaccurate leakage testing. Attaching the water-resistant cap that is wet inside may cause malfunction of the endoscope.
- When attaching the endoscope side connector of the ALT-Pro leak test air tube to the venting connector of the endoscope or water-resistant cap, check if the O ring inside the endoscope connector is present and free from scratches or flaws. Otherwise, accurate leakage testing cannot be conducted.
- When attaching the endoscope connector of the ALT-Pro leak test air tube to the venting connector of the endoscope or water-resistant cap, wipe out any water drop inside the endoscope connector and outside the venting connector. Such a water drop may enter the endoscope, resulting in malfunction of the endoscope.
- When attaching the endoscope connector of the ALT-Pro leak test air tube to the venting connector of the endoscope or water-resistant cap, wipe out any dust, dirt, or stain inside the endoscope connector and outside the venting connector. Otherwise, accurate leakage testing cannot be conducted.
- Be sure to attach the endoscope side connector of ALT-Pro leak test air tube to the venting connector of the endoscope or water-resistant cap. Otherwise, the inside of endoscope is not pressurized, resulting in inaccurate leakage testing.
- Do not place two endoscopes in the basin with any part of them stacked with each other. Accurate leakage testing cannot be conducted.
- Place the endoscope in the basin with the bending section of the endoscope straight and the insertion section and universal cord coiled into a diameter of 17 cm or over in performing automated leakage testing. Otherwise, accurate leakage testing cannot be conducted.

1 Confirm that the basin does not contain any water or disinfectant solution.

2 Place the endoscope in the basin. (When two endoscopes are tested simultaneously, place each endoscope into a different basin.)
Make the bending section of the endoscope straight and the insertion section and universal cord coiled into a diameter of 17 cm or over.

3 Visually check that there is no water drop, dust, dirt, nor stain inside the endoscope side connector of the ALT-Pro leak test air tube and outside the venting connector of either endoscope or water-resistant cap.

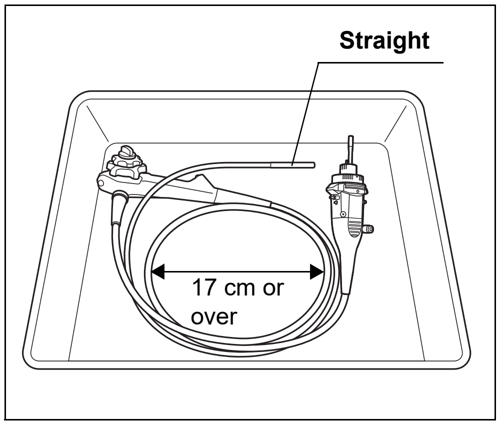


Figure 5.11

4 Turn the endoscope side connector of the ALT-Pro leak test air tube clockwise until it stops, pushing it to the venting connector as shown in Figure 5.12.

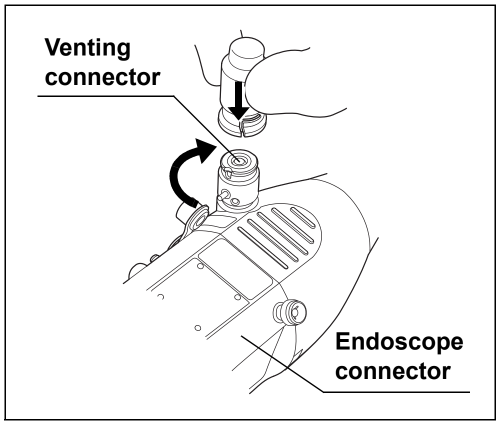


Figure 5.12

5 When the ALT-Pro leak test air tube is connected to the endoscope, the tube connector Indicator stops blinking and starts to light.

Ch.5

5.7 Recognition of user ID

To keep an endoscope leak test log, have this equipment recognize the individual User ID that identifies the reprocessing operator.

NOTE

- The START button on the main control panel does not function unless the User ID has been recognized.
- For addition or re-issue of the user ID card, contact Olympus.
- If this equipment fails to detect the user ID, apply the provided user ID master card to the RFID reader so that this equipment recognizes the user ID as the master ID. If a problem with the user ID is observed, contact Olympus.

1 Bring the user ID card of the leakage testing worker close to the RFID reader of this equipment until a short beep sounds.

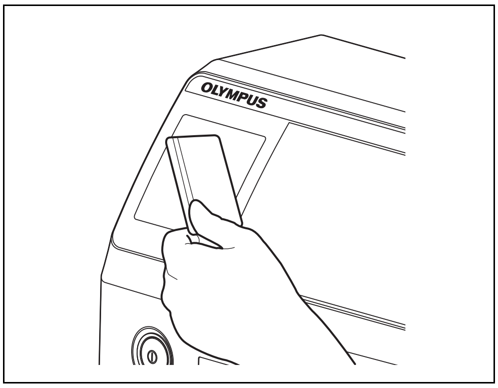


Figure 5.13

2 Confirm that the User ID detection indicator on the control panel is turned ON.

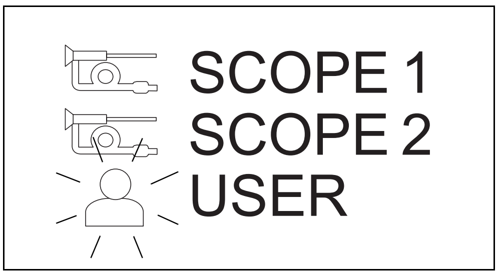


Figure 5.14

NOTE

- When the User ID detection indicator does not illuminate on the control panel, the user ID card has not been recognized. Bring the user ID card again close to the RFID reader of equipment.
- To replace the endoscope to be tested, press the STOP/BACK button and disconnect the endoscope from the ALT-Pro leak test air tube. Go back to the beginning of Section 5.5 and repeat the steps.

5.8 Automated leakage testing

This equipment features the following functions for the automated leakage testing:

- This equipment can perform a leakage test in air in an ALT; you do not need to store water in the basin.
- This equipment performs automated leakage testing of the endoscope.
- This equipment can store the records of leakage testing.

WARNING

Be sure to wear protective gear such as moisture-resistant clothes, goggles, a face mask, chemical-resistant gloves that fit properly and are long enough so that your skin is not exposed. Otherwise, potentially infectious materials attached to the endoscope, such as blood and/or mucus of the patient, may cause an infection.

CAUTION

- Do not immerse the endoscope in water nor touch it during automated leakage testing. Accurate leakage testing cannot be conducted.
- Some endoscope models need to be tested alone in automated leakage testing. To find these models, refer to “List of Compatible Endoscopes <ALT-Pro>”. If an endoscope of this model is tested simultaneously with another endoscope, the result of the automated leakage test may be inaccurate.
- In some endoscope models, however, the Immediate Mode needs to be selected even if 15 minutes or more has passed since their removal from the body. To find these models, refer to “List of Compatible Endoscopes <ALT-Pro>”. Selecting the Standard Mode to perform an automated leakage test of these models may result in an inaccurate test.

CAUTION

- With some endoscope models (Additional Wait Time Scopes = AWT-Scopes), short beeps are generated and the monitor displays the screen as shown in Figure 5.15 when the scope ID recognition is performed. When an AWT scope is used, do not proceed to the automated leakage test until 9 minutes or more have elapsed after withdrawal of the endoscope from the patient body. Otherwise, the test results may become inaccurate due to the temperature change. For the mode of automatic leak test of the AWT-Scope, select the Immediate Mode if the time elapsed from the endoscope withdrawal is less than 27 minutes or the Standard Mode when it is 27 minutes or more.

```
Caution
Additional Wait Time
Required Endoscope
See Instructions
```

Figure 5.15

Scope ID recognition	Time elapsed after withdrawal from body	Test Mode
Figure 5.15 is displayed.	≥9 min., <27 min.	Immediate Mode
	≥27 min.	Standard Mode

Table 5.4

NOTE

- Automated leakage testing cannot start without recognition of both scope ID and user ID. Read the both scope ID and user ID, referring to Section 5.5, “Recognition of scope ID” and Section 5.7, “Recognition of user ID”.
- For information on how to print past records refer to Section 7.10, “Manual printing”.
- The air fed into the endoscope may slightly expand the rubber cover of the bending section. This is not an irregularity.
- When a leakage test ends, the result is automatically recorded in this equipment.
- Up to 13,000 records of leakage testing can be stored in this equipment. If more than 13,000 records are stored, the oldest record is overwritten.
- To stop the leakage test during operation, press the STOP/BACK button.
- A very small pin hole may not be detected in automated leakage testing.
- Endoscopes not listed in the “List of Compatible Endoscopes <ALT-Pro>” cannot be automatically leak-tested. If an incompatible endoscope is connected, only the MANUAL light lights up when the SELECT button is pressed.

NOTE

- If automated leaking test process stops with the error code [E115] displayed on the monitor, there is either a large leak in the endoscope or improper connection of the ALT-Pro leak test air tubes or water-resistant cap is suspected. Press the ENTER button and confirm that the ALT-Pro leak test air tube(s) and water-resistant cap are properly connected to the ALT-Pro before perform an automated leaking test again. If the automated leaking test process still stops with the error code [E115] displayed on the monitor, a significant leak is suspected on the endoscope(s). If the error occurs while only one endoscope is being automatically leak tested, perform a manual leaking test of the endoscope to locate the leak. If the error occurs while two endoscopes are being automatically leak tested, automatically leak test the endoscopes one by one. If the error (code [E115]) is still displayed, perform a manual leaking test to locate the leak.
- If the error code [E115] occurs in succession, contact Olympus.

1 Check the LCD to see if the scope ID and user ID have been recognized.

```
[START]Begin AUTO S
GIF-XXX     1234567
GIF-YYY     7654321
John Smith
```

Figure 5.16

NOTE

"S" displayed at the upper left of the LCD indicates the Standard Mode of the automated leakage testing mode.
When the Immediate Mode is selected, "I" is displayed. Refer to Section 7.5, "Setting automated leakage test mode" for setting the modes.

2 Make sure that the AUTO light is turned ON. When the MANUAL light is turned ON, press the SELECT button to turn ON the AUTO light.

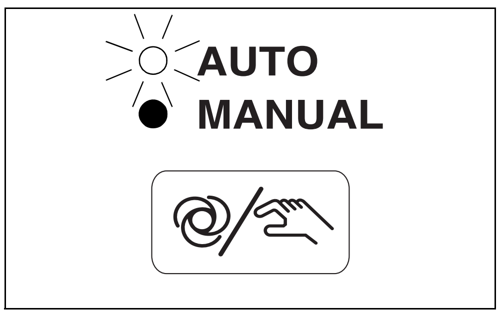


Figure 5.17

3 Press the START button. An automated leakage test will start. The number of mark "■" of the progress bar will increase as the process of automated leakage testing goes on to show how much time is left before the test is completed.

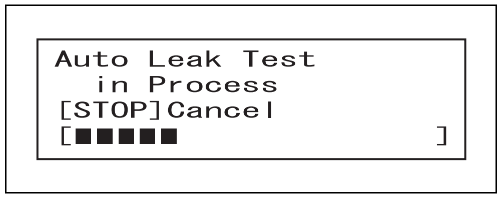


Figure 5.18

NOTE

An automated leakage test is completed in about 140 seconds (2 minutes and 20 seconds).

4 The automated leakage testing is automatically completed. When no leak is detected, a short beep sounds, and The LCD alternately displays the two ALT Results indicators (A) and (B) as shown in Figure 5.19. When any leakage is detected in the testing, long beeps sound, and the ALT Results (see C in Figure 5.19) indicator is displayed.

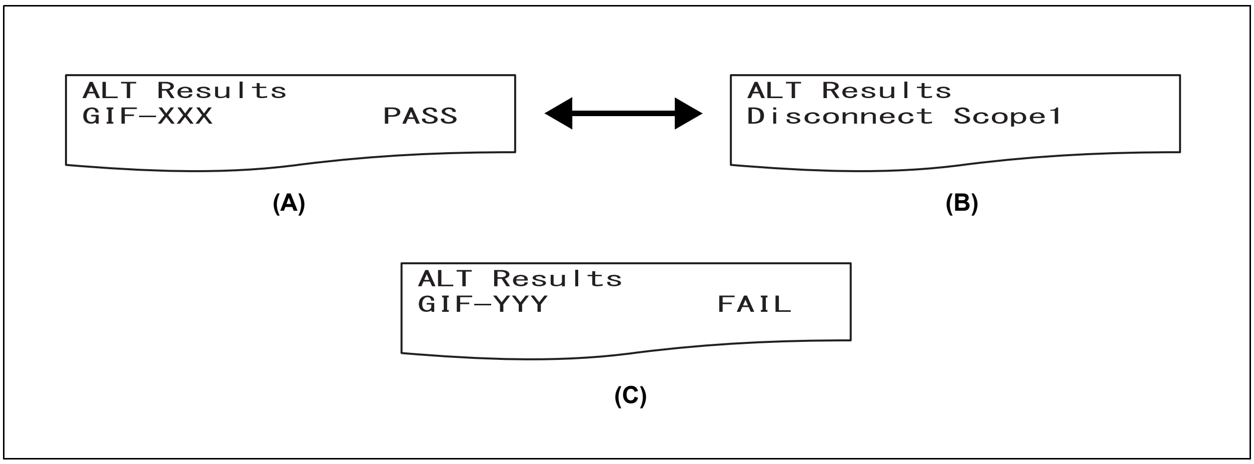


Figure 5.19

NOTE

- When the both endoscopes pass the leakage test, the test data is automatically sent to the connected printer for printing. When at least one endoscope fails to pass the leakage test, the data is not printed at this time. When you press the ENTER button to finish the automated leakage test, the result of automated leakage test is printed. When you perform a manual test immediately after the automated leakage test, the result of automated leakage test will be printed when the manual test is completed.
- If an automated leakage test in the Standard Mode starts more than 3 minutes but less than 15 minutes after the endoscopes withdrawal from the body, the result of the leakage test may be FAIL although the endoscope has no pin hole or damaged area. To select the appropriate mode, refer to Section 7.5, "Setting automated leakage test mode".

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NOTE

- With some endoscope models (Additional Wait Time Scopes = AWT-Scopes), short beeps are generated and the monitor displays the screen as shown in Figure 5.20 when the scope ID recognition is performed. When the automated leakage test in the Standard Mode is started with such an endoscope in 9 to less than 27 minutes after the endoscope withdrawal, the results of an endoscope without leak may sometimes become FAIL. To select the appropriate mode, refer to Section 7.5, “Setting automated leakage test mode”.

```
Caution
Additional Wait Time
Required Endoscope
See Instructions
```

Figure 5.20

- If the Standard Mode is selected to perform an automated leakage test of the endoscopes that should be tested in the Immediate Mode, the result of the automated leakage test may be FAIL even if the endoscopes have no pin hole or damaged area. To select the appropriate mode, refer to “List of Compatible Endoscopes <ALT-Pro> and Section 7.5, “Setting automated leakage test mode”.

5 Perform the following steps depending on the results in the leakage testing.

■ *The results in the automated leakage testing are all PASS:*

1 Press the ENTER button if all the results are PASS. The screen will be returned to the Standby screen.

```
Scan Scope ID
        Or User ID
[MENU]Go to Menu
07/21/2011 14:13
```

Figure 5.21

2 Rotate the endoscope side connector of the ALT-Pro leak test air tube counterclockwise to disconnect it.

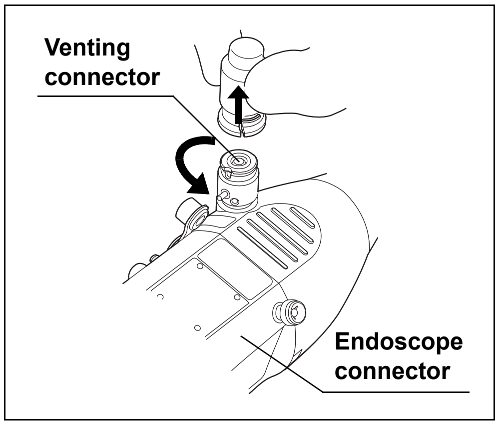


Figure 5.22

NOTE

When the endoscope requires to be connected with the water-resistant cap, remove the endoscope side connector of the ALT-Pro leak test air tube from the venting connector of the water-resistant cap.

3 Maintain the ALT-Pro leak test air tubes according to Section 6.2, “Care of ALT-Pro leak test air tubes”.

Ch.5

■ *If the result of leakage testing for at least one endoscope is FAIL*

If at least one endoscope fails to pass the automated leakage testing, check that the following set up conditions are met:

- Endoscope coil diameter (see Figure 5.11 in Section 5.6, “Connecting ALT-Pro leak test air tubes and endoscope”)
- ALT-Pro leak test air tube connections.
- Integrity of the ALT-Pro leak test air tube O-rings.
- Elapsed time of the endoscope's removal from the body (see Table 7.5 in Section 7.5, “Setting automated leakage test mode”)
- Water-resistant cap alignment (180 model or before) as described in the instruction manual for the endoscope
- Integrity of water-resistant cap packing (180 model or before)

Then perform an automated leakage testing again. If the result is still FAIL, a leak is suspected on the endoscope(s). Perform a manual leakage testing to locate the leak referring to Section 5.9, “Manual leakage testing”.

NOTE

When any leakage is detected in the testing, long beeps keep sounding. Press the ENTER button or START button to turn off the beep sound. To perform an automated leakage test again, press the ENTER button; or to perform a manual leakage test to locate the leak, press the START button.

1 Press the ENTER button. The standby monitor is displayed.

```
Scan Scope ID
        Or User ID
[MENU]Go to Menu
07/21/2011 14:13
```

Figure 5.23

2 Rotate the endoscope side connector of the ALT-Pro leak test air tube counterclockwise to disconnect it.

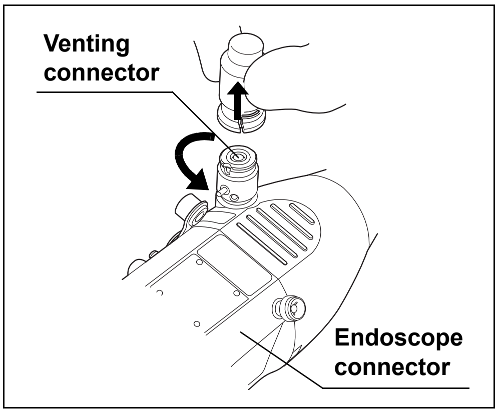


Figure 5.24

NOTE

When the endoscope requires to be connected with the water-resistant cap, remove the endoscope side connector of the ALT-Pro leak test air tube from the venting connector of the water-resistant cap.

3 If the result is FAIL, perform automated leakage testing procedure setup starting from Section 5.5, “Recognition of scope ID” through Section 5.8, “Automated leakage testing”.

4 If the results of the automated leakage test are all PASS, press the ENTER button and rotate the endoscope side connector of the ALT-Pro leak test air tube counterclockwise to disconnect the endoscope.

5 If the result of at least one endoscope is FAIL, perform the following steps without pressing the ENTER button.

- If one of the endoscopes has passed the automated leakage test, disconnect the endoscope from the ALT-Pro leak test air tube.
- If the endoscope fails the automated leakage test, go to Step 4 of Section 5.9, “Manual leakage testing” to perform a manual leakage test to locate the leak of the endoscope.

CAUTION

If you fail to disconnect the endoscope that has passed the automatic leakage testing before starting a manual test, do not disconnect the endoscope until the manual leakage test is completed.

NOTE

- If Standby screen is shown by pressing ENTER button, perform manual leakage testing referring to Section 5.9, “Manual leakage testing”.
- Contact Olympus if a manual leakage test cannot locate the leak of the endoscope that has failed twice in a row in the automated leakage testing.

If the error code [E115] is displayed on the screen

If the automated leaking test process stops with the error code [E115], the error is attributed to two possible causes: (1) A significant leak on the endoscope, and (2) improper connection of the ALT-Pro leak test air tube(s) and water-resistant cap to the ALT-Pro. Confirm that the ALT-Pro leak test air tube(s) and water-resistant cap are properly connected to the ALT-Pro and start an auto leakage testing again. If the error code [E115] is displayed again, a significant leak is suspected on the endoscope(s). The process to determine which endoscope has a leak differs depending on the number of the endoscopes automatically leak tested.

1 Press the ENTER button. The standby monitor is displayed.

```
Scan Scope ID
        Or User ID
[MENU]Go to Menu
07/21/2011 14:13
```

Figure 5.25

2 Confirm that the ALT-Pro leak test air tube(s), water-resistant cap, and endoscope(s) are properly connected to the ALT-Pro.

3 Rotate the endoscope side connector of the ALT-Pro leak test air tube counterclockwise to disconnect it.

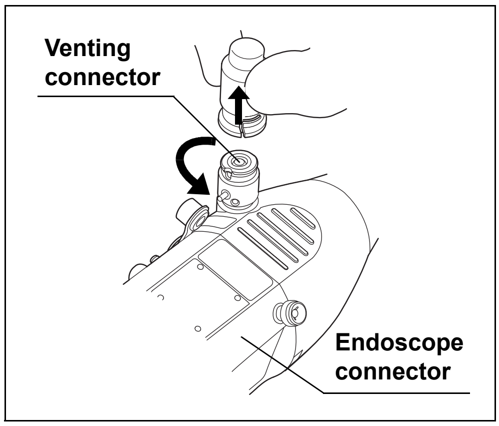


Figure 5.26

NOTE

When the endoscope requires to be connected with the water-resistant cap, remove the endoscope side connector of the ALT-Pro leak test air tube from the venting connector of the water-resistant cap.

4 Perform an automated leakage test as described below. The process differs depending on the number of the endoscopes to be automatically leak tested.

- When only one endoscope is being tested: If the error [E115] occurs during automated leakage test, perform an automated leakage test again according to the steps from Section 5.5, "Recognition of scope ID" through Section 5.8, "Automated leakage testing".
- When two endoscopes are being tested: If the error [E115] occurs during automated leakage test perform an automated leakage test of the endoscopes one by one according to the steps from Section 5.5, "Recognition of scope ID" through Section 5.8, "Automated leakage testing".

5 If the error [E115] still occurs in Step 4, perform a manual leakage test to locate the leak referring to Section 5.9, "Manual leakage testing".

NOTE

Contact Olympus if the errors [E115] occur twice in a row in automated leakage tests and the leak cannot be located in the manual leakage test.

5.9 Manual leakage testing

The manual leakage testing features the following functions:

- This equipment allows visual inspection of the immersed endoscope for any small hole on the surface of the endoscope that causes an air leak.
- The history of manual leakage tests can be recorded.

WARNING

Be sure to wear protective gear such as moisture-resistant clothes, goggles, a face mask, chemical-resistant gloves that fit properly and are long enough so that your skin is not exposed. Otherwise, potentially infectious materials attached to the endoscope, such as blood and/or mucus of the patient, may cause an infection.

CAUTION

- If continuous series of air bubbles come out of the endoscope during a manual leakage test, water may enter the endoscope from where such air bubbles come out. If continuous series of air bubbles come out, remove the endoscope from the water without pressing the STOP/BACK button. Remove the endoscope and contact Olympus for how to reprocess and send the endoscopes with a leak for repair.
- Do not attach and detach the endoscope connector of the water-resistant cap or ALT-Pro leak test air tube in water. Doing so may cause water to enter the endoscope, resulting in a malfunction of the endoscope.
- Be sure to attach the endoscope side connector of the ALT-Pro leak test air tube to the venting connector of the water-resistant cap. Otherwise, the inside of endoscope is not pressurized; the leakage testing cannot be conducted.
- When attaching the endoscope side connector of the ALT-Pro leak test air tube to the water-resistant cap, thoroughly wipe any residual water out of the inside of the endoscope side connector of the ALT-Pro leak test air tube and the outside of the venting connector of the water-resistant cap. Otherwise, water may penetrate the endoscope, resulting in a malfunction of the endoscope.
- To interrupt the manual leakage test, take out the endoscope from the basin before pressing the STOP/BACK button.
- Using water that is significantly different from room temperature (hot or warm) may cause the error code [E117]. Use only water that is close to room temperature.

NOTE

- Unless the scope ID and the user ID are detected, the manual leakage testing cannot start. To read the scope ID and the user ID, refer to Section 5.5, “Recognition of scope ID” and Section 5.7, “Recognition of user ID”.
- Refer to Section 7.10, “Manual printing” for how to set Auto Print and print out the log.
- The air fed into the endoscope may slightly expand the bending section rubber cover during leakage testing. This is not an irregularity.
- In a manual leakage test, a short beep keeps sounding in 55 minutes after the start of the manual leakage test to show that pressurization will end soon. The pressurization automatically ends in 60 minutes. When continuous series of air bubbles are found emerging from the endoscope, remove the endoscope from the water. Pressing the STOP/BACK button can terminate the test.
- When a manual leakage test ends, this equipment automatically records the test as history.
- The manual leakage testing can be usually applied to two endoscopes simultaneously; however, it may be applied only to one endoscope depending on the compatibility of the endoscopes. If the scope ID master card is used, the manual leakage testing can be applied only to one endoscope.

1 Use clean water that is close to room temperature and fill the basin deep enough to submerse the endoscope completely. (Do not use hot or warm water.)

2 Confirm that the scope ID and the user ID have been read with the numbers indicated on the LCD of the control panel.

Example) The indicator of the LCD when two endoscopes have been recognized:

```
[START]Begin Manual
GIF-XXX      1234567
GIF-YYY      7654321
John Smith
```

Figure 5.27

3 Confirm that the MANUAL light is turned ON. When the AUTO light is turned ON, press the SELECT button to turn ON the MANUAL light.

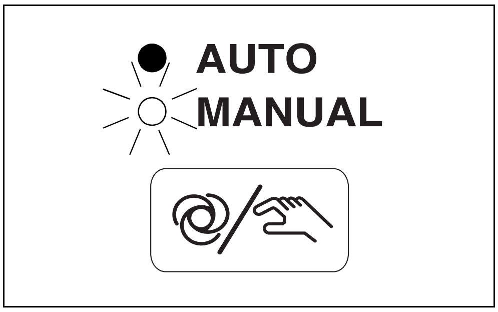


Figure 5.28

4 Press the START button, and the indicator shown in Figure 5.29 will be displayed. Do not immerse the endoscope in water while the indicator is being displayed because the endoscope has not been sufficiently pressurized.

```
Manual Leak Test
Please wait...
```

Figure 5.29

CAUTION

Do not immerse in water the endoscope not sufficiently pressurized. Otherwise, water may penetrate the endoscope, resulting in a malfunction of the endoscope.

5 Confirm that the indicator as shown in Figure 5.30 is displayed on the LCD. Immerse the endoscope in a basin that contains clean water.

```
Manual Leak Test
  in Process

[STOP]Stop Leak Test
```

Figure 5.30

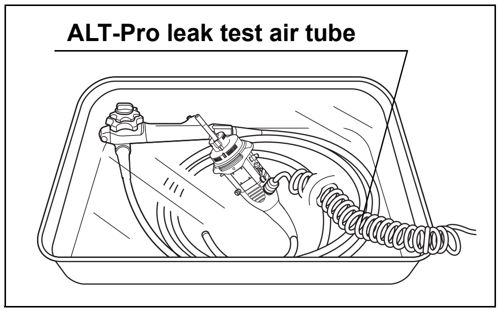


Figure 5.31

Ch.5

CAUTION

Once a manual leakage test starts, do not press the pin on the endoscope connector of the ALT-Pro leak test air tube and avoid connecting and disconnecting of the endoscope to the endoscope connector. Doing so may reduce air pressure within this equipment and cause water ingress into the endoscope being immersed in water, resulting in malfunction of the endoscope.

6 Observe the endoscope for about 30 seconds to inspect for a series of bubble being emitted from the endoscope while bending the bending section by turning the UD and RL angle knobs of the endoscope. (For the details, refer to the instruction manual of the endoscope.)

NOTE

A continuous series of air bubbles emerging from any location on the endoscope indicates a leak at that location. If there is a leak in the instrument channel or suction channel of the endoscope, a continuous series of air bubbles will emerge from one or more channel openings (e.g., distal end, suction connector, suction cylinder, instrument channel port) on the immersed endoscope.

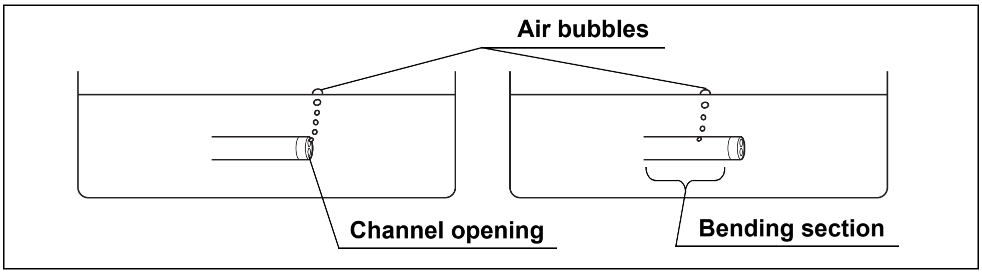


Figure 5.32

7 Remove the endoscope from the water.

CAUTION

Do not press the STOP/BACK button until the endoscope has been fully removed from the water. Pressing the STOP / BACK button while the endoscope is immersed increases the potential of fluid entry into the endoscope through the leakage site. Contact Olympus for instructions regarding the proper reprocessing of leaking endoscopes, and then return the endoscope for repair.

8 When the STOP/BACK button is pressed, a short beep sounds, and the indicator shown in Figure 5.33 is displayed on the LED monitor. The pressurized air is released from the endoscope.

```
Stopping Leak Test
Please wait···
```

Figure 5.33

9 After the completion of air purge and the manual leakage test, the LCD alternately displays two indicators, one showing the type of the endoscope and the other showing the next step as shown in Figure 5.34.

Example) Indicators when two endoscopes have been recognized

```
MLT Stopped
GIF-XXX
GIF-YYY
[↵]Exit
```

```
MLT Stopped
Disconnect Scope1
Disconnect Scope2
[↵]Exit
```

Figure 5.34

NOTE

When Auto Print is set, the history of manual leakage testing is printed. Refer to Section 7.10, “Manual printing” for how to set the print mode.

10 Press the ENTER button. The screen will go back to Standby screen.

```
Scan Scope ID
        Or User ID
[MENU]Go to Menu
07/21/2011 14:13
```

Figure 5.35

11 Rotate the endoscope side connector of the ALT-Pro leak test air tube counterclockwise to disconnect it.

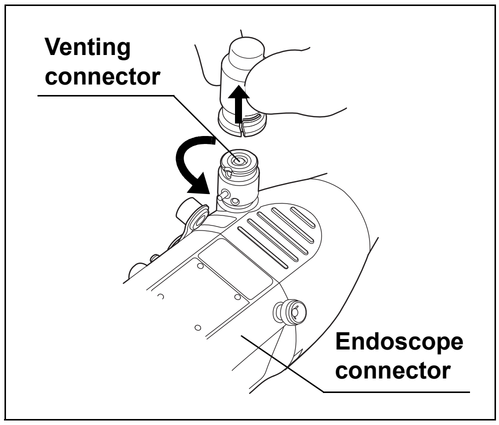


Figure 5.36

NOTE

When the endoscope requires to be connected with the water-resistant cap, remove the endoscope side connector of the ALT-Pro leak test air tube from the venting connector of the water-resistant cap.

12 Maintain the ALT-Pro leak test air tubes according to Section 6.2, “Care of ALT-Pro leak test air tubes”.

Chapter 6 Routine Maintenance

6.1 Summary

To ensure safety operation of this equipment, inspect and clean this equipment regularly.

❍ Inspection to be performed after each use

→Refer to Section 6.2, “Care of ALT-Pro leak test air tubes”.

❍ Inspection to be performed daily after use

→Refer to Section 6.3, “Turning power OFF and cleaning outer surface”.

❍ Work to be performed as required

→Refer to Section 6.4, “Installing the printer paper roll”.

→Refer to Section 6.5, “Replacing the fuse”.

→Refer to Section 6.6, “Storage”.

WARNING

- Be sure to inspect and clean this equipment as described in this chapter. Otherwise, the functions and performance of this equipment may not be performed properly.
- Should any irregularity be observed, do not use this equipment and contact Olympus. Damage or irregularity may compromise patient or user safety, and more severe equipment damage may result.
- Be sure to wear protective equipment such as protective eye wear, face mask, moisture-resistant gloves that fit properly and are long enough so that your skin is not exposed. Otherwise, dangerous chemicals and/or potentially infectious material such as blood and/or mucus of the patient may cause an infection.

6.2 Care of ALT-Pro leak test air tubes

Check	Required items
	Clean lint-free cloth
	Clean Sponge
	70% ethyl alcohol or isopropyl alcohol
	Detergent
	Two clean basins (for cleaning and rinsing)

Table 6.1

WARNING

- Both 70% ethyl alcohol and isopropyl alcohol are combustible. Handle them with care.
- Excessive detergent foaming can prevent fluid from adequately contacting surfaces and compromise the cleaning performance.
- Do not reuse the detergent solution.

CAUTION

- Do not sterilize the ALT-Pro leak test air tubes by autoclaving or gas. Doing so may damage the ALT-Pro leak test air tubes.
- Do not clean the ALT-Pro leak test air tubes by ultrasonic waves. Doing so may damage the ALT-Pro leak test air tubes.
- Do not push the pin in the endoscope side connector of the ALT-Pro leak test air tube while submerged in water. The water may enter the ALT-Pro leak test air tube. Performing a leak test using the ALT-Pro leak test air tube with water in it may cause malfunction of the endoscope.
- Check that the outer surface of the ALT-Pro leak test air tubes is free from scratches and flaws. The water may enter the ALT-Pro leak test air tube. Performing a leak test using the ALT-Pro leak test air tube with water in it may cause malfunction of the endoscope.

Use a low-foaming and neutral pH detergent that is labeled for use on medical device, that may or may not contain enzymes and follow the manufacturer’s dilution, temperature, and immersion time recommendations and the use-by-date of each detergent. Contact Olympus for the names of specific brands that have been tested for compatibility with the ALT-Pro leak test air tube.

1 Wipe the external surface of the ALT-Pro leak test air tubes with a piece of clean lint-free cloth.

2 Visually inspect for any debris remaining on the ALT-Pro leak test air tubes. The next step differs depending on whether the debris remains on the external surface.

■ *When no debris remains on the surface*

1 Turn OFF this equipment.

2 Wipe the external surface of the ALT-Pro leak test air tubes with a piece of lint-free cloth moistened with 70% ethyl alcohol or isopropyl alcohol and dry it thoroughly.

3 Connect the ALT-Pro leak test air tubes to this equipment according to the instruction in Section 4.1, “Installing the ALT-Pro leak test air tubes and turning ON the power” and put the tube on the tube hanger.

■ *When any debris remains on the surface*

1 Turn OFF this equipment.

2 While holding this equipment with one hand to stay it in place, hold the removing grip of the ALT-Pro leak test air tube and pull out the tube straight toward you.

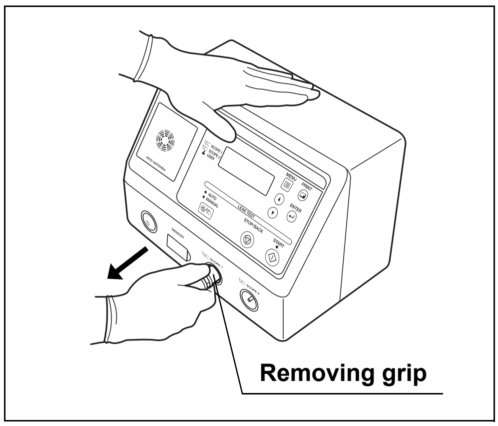


Figure 6.1

3 Fill a basin with a detergent solution at a concentration recommended by the manufacturer.

4 Immerse the ALT-Pro leak test air tube in the basin of the detergent solution.

5 Wipe all debris from the external surface of the ALT-Pro leak test air tube with a piece of clean lint-free cloth or sponge while ALT-Pro leak test air tube is immersed in the detergent solution.

6. Soak the ALT-Pro leak test air tube for the amount of time and at the temperature recommended by the detergent manufacturer.
7. Take out the ALT-Pro leak test air tube from the detergent solution.
8. Fill a basin with clean water and immerse the ALT-Pro leak test air tube in it.
9. Rinse the ALT-Pro leak test air tube thoroughly in the water by moving it slowly.
10. Take out the ALT-Pro leak test air tube from the water.
11. Visually inspect for any debris left on the ALT-Pro leak test air tube. If debris is found on the ALT-Pro leak test air tube, repeat Step 4 through 10 of the cleaning procedure until all debris is removed.
12. Wipe the external surface of the ALT-Pro leak test air tube with a piece of clean lint-free cloth.
13. Wipe the external surface of the ALT-Pro leak test air tube with a piece of lint-free cloth moistened with 70% ethyl alcohol or isopropyl alcohol and dry it thoroughly.
14. Connect the ALT-Pro leak test air tube to this equipment according to the instruction in Section 4.1, “Installing the ALT-Pro leak test air tubes and turning ON the power” and put the tube on the tube hanger.

6.3 Turning power OFF and cleaning outer surface

Check	Required items
	70% ethyl alcohol or isopropyl alcohol
	Neutral detergent
	Clean lint-free cloth

Table 6.2

WARNING

- After wiping with a piece of moistened clean lint-free cloth, dry this equipment thoroughly before using it again. Otherwise, an electric shock may result.
- Both 70% ethyl alcohol and isopropyl alcohol are combustible. Handle them with due care.
- Do not directly spray agent such as spray-type 70% ethyl alcohol or isopropyl alcohol over this equipment. The agent may enter this equipment through the vent hole, causing malfunction of this equipment.

CAUTION

- Do not clean the connectors including the AC Power Inlet of the power cord. Cleaning them can deform or corrode the contact, which may cause malfunction of this equipment.
- Do not expose this equipment to water, a cleaning or disinfection solution, or autoclave gas for cleaning and sterilization. Doing so may damage this equipment.
- Do not wipe the external surface with a hard or abrasive wiping material. The surface may be scratched.

After using this equipment, immediately perform the following cleaning procedure.

1. Turn OFF this equipment and disconnect the power cord from AC Power Inlet.
2. Wipe the external surface of this equipment with a piece of clean lint-free cloth moistened with neutral detergent solution and dry it with a piece of clean lint-free cloth. Remove in particular dust, dirt, or stain on the surface by wiping it thoroughly. For the usage of the detergent, follow the instructions of the detergent manufacturer.
3. Wipe the external surface of this equipment with a piece of lint-free cloth moistened with 70% ethyl alcohol or isopropyl alcohol and dry it thoroughly.

6.4 Installing the printer paper roll

WARNING

Do not touch the printer or the area around it during and immediately after printing. They will be very hot and may cause burns.

CAUTION

- Always use the Olympus-designated printer paper roll. Otherwise, incorrect printing or equipment failure may result.
- To prevent printer failure or printer paper roll discoloration, do not touch the printer or printer paper roll with wet hands.
- Always keep the paper cover closed. Otherwise, the printer and/or printer paper roll may get wet and malfunction.
- To avoid damage to or deterioration of the printout, do not allow the paper to make contact with the following.
  - Alcohol or detergent
  - Grease, organic solvents, chemicals (medical, industrial, or cosmetic use)
  - Stamp ink
- To prevent accelerated print deterioration, do not allow the paper to make contact with the following.
  - Water
  - Materials containing plasticizer (PVC film, desk mat, leather products, journal cover, etc.)
  - Certain stationery (plastic tape, mending tape, fluorescent-ink pen, oil-ink pen, adhesives other than starchy paste
- To prevent discoloration of unused paper, store the printer paper roll without opening it in a place meeting the following conditions.
  - Dark, cool place
  - Environment not exposed to NO_x, So_x, or O_3 (Ozone)
- When red lines appear on both sides of the printer paper roll during printing, replace the printer paper roll.

Ch.6

1 Take out the printer paper roll from the package and peel off the tape holding the end of the paper.

2 Press the cover open button of the printer to open the paper cover.

CAUTION

Make sure the printing surface is correct. Otherwise, it does not print properly. Do not touch the paper cutter of the printer. It may injure your hand.

3 Remove the printer paper roll remaining in the printer.

4 Put the printer paper roll on the paper roll holder. Make sure the printer paper roll feed direction is set as shown in below.

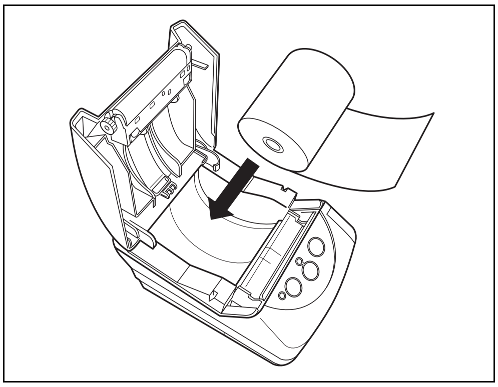


Figure 6.2

5 Pull the edge of the printer paper roll to feed the paper by about 5 cm.

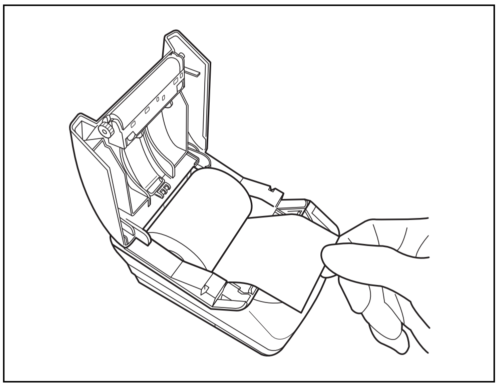


Figure 6.3

6 Close the paper cover by pushing the edges of it until it clicks.

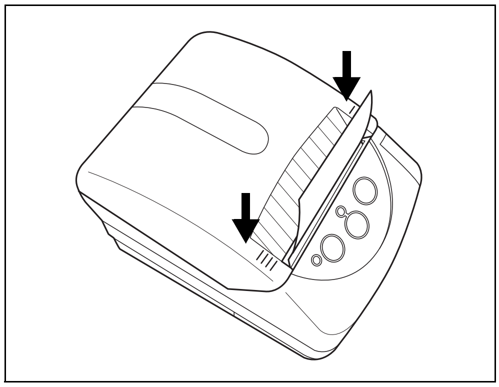


Figure 6.4

7 Turn ON the printer by holding down the POWER button.

8 Press the FEED button on the operational panel of the printer and confirm that the printer paper roll is fed smoothly.

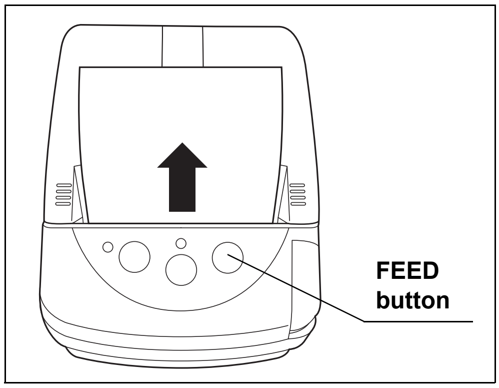


Figure 6.5

NOTE

If the printer paper cannot be fed neatly, press the cover open button of the printer to open the paper cover, take the printer paper roll out of the printer and retry insertion.

9 Cut the excess paper coming out of the slit.

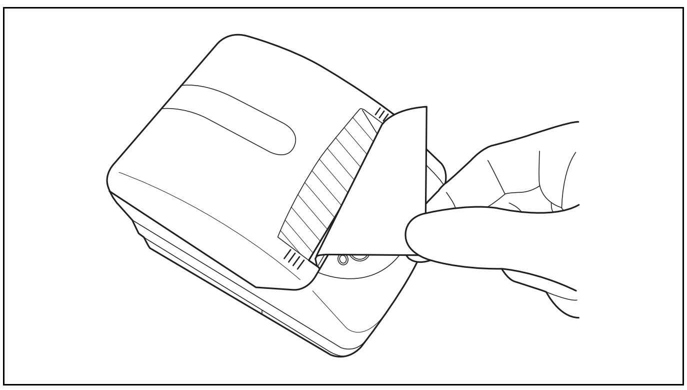


Figure 6.6

6.5 Replacing the fuse

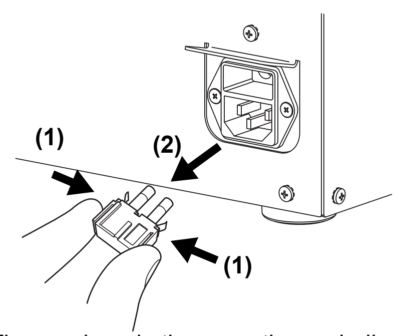
WARNING

- Be sure to turn OFF this equipment and unplug the power cord from the wall mains outlet and the AC Power Inlet of this equipment before removing the fuse box from this equipment. Otherwise, a fire or an electric shock may occur.
- To prevent an electric shock, do not touch this equipment and the power cord with wet hands.

1 Turn OFF this equipment and unplug the power plug from the wall mains outlet.

2 Unplug the power cord from the AC Power Inlet of this equipment.

3 Push the projections on the fuse box in the direction of the arrows (1) in Figure 6.7 and remove the fuse.



The numbers in the parentheses indicate the sequence of fuse box removal.

Figure 6.7

Ch.6

WARNING

Always use the Olympus-designated fuse. Otherwise, malfunction or failure of this equipment may occur, resulting in a fire or an electric shock.

4 Replace the two fuses in the fuse box with new ones.

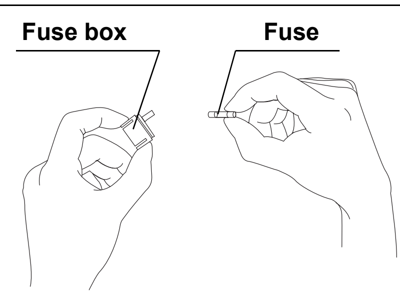


Figure 6.8 Replacing fuses

5 Insert the fuse box into this equipment until it clicks. Confirm that the fuse box is properly installed, referring to Figure 6.9.

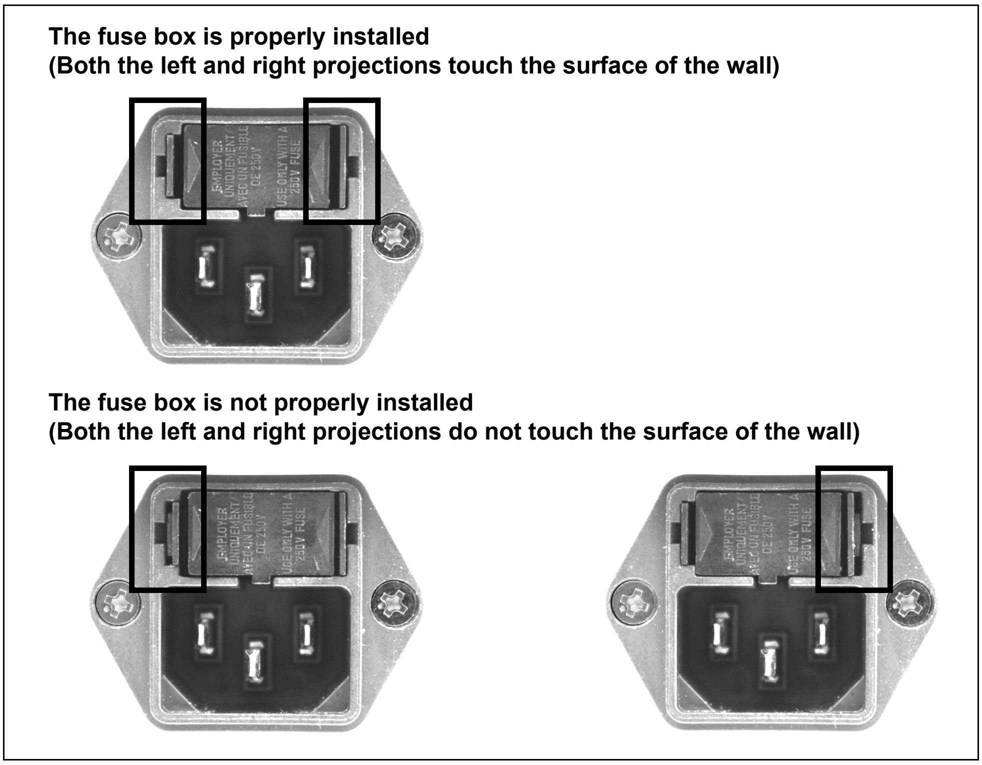


Figure 6.9

6 Plug the power cord and turn the power switch ON. Confirm the control panel and power indicator light up.

WARNING

If the operation panel and power indicators fail to light on after fuses are replaced with new ones, unplug the power cord immediately from the wall mains outlet. Otherwise, an electric shock may result.

7 If the power fails to turn ON even after fuses are replaced with new ones, contact Olympus.

6.6 Storage

When this equipment is not used for a long time, follow the steps below.

CAUTION

Do not store this equipment in a location exposed to direct sunlight, X-rays, radio activity, or strong electromagnetic radiation (e.g., near microwave medical treatment equipment, shortwave medical treatment equipment, MRI, radio, or mobile phone). Otherwise, this equipment may be damaged.

1. Turn OFF this equipment and disconnect the power cord from the wall mains outlet.
2. Disconnect the ancillary equipment connected to this equipment.
3. Store this equipment on a flat, level floor surface in a clean, dry, and stable location.

Chapter 7 Other Functions

7.1 Menu components

Various menus can be selected by pressing the Menu button and operating the ▲ and ▼ buttons. The display on the monitor changes as shown in Figure 7.1.

LCD display	Description of each menu
Menu(1/4) >History USB Download [↵]Select [STOP]Back	**<<History>>** Select to display and check the leakage test log on the LCD. →Refer to Section 7.2, "Data display".
▲ and ▼ buttons	
Menu(1/4) History >USB Download [↵]Select [STOP]Back	**<<USB Download>>** Select to download the leakage test log to the portable memory (MAJ-1925). →Refer to Section 7.3, "Download to the portable memory".
▲ and ▼ buttons	
Menu(2/4) >Configuration Information [↵]Select [STOP]Back	**<<Configuration>> Select to set the following:** • Automated leakage test mode →Refer to Section 7.5, "Setting automated leakage test mode". • Date →Refer to Section 7.6, "Setting date". • Time →Refer to Section 7.7, "Setting time". • Print mode →Refer to Section 7.11, "Setting auto printing".
▲ and ▼ buttons	
Menu(2/4) Configuration >Information [↵]Select [STOP]Back	**<<Information>>** Selected to display the total operation time of the pump, the number of recorded history, and software versions. →Refer to Section 7.8, "Checking equipment status".
▲ and ▼ buttons	
Menu(3/4) >Self Check Format USB Memory [↵]Select [STOP]Back	**<<Self Check>>** Select to perform a self check. →Refer to Section 7.9, "Performing only self-check".
▲ and ▼ buttons	
Menu(3/4) Self Check >Format USB Memory [↵]Select [STOP]Back	**<<Format USB Memory>>** Select to format the portable memory (MAJ-1925). →Refer to Section 7.4, "Management of portable memory data in personal computer".
▲ and ▼ buttons	
Menu(4/4) >Network Setting [↵]Select [STOP]Back	**<<Network Setting>>** Select to set the network. →Refer to Section 7.12, "Setting the network".

Figure 7.1

Ch.7

7.2 Data display

The results of leakage testing or self-check (Leak Test Log) can be displayed on the LCD by pressing the MENU button to select “History”. The records of error (Error Log) can be displayed on the LCD, too.

Indicator	Description
Leak Test Log	The LCD displays the results of leakage testing or self-check. The past data can be traced back by operating the ▲ and ▼ buttons.
Error Log	The LCD displays the record of errors. The past data can be traced back by operating the ▲ and ▼ buttons.

Table 7.1

■ *Leak Test Log Display*

The results of leakage testing or self-check are displayed on the LCD. Up to 700 newest logs can be displayed.

1 Press the MENU button in the Standby state.
“Menu (1/4)” is displayed on the LCD.

```
Menu(1/4)
>History
 USB Download
[↵]Select [STOP]Back
```

Figure 7.2

2 Select “History” by operating the ▲ and ▼ buttons, and press the ENTER button.

```
History Menu
>Leak Test Log
 Error Log
[↵]Select [STOP]Back
```

Figure 7.3

Ch.7

3 Select “Leak Test Log” by operating the ▲ and ▼ buttons and press the ENTER button. The LCD display the leak test log. The model name or serial number of the endoscopes used in the leakage testing is displayed alternately as shown in Figure 7.4 (A) or Figure 7.4 (B). The result of self-check is also displayed as shown in Figure 7.4 (C)

```
07/21/2011      S  #001
GIF-H180        PASS
PCF-2TQ140ZI    FAIL(*)
Press ▲▼    [↵]Exit
```

(A)

```
10:05           S  #001
1234567         PASS
7654321         FAIL(*)
Press ▲▼    [↵]Exit
```

(B)

```
07/21/2011         #002
09:30
Self Check      PASS
Press ▲▼    [↵]Exit
```

(C)

Figure 7.4

Indicator	Description
PASS	When the result of the automated leakage test or self-check was PASS.
FAIL	• When the result of the automated leakage test was FAIL, but a manual leakage test was not performed immediately after the automated leakage test. • When the result of the self-check is FAIL.
FAIL (*)	When the result of the automated leakage test was FAIL and a manual leakage test was performed immediately after the automated leakage test.
MANUAL	When a manual leakage test was performed.
S	When the “Standard Mode” is selected for the automated leakage testing mode. (“I” is displayed when the “Immediate Mode” is selected.

Table 7.2

NOTE

If there is no data in the memory in this equipment, “No History” is displayed.

4 The data can be tracked back by operating the ▼ and ▲ buttons.

5 The Standby screen appears by pressing the ENTER button.

■ *Error Log Display*

The record of errors is displayed on the LCD if an error occurs. Up to 1000 newest logs can be displayed.

1 Press the MENU button in the Standby state to display “Menu (1/4)” on the LCD.

```
Menu(1/4)
>History
 USB Download
[↵]Select [STOP]Back
```

Figure 7.5

2 Select “History” by pressing the ▲ and ▼ buttons and press the ENTER button.

```
History Menu
>Leak Test Log
 Error Log
[↵]Select [STOP]Back
```

Figure 7.6

3 Select “Error Log” by pressing the ▲ and ▼ buttons and press the ENTER button; the LCD displays the latest record of an error.

```
E030              #0001
Date 07/21/2011
Time 16:17
Press ▲▼    [↵]Exit
```

Figure 7.7

NOTE

If there is no data in the memory in the inside of this equipment, “No History” is displayed.

4 The data can be tracked back by operating the ▼ and ▲ buttons.

5 Press the ENTER button to go back to the Standby screen.

7.3 Download to the portable memory

When the portable memory is connected, the following data can be downloaded to the portable memory.

Download option	Type of download log	Description
Today	Leak Test Log	Downloads the results of all leakage tests and self-checks performed that day.
	Error Log	Downloads all the records of errors that occurred that day.
	Total Log	Downloads the results of all leakage tests and self-checks performed and the records of errors that occurred on that day.
Select Date	Leak Test Log	Downloads the results of all leak tests and self-checks performed on a selected day.
	Error Log	Downloads the records of all errors that occurred on a selected day.
	Total Log	Downloads the results of all leakage tests and self-checks performed and the records of errors that occurred on a selected day.
All Data	Leak Test Log	Downloads the results of all leakage tests and self-checks stored in this equipment.
	Error Log	Downloads the records of all errors stored in this equipment.
	Total Log	Downloads the results of leakage tests and self-checks and records of errors stored in this equipment

Table 7.3

CAUTION

- Always use the portable memory (MAJ-1925) listed in System Chart of Appendix. Using other kinds of portable memory may result in improper downloading.
- Be sure to format the portable memory before downloading. Refer to “■ Formatting the portable memory” on page 91.
- Be sure to format the portable memory by this equipment. If it is formatted by other equipment such as personal computers, downloading cannot be performed.
- Avoid exposing the portable memory to high-powered static electricity, electromagnetic wave, magnetism, high temperature, high humidity, or corrosive circumstances. The portable memory or data may be damaged.
- The portable memory is designed to save data temporarily. To save the data for a long time, use a personal computer.

CAUTION

- To download data to the portable memory, do not disconnect the portable memory until downloading data is completed. Otherwise, the saved data may be deleted.
- Do not insert the portable memory when turning ON the power. It may not work properly.
- Do not touch the portable memory with wet hands. It may fail.
- Make sure that the surface of the portable memory port is free from water. Otherwise, this equipment or portable memory may fail.
- Do not allow a foreign object to penetrate the inside of the portable memory port. Otherwise, this equipment may be damaged.
- The portable memory is inserted to the portable memory port slightly protruding from this equipment. Be careful not to exert an impact to the portable memory. The portable memory data may be damaged. When it is not used, remove the portable memory.
- Refer to the instruction manual for the portable memory for the use.

NOTE

- The results of leakage tests or self-check and the records of errors are saved from the latest data.
- Up to 13,000 records of leakage testing can be stored in this equipment. If more than 13,000 records are stored, the oldest record is overwritten. Periodically download the records in the portable memory.
- When “Today” is selected as the scope of data for download, not only log data recorded on that day but also those on previous days are downloaded if stored in this equipment without being downloaded (see Figure 7.8). In this case, daily files are created in the folder.

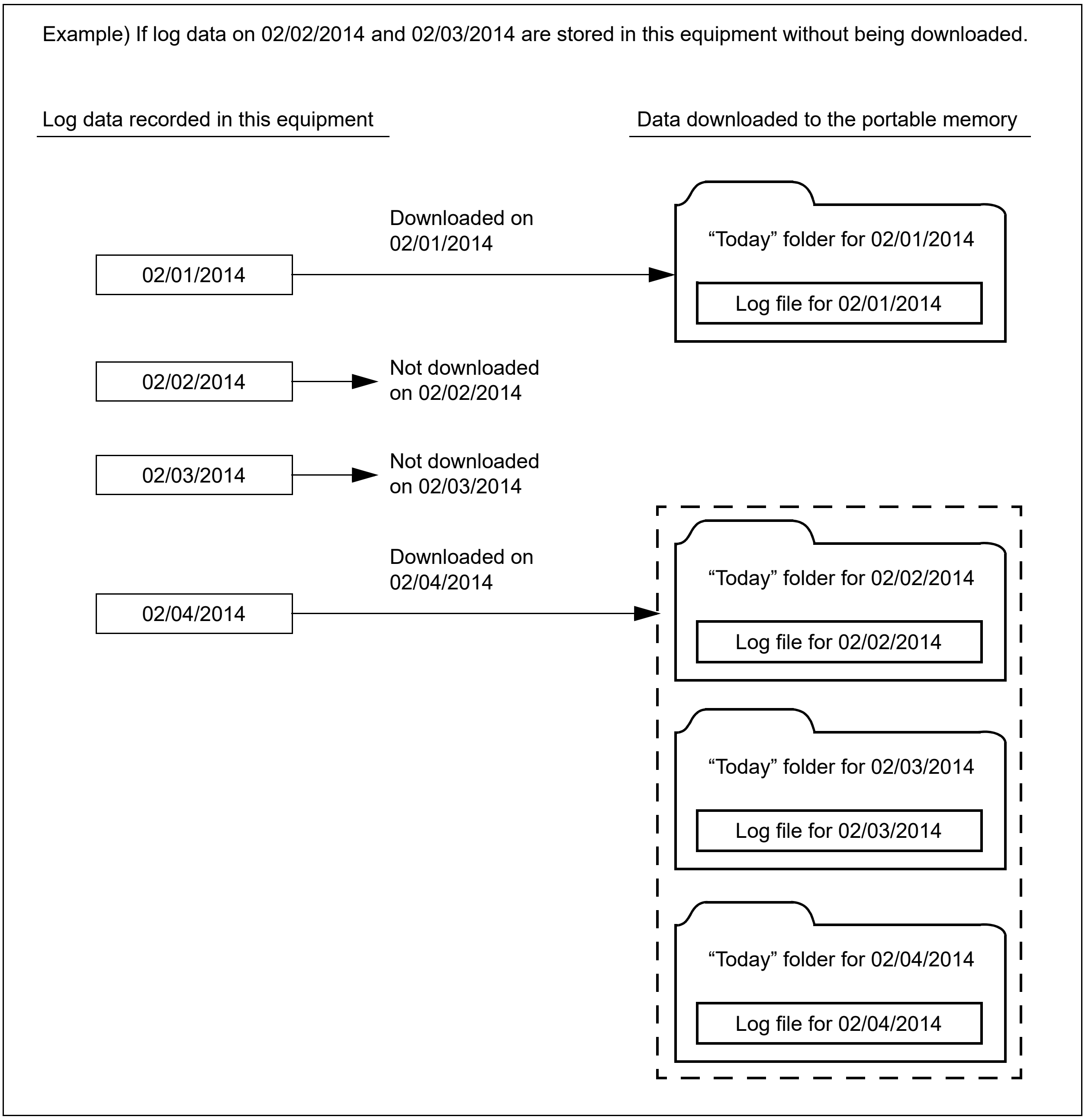

Example) If log data on 02/02/2014 and 02/03/2014 are stored in this equipment without being downloaded.
Log data recorded in this equipment
Data downloaded to the portable memory
02/01/2014
Downloaded on 02/01/2014
"Today" folder for 02/01/2014
Log file for 02/01/2014
02/02/2014
Not downloaded on 02/02/2014
02/03/2014
Not downloaded on 02/03/2014
02/04/2014
Downloaded on 02/04/2014
"Today" folder for 02/02/2014
Log file for 02/02/2014
"Today" folder for 02/03/2014
Log file for 02/03/2014
"Today" folder for 02/04/2014
Log file for 02/04/2014

Figure 7.8

■ *Downloading data*

1 Open the memory port cap in this equipment.

2 Insert the portable memory into the portable memory port until it stops.

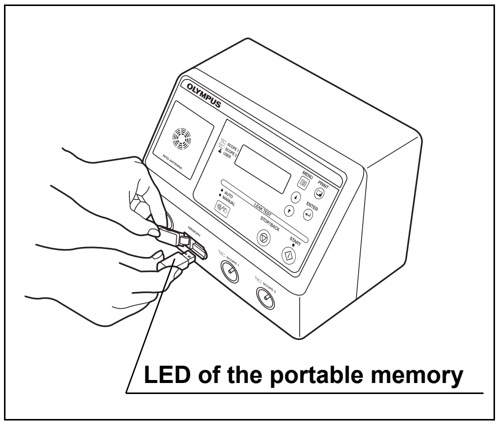


Figure 7.9

NOTE

The portable memory cannot be connected through a USB HUB.

3 Press the MENU button and display “Menu (1/4)” on the LCD by operating the ▲ and ▼ buttons.

```
Menu(1/4)
 History
>USB Download
[↵]Select [STOP]Back
```

Figure 7.10

4 Select “USB Download” by operating the ▲ and ▼ buttons and press the ENTER button.

```
Download Menu(1/2)
>Today
 Select Date
[↵]Select [STOP]Back
```

Figure 7.11

5 Select “Today”, “Select Date”, or “All Data” by operating the ▲ and ▼ buttons and press the ENTER button. If “Select Date” is selected, select the date by operating the ▲ and ▼ buttons. Press the ENTER button.

```
Today (1/2)
>Leak Test Log
 Error Log
[↵]Select [STOP]Back
```

Figure 7.12

6 Select “Leak Test Log”, “Error Log”, or “Total Log” by operating the ▲ and ▼ buttons and press the ENTER button. The screen shown in Figure 7.13 (A) is displayed. If the portable memory is not connected, the screen shown in Figure 7.13 (B) is displayed. Insert the portable memory into the portable memory port until it stops. When connection of the portable memory has been confirmed, the message shown in Figure 7.13 (A) is displayed, and downloading starts automatically.

```
USB Download
USB Memory Detected

[STOP]Cancel
```

(A)

```
USB Download
Insert USB Memory

          [STOP]Back
```

(B)

Figure 7.13

7 When the data download to the portable memory starts automatically, the LCD displays the screen.

```
Downloading Data...
Do NOT Remove Memory
Do NOT Turn OFF
[STOP]Cancel
```

Figure 7.14

8 When the data download to the portable memory is complete, the LCD displays “Data Download Completed”.

```
Data Download
Completed
Remove USB Memory
[↵]Exit
```

Figure 7.15

9 Pull the portable memory straight out.

CAUTION

Do not pull the portable memory from the portable memory port while the access lamp of the portable memory is blinking. The data within the portable memory may get corrupted.

10 Press the ENTER button to return to the standby screen.

11 Close the memory cap on this equipment.

■ *Formatting the portable memory*

CAUTION

- Be sure to format the portable memory by this equipment. If it is formatted by another equipment, such as a personal computer, downloading cannot be performed. The portable memory formatted by the personal computer needs to be formatted by the Olympus product again.
- Do not format the portable memory in which data has been saved. The data will be all deleted.

1 Open the memory port cap of this equipment.

2 Insert the portable memory into the portable memory port until it clicks.

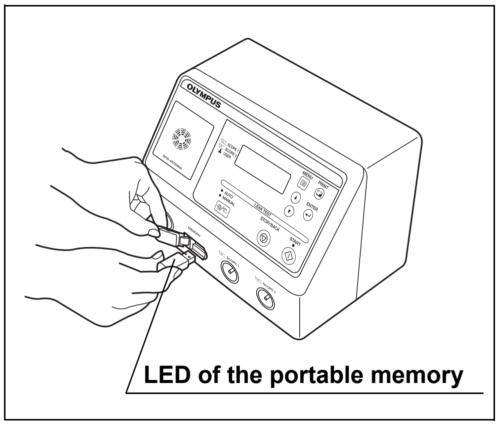


Figure 7.16

3 Press the MENU button and display "Menu (3/4)" on the LCD by operating the ▲ and ▼ buttons.

```
Menu(3/4)
 Self Check
>Format USB Memory
[↵]Select [STOP]Back
```

Figure 7.17

4 Select "Format USB Memory," and press the ENTER button.

```
Press ▲▼ to Select
>Cancel Format
 Begin Format
[↵]Enter [STOP]Back
```

Figure 7.18

5 Select "Begin Format" by operating the ▲ and ▼ buttons and press the ENTER button.

6 Formatting the portable memory starts.

```
Format
Please Wait...
Do NOT Remove Memory
Do NOT Turn OFF
```

Figure 7.19

NOTE

If the portable memory is not connected, the error code [E136] is displayed. Insert the portable memory into the portable memory port until it clicks.

7 When formatting the portable memory is completed, the LCD displays “Format Completed”.

```
Format Completed

[↵]Exit
```

Figure 7.20

8 Pull the portable memory straight out.

CAUTION

Do not pull the portable memory from the portable memory port while the access lamp of the portable memory is blinking. The data within the portable memory may get corrupted.

9 Press the ENTER button to return to the standby screen.

10 Close the memory cap on this equipment.

7.4 Management of portable memory data in personal computer

The data in the portable memory can be managed in a commercial personal computer.

CAUTION

The data transferred to a personal computer is master data. Do not make a change in the master data. To manage the history of leakage testing in the facility, make downloads of the master data.

NOTE

Refer to the instruction manual for the portable memory for the operation environment. When processing the reprocessing data in your facility, it is recommended to use a spreadsheet program that is installed on a PC that is capable of editing CSV format files.

1. Set the portable memory in a personal computer.
2. Open the drive where the portable memory is connected. Download the ALT-Pro folder within the portable memory as shown in Figure 7.21 or a CSV file in the folder to a personal computer.
3. Disconnect the portable memory when downloading has been completed and store it.

Folder structure

The portable memory has the following folder structure for the data downloaded in the portable memory according to Section 7.3, “Download to the portable memory”.

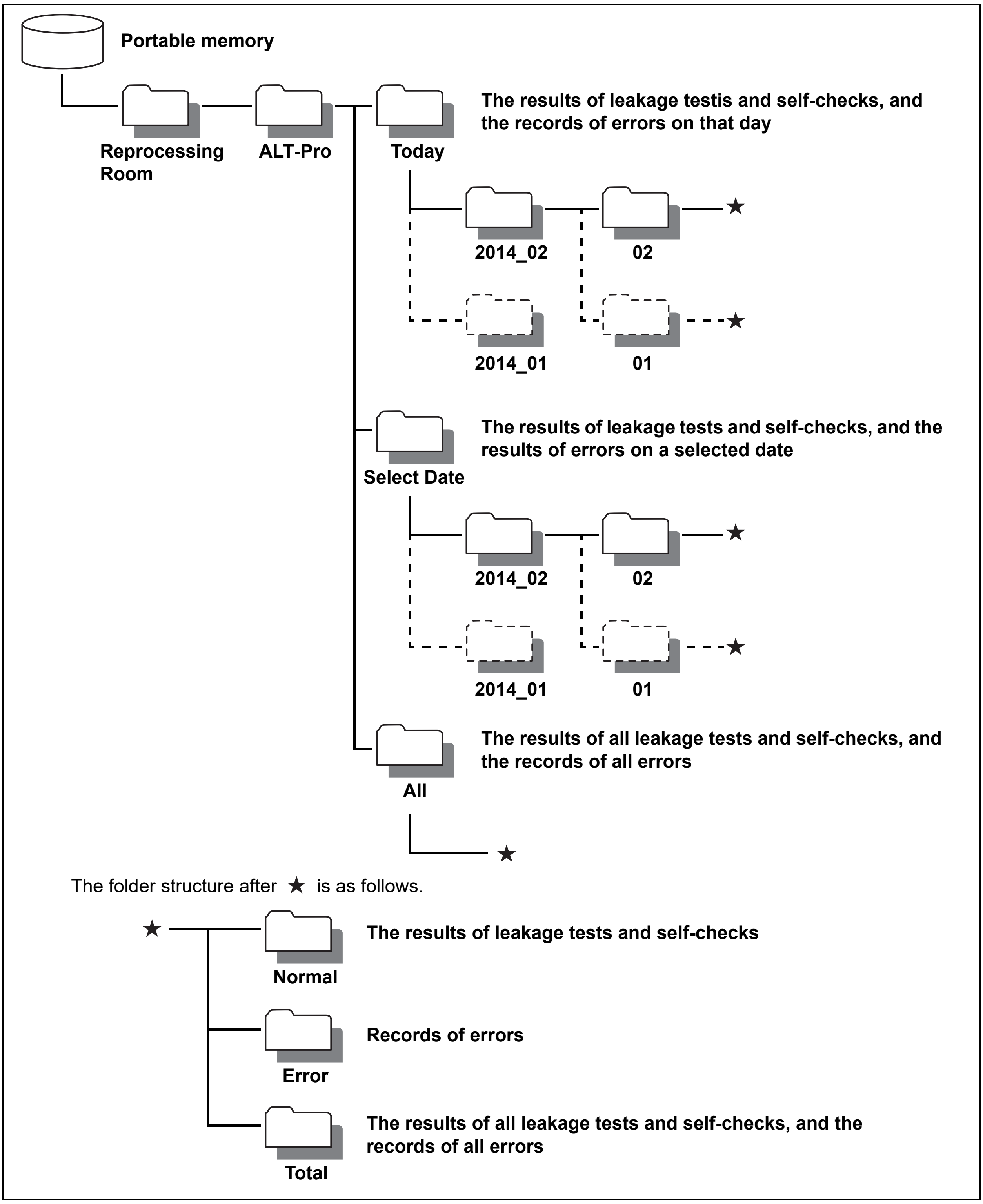


Figure 7.21

❍ File name

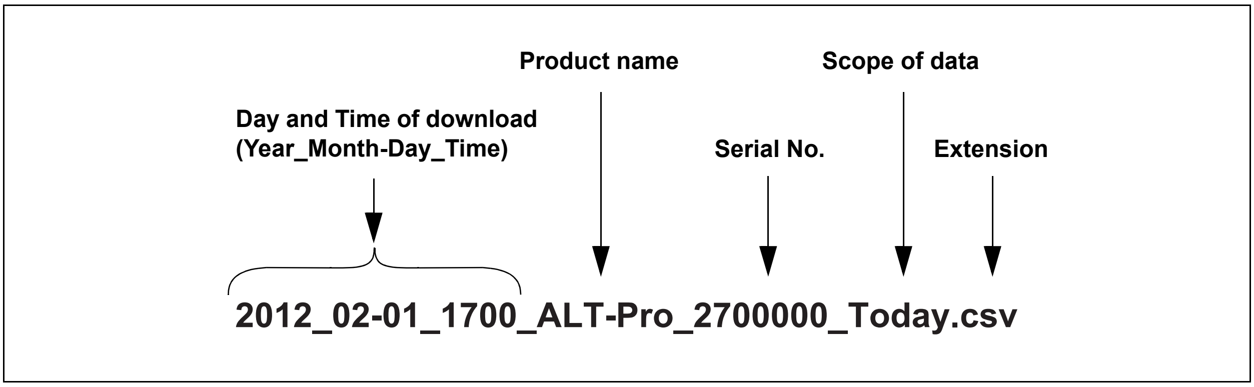


Figure 7.22

The scope of data in Figure 7.22 is indicated as shown in Table 7.4 below.

	Indicator	Description
Scope of data	Today	Records on that day.
	Select_Date	Records in a selected day.
	All	All records

Table 7.4

NOTE

The file name includes “Error” after the scope of data when the data is an error record. (e.g.,2012_02-01_1700_ALT-Pro_2700000_Today_error.csv)

❍ Image view of the file

The file is CSV format. The image view using Microsoft® Excel® is shown below.

ALT−Pro	XXXXXX															
01/01/2012	22:56:22															
Today																
Total Log																
No.	Leak Test	Start(date)	Start(time)	Complete(date)	Complete(time)	Mode	Scope1 Name	Scope1 ID	Scope1 Result	Scope2 Name	Scope2 ID	Scope2 Result	User Name	User ID	Process No.	
1	Complete	01/01/2012	22:52:30	01/01/2012	22:54:54	Auto	GIF−Q160	1	PASS	GIF−Q160	2	FAIL	John Smith			
2	Complete	01/01/2012	22:47:37	01/01/2012	22:50:01	Auto	GIF−Q160	3	PASS	GIF−Q160	2	PASS	John Smith			
3	Complete	01/01/2012	22:45:15	01/01/2012	22:46:41	Manual	GIF−Q160	1		GIF−Q160	2		John Smith			
4	Complete	01/01/2012	22:43:13	01/01/2012	22:43:57	Self Check			PASS							
5	E115			01/01/2012	22:41:50		GIF−Q160	1		GIF−Q160	2		John Smith		201	
6	Complete	01/01/2012	22:37:42	01/01/2012	22:38:25	Self Check			PASS							
End of data. Check code : CB4																

Figure 7.23

7.5 Setting automated leakage test mode

The automated leakage test mode has two options, “Standard Mode” and “Immediate Mode”.

Test Mode	Description
Standard Mode	This mode is used to start an automated leakage test of endoscopes more than 15 minutes after their withdrawal from the body.
Immediate Mode	This mode is used to start an automated leakage test of endoscopes more than 3 minutes but less than 15 minutes after their withdrawal from the body. (Some endoscope models need to be tested in this mode even if more than 15 minutes have passed since their removal of from the body.)

* With some endoscope models, the automated leakage test should be started in 9 minutes or more have elapsed after the endoscope withdrawal from the body.

Table 7.5

CAUTION

- Perform an automated leakage test more than three minutes after withdrawing the endoscope from the body. The endoscopes upon removal from the body may be exposed to a sudden temperature change, resulting in an inaccurate test.
- If automated leakage testing is required within 15 minutes after removal of the endoscope from the body, select the Immediate Mode. Selecting the Standard Mode may result in an inaccurate test because the endoscope upon removal from the body may have a temperature greatly different from ambient temperature and undergo a sudden temperature change.
- In some endoscope models, however, the Immediate Mode needs to be selected even if 15 minutes or more has passed since their removal from the body. To find these models, refer to “List of Compatible Endoscopes <ALT-Pro>”. Selecting the Standard Mode to perform an automated leakage test of these models may result in an inaccurate test.

CAUTION

- With some endoscope models (Additional Wait Time Scopes = AWT-Scopes), short beeps are generated and the monitor displays the screen as shown in Figure 7.24 when the scope ID recognition is performed. When an AWT scope is used, do not proceed to automated leakage test until 9 minutes or more have elapsed after withdrawal of the endoscope from the patient body. Otherwise, the test results may become inaccurate due to the temperature change. For the mode of automated leakage test of the AWT-Scope, select the Immediate Mode if the time elapsed from the endoscope withdrawal is less than 27 minutes or the Standard Mode when it is 27 minutes or more.

```
Caution
Additional Wait Time
Required Endoscope
See Instructions
```

Figure 7.24

Scope ID recognition	Time elapsed after withdrawal from body	Test Mode
Figure 7.24 is displayed.	≥9 min., <27 min.	Immediate Mode
	≥27 min.	Standard Mode

Table 7.6

NOTE

Once the automated leakage test mode has been set, it remains unless changed.

Ch.7

1 Press the MENU button in the Standby status of this equipment to display “Menu (1/4)” in the LCD.

```
Menu(1/4)
>History
 USB Download
[↵]Select [STOP]Back
```

Figure 7.25

2 Select “Configuration” by operating the ▲ and ▼ buttons and press the ENTER button.

```
Menu(2/4)
>Configuration
 Information
[↵]Select [STOP]Back
```

Figure 7.26

3 Select “Select AUTO mode” by operating the ▲ and ▼ buttons and press the ENTER button.

```
Configuration(1/3)
>Select AUTO mode
 (Standard Mode)
[↵]Select [STOP]Back
```

Figure 7.27

4 The LCD displays the screen as shown in Figure 7.28. Select the Immediate Mode by operating the ▲ and ▼ buttons if 15 minutes have not passed after removal of the endoscope from the body. Select the Standard Mode by operating the ▲ and ▼ buttons if more than 15 minutes have passed after removal of the endoscope from the body.

```
Press ▲▼ to Select
 Select AUTO Mode
 (Standard Mode)
[↵]Exit [STOP]Back
```

Figure 7.28

5 Press the ENTER button to display the Standby screen.

7.6 Setting date

To save the correct date of leakage testing, adjust the date in this equipment.

CAUTION

Set the date correctly. An incorrect date may result in improper data storage or unexpected performance of the functions to download the data to the portable memory or print the data.

NOTE

You can go back to the previous screen by pressing the STOP/BACK button.

1 Press the MENU button in the Standby status of this equipment to display “Menu (2/4)” on the LCD by operating the ▲ and ▼ buttons.

2 Select “Configuration” and press the ENTER button.

```
Menu(2/4)
>Configuration
 Information
[↵]Select [STOP]Back
```

Figure 7.29

3 Select “Date” by operating the ▲ or ▼ buttons, and confirm the selection by pressing the ENTER button.

```
Configuration(2/3)
>Date 07/21/2011
 Time 10:17
[↵]Select [STOP]Back
```

Figure 7.30

4 Select the date format (the order in which year, month, and day are displayed) by operating the ▲ and ▼ buttons and confirm the format by pressing the ENTER button.

```
Press ▲▼ to Select
 Date MM/DD/YYYY

[↵]Enter  [STOP]Back
```

Figure 7.31

NOTE

The date format can be selected from “[MM/DD/YYYY], [DD/MM/YYYY], or [YYYY/MM/DD]”.

5 Set the date in the selected order by operating the ▲ and ▼ buttons while the digits are blinking and confirm the date by pressing the ENTER button.

6 When the date is confirmed, the standby screen is displayed.

7.7 Setting time

Set the time in this equipment to save the correct time of leakage testing.

1 Press the MENU button in the Standby status of this equipment to display “Menu (2/4)” on the LCD by operating the ▲ and ▼ buttons.

2 Select “Configuration” and press the ENTER button.

```
Menu(2/4)
>Configuration
 Information
[↵]Select [STOP]Back
```

Figure 7.32

3 Select “Time” by operating the ▲ and ▼ buttons, and confirm the selection by pressing the ENTER button.

```
Configuration(2/3)
 Date 07/21/2011
>Time 10:17
[↵]Select [STOP]Back
```

Figure 7.33

4 Set the hour by operating the ▲ and ▼ buttons while the digits are blinking and confirm it by pressing the ENTER button.

```
Press ▲▼ to Select

 Hour 10:17
[↵]Enter  [STOP]Back
```

Figure 7.34

5 Set the minute by operating the ▲ and ▼ buttons while the digits are blinking and confirm it by pressing the ENTER button. The standby display is displayed.

7.8 Checking equipment status

The total operation hours of the pump, the number of recorded logs, and the software version can be checked.

1 Press the MENU button in the Standby status and operate the ▲ and ▼ buttons to display “Menu (2/4)” on the LCD.

2 Select “Information” and press the ENTER button.

```
Menu(2/4)
 Configuration
>Information
[↵]Select [STOP]Back
```

Figure 7.35

3 Operate the ▲ and ▼ buttons to check the total operation hours of the pump, the number of recorded logs, and the software version.

```
Information(1/3)
Pump: ***hours

[↵]Exit [STOP]Back
```

▲ and ▼ buttons

```
Information(2/3)
Leak Test Log:#13000
Error Log    :#1000
[↵]Exit [STOP]Back
```

▲ and ▼ buttons

```
Information(3/3)
Software Ver.1.00

[↵]Exit [STOP]Back
```

Figure 7.36

4 Press the ENTER button to display the Standby screen.

7.9 Performing only self-check

The same as the self-check described in Section 4.2, “Self-check” can be performed on the MENU screen.

CAUTION

- Do not forcibly bend nor put anything on the ALT-Pro leak test air tubes to be connected. Doing so may result in not only an inaccurate self-check result but also damage to the ALT-Pro leak test air tubes.
- If the error code [112] is displayed on the LCD as a result of self-check, follow the instruction described in Section 8.1, “Troubleshooting guide”.

NOTE

When the self-check is completed, the result is automatically recorded in this equipment.

1 Press the MENU button in the Standby status and operate the ▲ and ▼ buttons to display “Menu (3/4)” on the LCD.

```
Menu(3/4)
>Self Check
 Format USB Memory
[↵]Select [STOP]Back
```

Figure 7.37

2 Select “Self check” by operating the ▲ and ▼ buttons and press the ENTER button.

3 The message as shown in Figure 7.38 below is displayed on the LCD. If an endoscope is connected, disconnect it.

```
Disconnect Endoscope
Press START Button

[STOP]Back
```

Figure 7.38

CAUTION

Do not perform a self-check when an endoscope or water-resistant cap is connected to this equipment. Doing so may result in incorrect self-check.

Ch.7

4 Press the START button. The LCD displays "Self Check In Process". The number of mark "■" of the progress bar increases to show the progress of the self-check until all the process is completed.

```
Self Check
  In Process
[STOP]Cancel
[■■■■■                ]
```

Figure 7.39

NOTE

Press the STOP/BACK button during self-check. The LCD displays the Standby screen.

5 A short beep sounds when the self-check is completed properly, and the result is displayed in the LCD.

```
Self Check Result
                PASS

[↵]Exit
```

Figure 7.40

6 Press the ENTER button to display the Standby screen.

7.10 Manual printing

Press the PRINT button in the Standby status of this equipment and confirm that the “Print Option Screen” is displayed on the LCD. Either “Last”, “Today”, or “Select Date” can be selected by operating the ▲ and ▼ buttons.

Print option	Print type log	Description
Last	Leak Test Log	Prints the results of the last leakage test or self-check.
	Error Log	Prints the log of the last error.
Today	Leak Test Log	Prints the results of all of the leakage tests and self-checks performed that day.
	Error Log	Prints all the logs of the errors that occurred that day.
	Total Log	Prints all the results of the leakage tests and self-checks performed and all the logs of the errors that occurred on that day.
Select Date	Leak Test Log	Prints all the results of leakage tests or self-checks performed on a selected day.
	Error Log	Prints all the logs of the error that occurred on a selected day.
	Total Log	Prints all the results of leakage tests and self-checks performed and the records of the error that occurred on a selected day.

Table 7.7

WARNING

Do not touch the thermal head of the printer or its surroundings immediately after printing as they will be very hot and may cause skin burns.

CAUTION

- Printed data may be lost as the paper ages and deteriorates. To store the data for a long period of time, transfer them to a medium with long-term storage capability.
- To prevent printer failure or printer paper roll discoloration, do not touch the printer or printer paper roll with wet hands.
- Always keep the paper cover closed. Otherwise, the printer and/or the printer paper roll may get wet and malfunction.

CAUTION

- Keep the printer paper roll away from the following items. Otherwise, the print may become black or disappear.
  - Alcohol or detergent
  - Oil, fat, organic solvents, or chemicals (medical or cosmetic)
  - Stamp ink
- Keep the printer paper roll away from the following items. Otherwise, the print may be deteriorated soon:
  - Water
  - Materials containing plasticizer (PVC film, desk mat, leather products, journal cover, etc.)
  - Certain stationery (plastic tape, mending tape, fluorescent-ink pen, oil-ink pen, adhesives other than starchy paste)
- To prevent discoloration of unused paper, store the printer paper roll without opening in a place meeting the following conditions.
  - Dark, cool space
  - Place not exposed to NO_x, SO_x, or O_3 (ozone)
- When red lines appear on both sides of the printer paper roll during printing, replace the printer paper roll.

NOTE

- If the printed data is not clear, try printing again by manual.
- When the print paper roll has run out during printing and is replaced by a new one, the printer restarts printing the result of the leakage test or self-check from the middle of the data.
- This equipment can store data on leak-test results of up to 13,000. Data exceeding the capacity 13,000 will be deleted from the log in the order of its test date.
- Manual printing cannot be performed even if the PRINT button is pressed when the printer is not connected.

1 Press the PRINT button in the Standby status to display the Print Option screen on the LCD.

```
Print Menu(1/2)
>Last
 Today
[↵]Select [STOP]Back
```

Figure 7.41

2 Select “Last”, “Today”, or “Select Date” by operating the ▲ and ▼ buttons. Press the ENTER button. If “Select Date” is selected, select the date operating the ▲ and ▼ buttons, and press the ENTER button.

```
Last
>Leak Test Log
 Error Log
[↵]Select [STOP]Back
```

Figure 7.42

3 Select “Leak Test Log”, “Error Log”, or “Total Log” by operating the ▲ and ▼ buttons and press the ENTER button. The monitor displays the screen as shown in Figure 7.43.

```
Print Last
      07/21/2011

[↵]Print  [STOP]Back
```

Figure 7.43

4 Press the ENTER button, while the monitor is displaying the screen as shown in Figure 7.43. Printing starts.

5 A buzzer beeps and the paper feed stops when printing completes. Cut the printed part of paper and ensure that information is printed correctly.

```
Print Completes


[↵]Exit
```

Figure 7.44

Ch.7

6 Press the ENTER button, and the standby monitor is displayed.

NOTE

For the printed information, see Figure 7.45 through 7.48.

❍ Printed information on automated leakage testing

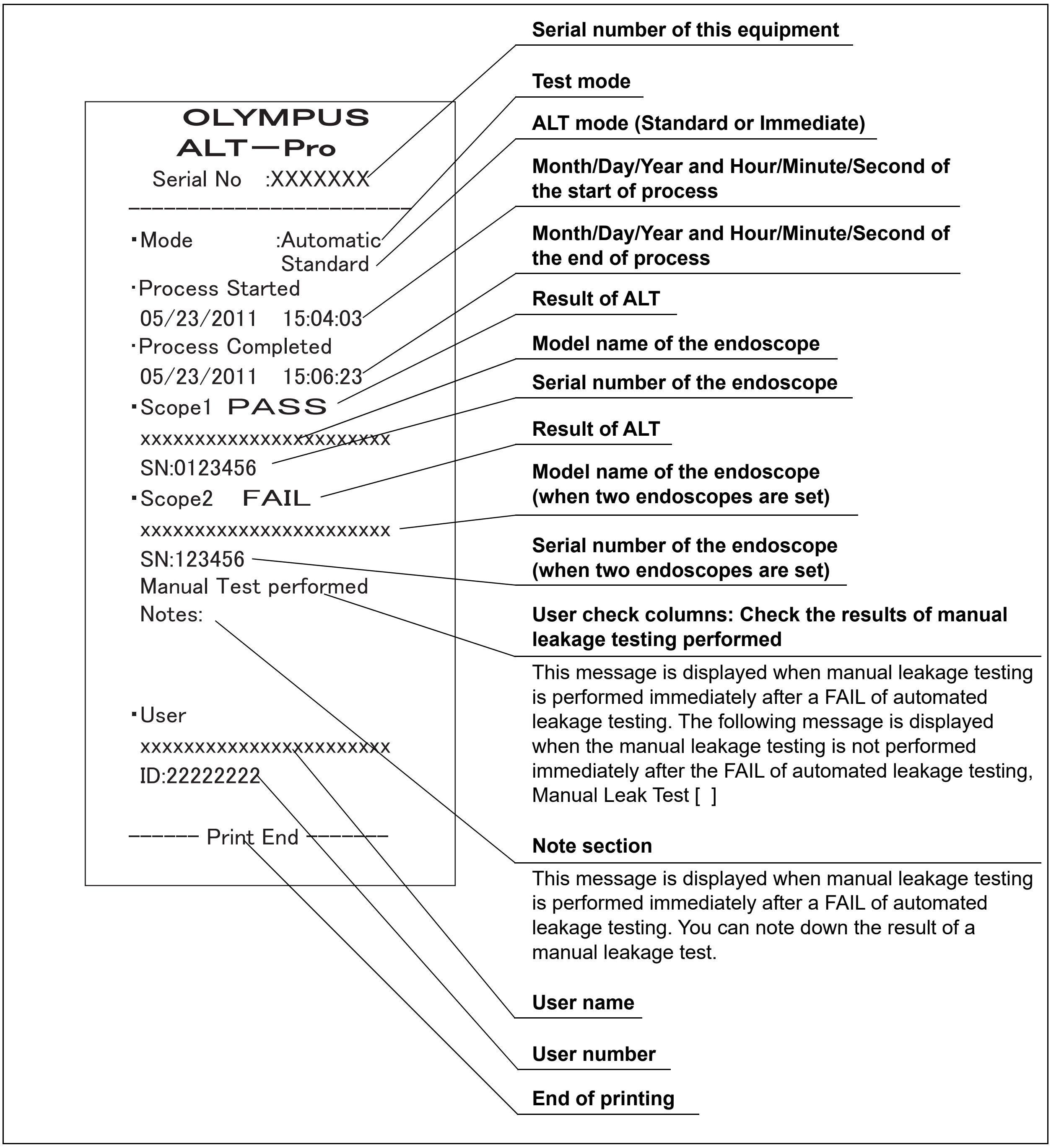


Figure 7.45

❍ Printed information on manual leakage testing

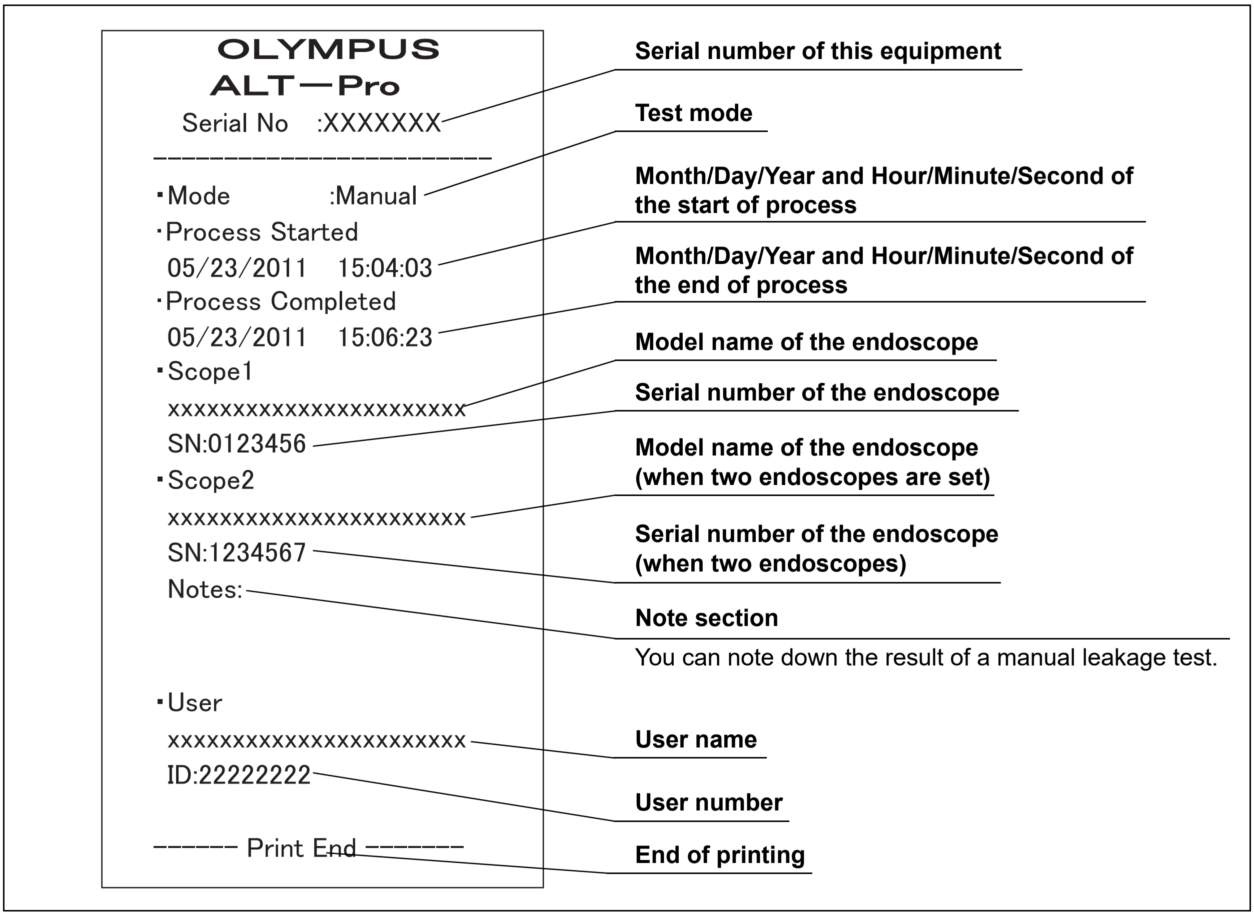


Figure 7.46

Ch.7

❍ Printed information on self-check

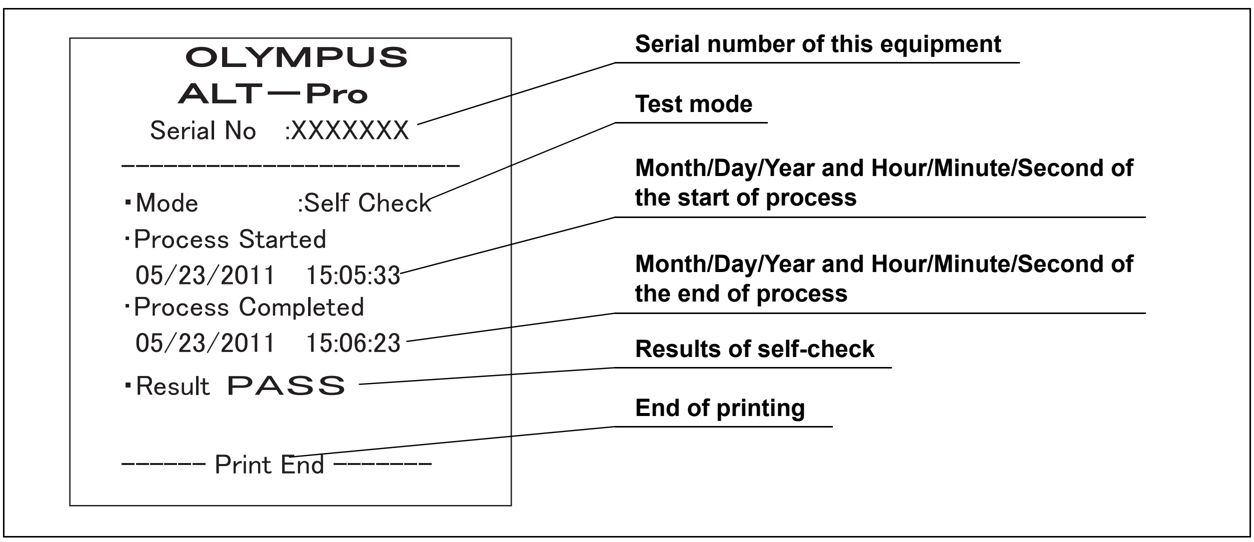


Figure 7.47

❍ Printed information on error code logs

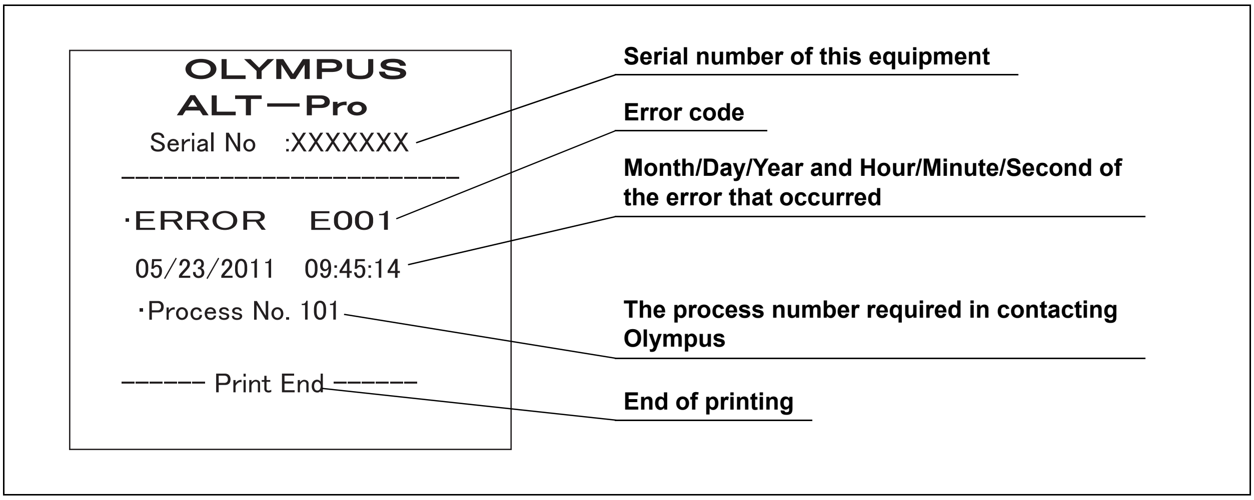


Figure 7.48

■ *When the printer roll paper runs short during printing*

1. When the printer paper roll runs short while printing, the error code [E094] is displayed in the LCD. Press STOP/BACK button on the control panel and cancel the error code.
2. Replace the printer paper roll according to Section 6.4, “Installing the printer paper roll”.
3. Press the MENU button when the printer paper roll has been replaced. The moment the MENU button is pressed, the printer prints unprinted data on the results of the leakage testing or self-check.

7.11 Setting auto printing

The results of leakage testing and self-check and records of errors can be automatically printed as long as a printer is connected to this equipment.

WARNING

Do not touch the thermal head of the printer or its surroundings immediately after printing as they will be very hot and may cause skin burns.

CAUTION

- Printed data may be lost as the paper ages and deteriorates. To store the data for a long period of time, transfer them to a medium with long-term storage capability.
- To prevent printer failure or printer paper roll discoloration, do not touch the printer or the printer paper roll with wet hands.
- Always keep the paper cover closed. Otherwise, the printer and/or the printer paper roll may get wet and malfunction.
- Keep the printer paper roll away from the following items. Otherwise, the print may become black or disappear.
  - Alcohol or detergent
  - Oil, fat, organic solvents, or chemicals (medical or cosmetic)
  - Stamp ink
- Keep the printer paper roll away from the following items. Otherwise, the print may be deteriorated soon:
  - Water
  - Materials containing plasticizer (PVC film, desk mat, leather products, journal cover, etc.)
  - Certain stationery (plastic tape, mending tape, fluorescent-ink pen, oil-ink pen, adhesives other than starchy paste)
- To prevent discoloration of unused paper, store the printer paper roll without opening in a place meeting the following conditions.
  - Dark, cool space
  - Place not exposed to NO_x, SO_x, or O_3 (ozone)
- When red lines appear on both sides of the printer paper roll during printing, replace the printer paper roll.

Ch.7

NOTE

- The Auto Print setting remains active once this equipment is turned OFF.
- To perform Auto Print, turn ON the printer before turning ON this equipment.
- If the printer is not connected, “Print NONE” is displayed and Auto Print cannot be set.

1 Press the MENU button in the Standby status, and display “Menu (2/4)” on the LCD by operating the ▲ and ▼ buttons.

2 Select “Configuration” by operating the ▲ and ▼ buttons, and press the ENTER button.

```
Menu(2/4)
>Configuration
 Information
[↵]Select [STOP]Back
```

Figure 7.49

3 Select “Print AUTO” by operating the ▲ and ▼ buttons and press the ENTER button.

```
Configuration(3/3)
>Print AUTO 1 Copy

[↵]Select [STOP]Back
```

Figure 7.50

4 Select the number of copies to be printed by operating ▲ and ▼ buttons, and press the ENTER button. After pressing the ENTER button, the screen will go back to standby.

NOTE

You can set the number of copies as specified in Table 7.8.

Indicator	Description
OFF	Auto Print is disabled.
1 Copy	Every time the leakage testing or self-check is completed, the result is printed out on one sheet. Also, when an error occurs, the error log is printed out on one sheet.
2 Copies	Every time the leakage testing or self-check is completed, the result is printed out on two sheets. Also, when an error occurs, the error log is printed out on two sheets.

Table 7.8

7.12 Setting the network

When this equipment is connected to the server, the results of leakage tests and the records of errors can be transmitted to the server.
Press the MENU button and select “Network Setting” to display the Network Setting screen. See Table 7.9 for the network setting items.

Screen	Setting item	Description
Network Setting (1/7)	NW Security Lock	Enable or disable the security of network setting. After enabling this setting, it is also required to set the password (4 numerals). When this setting is enabled, the password entry is required for shifting from “Menu” to “Network Setting”.
	NW Security PW	Change the password for NW Security Lock. The password change is unavailable when NW Security Lock is disabled.
Network Setting (2/7)	Communicate	Enable or disable the network communication function of the server.
	IP Address(Server)	Set the IP address of the server to hold communication.
Network Setting (3/7)	Subnet Mask	Set the same Subnet Mask value as the server to communicate with.
	Gateway	Set the same Gateway value as the server to communicate with.
Network Setting (4/7)	IP Address(ALT-Pro)	Set the IP address of ALT-Pro.
	MAC Address	View the MAC address of ALT-Pro. (It cannot be changed.)
Network Setting (5/7)	Port No.(Server)	Do not change this. (Initial setting: 51800)
	Port No.(ALT-Pro)	Do not change this. (Initial setting: 51800)
Network Setting (6/7)	COMM Mode	One of the communication modes can be selected from the following: • Auto negotiation (Initial setting) • 100Mbps full duplex • 100Mbps half duplex • 10Mbps full duplex • 10Mbps half duplex
	Time out(Ctrl layer)	Do not change this. (Initial setting: 8)
Network Setting (7/7)	Time out(App Layer)	Do not change this. (Initial setting: 15)

Table 7.9

WARNING

Make sure that each connector or terminal is free from water on it before connecting the LAN cable. Any water may damage this equipment or server.

CAUTION

Do not change the initial settings at the receipt of the equipment for Port No. (Server), Port No. (ALT-Pro), Time out (Ctrl Layer), and Time out (App Layer). Otherwise, the equipment may not be able to communicate with the server.

NOTE

Turning the power OFF of this equipment will not change the network setting.

■ *Security settings*

1 Press the MENU button in the Standby status, display “Menu (4/4)” on the LCD by operating the ▲ and ▼ buttons, select “Network Setting”, and press the ENTER button.

```
Menu(4/4)
>Network Setting

[↵]Select [STOP]Back
```

Figure 7.51

2 Display “Network Setting (1/7)” on the LCD by operating the ▲ and ▼ buttons, select “Security Lock”, and press the ENTER button.

```
Network Setting(1/7)
>NW Security Lock
 NW Security PW
[↵]Select [STOP]Back
```

Figure 7.52

3 When “ON” is selected by operating the ▲ and ▼ buttons and the ENTER button is pressed, the security of the Network Setting is enabled and the LCD displays the password setting screen shown in Figure 7.54.

```
Select lock setting
>ON
 OFF
[↵]Enter  [STOP]Back
```

Figure 7.53

```
Input new Password
Press ▲▼ to Select
1234
[↵]Next   [STOP]Back
```

Figure 7.54

4 To set the password (4 numerals), select the value of each flashing digit by operating the ▲ and ▼ buttons and press the ENTER button to enter the numeral. When the fourth numeral has been entered, the LCD displays the screen as shown in Figure 7.52.

NOTE

If you want to change the password, open the screen shown in Figure 7.52, select “NW Security PW” by operating the ▲ and ▼ buttons and press the ENTER button. When the LCD displays the password setting screen shown in Figure 7.54 start setting the new password.

5 Press the STOP/BACK button. When the LCD displays the “Menu (4/4)” screen (see Figure 7.51), press the STOP/BACK button again to return to the standby screen.

■ *Network settings*

1 Press the MENU button in the Standby status, display “Menu (4/4)” on the LCD by operating the ▲ and ▼ buttons, select “Network Setting”, and press the ENTER button.

```
Menu(4/4)
>Network Setting

[↵]Select [STOP]Back
```

Figure 7.55

2 The LCD displays the network setting screen shown in Figure 7.56.

```
Network Setting(1/7)
>NW Security Lock
 NW Security PW
[↵]Select [STOP]Back
```

Figure 7.56

NOTE

The LCD displays the screen shown in Figure 7.57 if the network security lock has already been enabled. Input the password (4 numerals) by operating the ▲ and ▼ buttons and press the ENTER buttons and press the ENTER button. The LCD displays the screen show in Figure 7.56.

```
Input user Password
Press ▲▼ to Select
 ****
[↵]Select [STOP]Back
```

Figure 7.57

3 Display “Network Setting (2/7)” on the LCD by operating the ▲ and ▼ buttons, select “Communicate”, and press the ENTER button.

```
Network Setting(2/7)
>Communicate
 IP address(Server)
[↵]Select [STOP]Back
```

Figure 7.58

4 Select “ON” by operating the ▲ and ▼ buttons and press the ENTER button. The network communication with the server is enabled and the LCD returns to the screen immediately before the current screen. (See Figure 7.58)

```
Select communicate
>ON
 OFF
[↵]Enter   [STOP]Back
```

Figure 7.59

5 Display “Network Setting (2/7)” on the LCD by operating the ▲ and ▼ buttons, select “IP address(Server)”, and press the ENTER button.

```
Network Setting(2/7)
 Communicate
>IP address(Server)
[↵]Select [STOP]Back
```

Figure 7.60

6 To set the IP address of the server to be communicated with, select the value of each flashing digit by operating the ▲ and ▼ buttons and press the ENTER button to enter the value. When the last value has been entered, the LCD returns to the screen immediately before the current screen. (See Figure 7.60)

```
IP Address Setting
Press ▲▼ to Select
192.168.119.001
[↵]Enter   [STOP]Back
```

Figure 7.61

7 Display “Network Setting (3/7)” on the LCD by operating the ▲ and ▼ buttons, select “Subnet Mask”, and press the ENTER button.

```
Network Setting(3/7)
>Subnet Mask
 Gateway
[↵]Select [STOP]Back
```

Figure 7.62

8 To set the subnet mask of the server to be communicated with, select the value of each flashing digit by operating the ▲ and ▼ buttons and press the ENTER button to enter the value. When the last value has been entered, the LCD returns to the screen immediately before the current screen. (See Figure 7.62)

```
Subnet Mask setting
Press ▲▼ to Select
192.168.119.001
[↵]Enter   [STOP]Back
```

Figure 7.63

Ch.7

9 Display “Network Setting (3/7)” on the LCD by operating the ▲ and ▼ buttons, select “Gateway”, and press the ENTER button.

```
Network Setting(3/7)
 Subnet Mask
>Gateway
[↵]Select [STOP]Back
```

Figure 7.64

10 To set the Gateway of the server to be communicated with, select the value of each flashing digit by operating the ▲ and ▼ buttons and press the ENTER button to enter the value. When the last value has been entered, the LCD returns to the screen immediately before the current screen. (See Figure 7.64)

```
Gateway setting
Press ▲▼ to Select
192.168.119.001
[↵]Enter  [STOP]Back
```

Figure 7.65

11 Display “Network Setting (4/7)” on the LCD by operating the ▲ and ▼ buttons, select “IP address (ALT-Pro)”, and press the ENTER button.

```
Network Setting(4/7)
>IP address(ALT-Pro)
 Mac address
[↵]Select [STOP]Back
```

Figure 7.66

12 To set the IP address of the ALT-Pro, select the value of each flashing digit by operating the ▲ and ▼ buttons and press the ENTER button to enter the value. When the last value has been entered, the LCD returns to the screen immediately before the current screen. (See Figure 7.66)

```
Gateway setting
Press ▲▼ to Select
192.168.119.001
[↵]Enter  [STOP]Back
```

Figure 7.67

Ch.7

13 Display “Network Setting (6/7)” on the LCD by operating the ▲ and ▼ buttons, select “COMM Mode”, and press the ENTER button.

```
Network Setting(6/7)
>COMM Mode
 Timeout(Ctrl Layer)
[↵]Enter  [STOP]Back
```

Figure 7.68

14 Select the communication mode by operating the ▲ and ▼ buttons and press the ENTER button. The communication mode is selected and the LCD returns to the screen immediately before the current screen. (See Figure 7.68)

```
Select COMM Mode(1/3)
>Auto negotiation
 100Mbps full duplex
[↵]Enter  [STOP]Back
```

Figure 7.69

15 Press the STOP/BACK button. The LCD displays the “Menu (4/4)” screen with “Sv” on the top right corner.

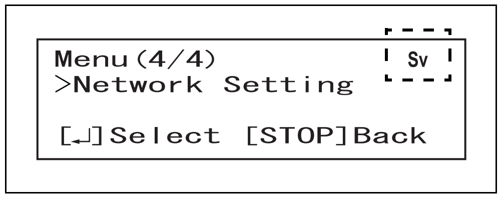


Figure 7.70

NOTE

- Characters “Sv” appear when the ALT-Pro is connected to the server.
- Characters “Sv” flash when the ALT-Pro is not connected to the server. In this case, characters “Sv” change from flashing to steady display when the ALT-Pro is connected to the server.
- For details, refer to the instructions manual for KE-BASE SYSTEM SOFTWARE (Ver.2.1.0 or later).

Ch.7

Chapter 8 Troubleshooting

WARNING

When the leakage testing process is interrupted due to malfunction of this equipment or other irregularities, the endoscopes may not be properly tested. In this case leakage test should be started again from the beginning.

If any irregularity is detected during an inspection, or if this equipment is clearly malfunctioning, do not use it and disconnect the power cord from AC Power Inlet. Contact Olympus for repair.

Some problems that appear to be malfunctions may be correctable by referring to Section 8.1, “Troubleshooting guide” below. If the problem cannot be resolved by the described remedial action, do not use this equipment and contact Olympus.

8.1 Troubleshooting guide

WARNING

When the leakage testing process is interrupted due to malfunction of this equipment or other irregularities, the endoscopes may not be properly tested. In this case leakage test should be started again from the beginning.

CAUTION

Do not turn OFF the power when an error code is displayed. This equipment will automatically start a process to address the error code. (The error code will blink during automatic processing.) After the error code stops blinking and lights steadily, follow the instructions in “Error codes and remedial actions” below.

❍ Error codes and remedial actions

Error code	Problem	Possible causes	Solution
E041	The power was lost and then restored during the process.	• Power supply was interrupted. • Power cord contact failure.	• Check the power cord for proper connection. • Disconnect the power cord from the power outlet and check that the power cord is free from scratches or damage. If any irregularity is detected, replace the power cord.
E083	This equipment does not work.	Irregularity inside this equipment	Contact Olympus
E084	Malfunction of the RFID reader.	Irregularity in the electrical circuitry inside this equipment	Turn OFF this equipment and turn it ON again. If the problem persists after turning this equipment ON again, contact Olympus.
E085	This equipment does not work.	Irregularity inside this equipment	Contact Olympus.
E089	A wrong date is displayed.	The battery in this equipment has run out.	Contact Olympus. Even if this error has occurred, manual leakage testing and self-check can be performed. However, note that the date and time recorded in the log may not be correct.
E094	Data cannot be printed.	• The interface cable is not securely connected. • Printer paper roll has run out. • The paper cover is open. • Irregularity in the printer. • The printer is turned OFF.	• Confirm that the interface cable is securely connected. • Check to see if the printer paper roll has been installed. • Check to see if the paper cover is closed. • If the problem still persists, contact Olympus.
E112	Self-check error.	• No ALT-Pro leak test air tubes are connected. • The water-resistant cap or endoscope is connected to the ALT-Pro leak test air tubes. • A leak has been detected. • Irregularity in the pump. • Only one ALT-Pro leak test air tube is connected. • An irregularity may have occurred in the ALT-Pro leak test air tubes.	Perform the steps described in Section "■ If the error code [E112] is displayed during self-check" on page 37.

Error code	Problem	Possible causes	Solution
E113	An endoscope is disconnected.	An endoscope is not connected to the ALT-Pro leak test air tube when the automated leakage testing starts.	Detach all the endoscopes from the scope side connector of the ALT-Pro leak test air tubes and press the ENTER button. Review Section 5.5, “Recognition of scope ID” and Section 5.7, “Recognition of user ID” and connect an endoscope to the ALT-Pro leak test air tube.
E115	Pressure cannot be applied for leak testing.	• The ALT-Pro leak test air tubes have not been properly connected. • A large hole in the endoscope. • Irregularity in the pump. • Irregularity in the valve.	Perform the steps described in Section “■ If the error code [E115] is displayed on the screen” on page 62.
E117	Overpressure in channels.	• Irregularity in the pump. • Irregularity in the valve. • Irregularity in the endoscope. • Use of hot or warm water in manual leakage testing. • A large difference between the temperature of the water used in manual leakage testing and room temperature.	• If the endoscope is immersed in water, remove it from the water. Wait 30 seconds or more after the occurrence of the error code to release air from the endoscope. Then, remove the endoscope from the ALT-Pro leak test air tube. Perform self-check following the instruction in Section 7.9, “Performing only self-check”. • Use only water that is close to room temperature. • If the problem still persists, contact Olympus.
E118	Failure to exhaust the air.	Irregularity in the valve.	Wait 30 seconds or more after the occurrence of the error code to release air from the endoscope. Then, remove the endoscope from the ALT-Pro leak test air tube and contact Olympus.
E119	Process control does not function properly.	Irregularity in the electrical circuit.	Turn OFF this equipment. Wait for a while before turning it ON again. If the problem persists after turning this equipment ON again, contact Olympus.
E122	Improper endoscope connection.	The endoscope was connected to a wrong endoscope connector.	Detach all the endoscopes from the scope side connector of the ALT-Pro leak test air tubes and press the OK button. Go back to Section 5.5, “Recognition of scope ID” and Section 5.7, “Recognition of user ID” to have this equipment recognize the scope ID and user ID again, and connect the endoscope to the proper port.

Error code	Problem	Possible causes	Solution
E123	The connection of endoscope cannot be checked after the recognition of scope ID.	• No ALT-Pro leak test air tubes are connected. • The water-resistant cap or endoscope is connected to the ALT-Pro leak test air tubes.	• Check that the ALT-Pro leak test air tubes are connected. • Check that the water-resistant cap or endoscope is not connected to the ALT-Pro leak test air tubes. • Press the Enter button and go back to Section 5.5, "Recognition of scope ID".
E124	Process control does not function properly.	The more noise occurred than expected, causing the malfunction of process control.	Turn OFF this equipment. Wait for a while before turning it ON again. If the problem persists after turning this equipment ON again, contact Olympus.
E125	Processing at startup does not work.	Faulty piece within this equipment.	Contact Olympus.
E131	Data cannot be downloaded to the portable memory.	The portable memory is removed while accessing to the data in this equipment.	Insert the portable memory.
E132	Data cannot be downloaded to the portable memory.	Faulty portable memory.	Exchange the portable memory.
E133	Data cannot be downloaded to the portable memory.	• Faulty format. • A portable memory not designated by Olympus is connected.	• Format the portable memory referring to "■ Formatting the portable memory" on page 91. • Connect the portable memory designated by Olympus.
E134	Data cannot be downloaded to the portable memory.	Irregularity in the system associated with the portable memory.	Contact Olympus.
E136	Data cannot be downloaded to the portable memory.	No portable memory is connected.	Insert the portable memory into the portable memory port properly.
E137	Data cannot be downloaded to the portable memory.	The free space in the portable memory is full.	• Replace the portable memory. • Delete unnecessary data.

❍ Other errors and remedial action

Problem	Possible causes	Solution
This equipment is not activated.	The power cord is not connected.	Connect this equipment to the hospital-grade wall mains according to the instruction described in Section 3.4, “Connecting the power supply”.
	This equipment is not turned ON.	Turn this equipment ON.
	A fuse has been blown.	Contact Olympus to purchase compatible fuses. Replace the fuses referring to Section 6.5, “Replacing the fuse”.
Nothing is displayed on the LCD.	An excessive static electricity has been generated under extremely dry conditions, causing liquid crystal display to malfunction.	Turn OFF this equipment and turn it ON again. If the problem persists after turning this equipment ON again, contact Olympus.
The ALT-Pro leak test air tube cannot be connected to this equipment.	The tube connector of this equipment is stained.	Clean the tube connector of this equipment. If the stain cannot be cleaned off, contact Olympus.
	A tube other than the provided ALT-Pro leak test air tube is connected.	Connect the ALT-Pro leak test air tubes provided with this equipment. Other water leak tester (MB-155) cannot be connected.

In a manual leakage test performed due to FAIL in a leakage test conducted following Section 5.8, “Automated leakage testing”, no leak can be detected.	• If the endoscope generates short beeps and displays the following message when the scope ID recognition is performed:
	`Caution`
	`Additional Wait Time`
	`Required Endoscope`
	`See Instructions`
	• The result has become FAIL because the automated leakage test was conducted in less than 9 minutes after the endoscope withdrawal from the patient body.
	• The result has become FAIL because the automated leakage test in the Standard Mode was conducted in less than 27 minutes after the endoscope withdrawal from the patient body.
		Perform an automated leakage test once again. Contact Olympus if the result is still FAIL and a manual leakage test cannot locate the leak.

Self-check cannot be started.	Scope ID or user ID has been detected.	Press the STOP/BACK button to return to the standby screen.

Problem	Possible causes	Solution
The endoscope cannot be connected to the ALT-Pro leak test air tube.	The endoscope is not compatible with this equipment.	Connect an endoscope described in the “List of Compatible Endoscopes <ALT-Pro>”.
An ALT or MLT cannot be started.	No scope ID or user ID has been detected.	Bring the scope ID tag or user ID seal to the RFID reader and scan the tag with the reader. Refer to Section 5.5, “Recognition of scope ID” and Section 5.7, “Recognition of user ID”.
“Time Out” is displayed.	An endoscope has not been connected to the ALT-Pro leak test air tube for a long time since the scope ID was recognized.	Review Section 5.5, “Recognition of scope ID” and Section 5.7, “Recognition of user ID” and connect an endoscope to the ALT-Pro leak test air tube.
Incorrect date and time is displayed on the monitor or printed.	The date and time is not properly set.	Set the date and time according to Section 7.6, “Setting date” and Section 7.7, “Setting time”.
Air bubbles come from the ALT-Pro leak test air tube during the MLT process.	There is a pin hole in the ALT-Pro leak test air tube.	Replace the ALT-Pro leak test air tubes.
“Low Battery” is displayed.	The battery for internal clock has run out after long-term use.	Contact Olympus.
Data cannot be downloaded to the portable memory.	The portable memory is not properly connected to the portable memory port.	Connect the portable memory properly.
	Available space of the portable memory is low.	Delete unnecessary data in the portable memory or use a new portable memory.
Self-check does not start at startup of this equipment.	• An error code is displayed. • If a printer is connected to this equipment, it is not properly recognized.	• If an error code appears, refer to the “Error codes and remedial actions”. • Refer to Table 4.2 on page 35.
The data recorded in the portable memory cannot be played back.	The data has been already edited with a personal computer or other equipment.	The data cannot be played back.
“Contact Olympus for Pump Replacement” is displayed.	It is time to replace the pump.*1	Contact Olympus for the pump replacement.*2
No beep sound can be heard.	The speaker has failed.	Contact Olympus.
This equipment is not connected to any of the peripheral equipment.	The connector has deformed.	Contact Olympus.
Data cannot be transmitted to the server.	The network setting is inappropriate.	Set properly by referring to Section 7.12, “Setting the network”. If the problem persists, contact Olympus.
	This equipment and server are not connected with a LAN cable.	Connect a LAN cable correctly.

Problem	Possible causes	Solution
The network setting cannot be confirmed.	A wrong password is entered.	Enter the right password. Contact Olympus if you do not remember the password.

*1 This indication is lit up when the total operation time of the pump reaches 700 hours.

*2 If the facility wants to continue to use this equipment before pump replacement, this equipment can be used continuously if there is no error in self-check. However, note that this indication appears every time turning on the power until the pump is replaced.

8.2 Return of this equipment for repair

CAUTION

- Olympus is not liable for any injury or damage that occurs as a result of repairs attempted by non-Olympus personnel.
- This equipment after repair may retain settings different from those before repair. Record the automated leakage test mode, auto printing mode and network setting before it is returned for repair and confirm the settings when it is returned. Refer to Section 7.5, “Setting automated leakage test mode”, Section 7.11, “Setting auto printing” and Section 7.12, “Setting the network” for how to confirm and set the settings.

When returning the automated endoscope leak tester for repair, contact Olympus. With the automated endoscope leak tester, include a description of the malfunction or damage and the name and telephone number of the individual at your location who is most familiar with the problem. Include a repair purchase order.

Appendix

Combination equipment

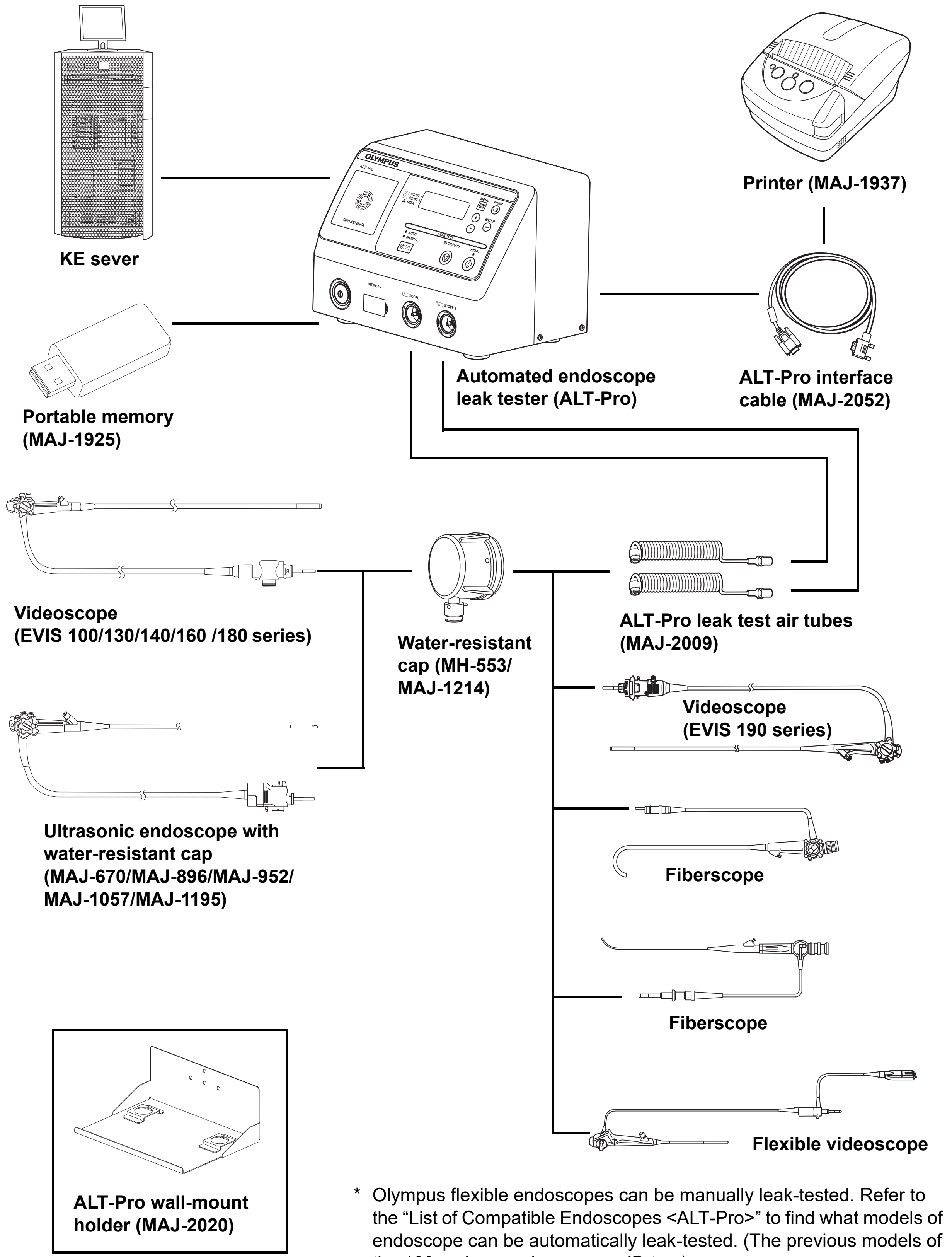
System chart

The system chart below lists recommended combinations of equipment and accessories that can be used with this equipment. If you have any questions, contact the persons in charge of this equipment indicated on the back cover of instructions.

WARNING

If combinations of equipment other than those shown below are used, the full responsibility is assumed by the medical treatment facility. Olympus cannot guarantee that this equipment will perform as expected. Nor can Olympus guarantee the safety of the patients and medical operators. Nor can the durability of this equipment be guaranteed when non-designated equipment is used. Any damage resulting from improper combinations will not be serviced or repaired free of charge.

❍ System chart



* Olympus flexible endoscopes can be manually leak-tested. Refer to the “List of Compatible Endoscopes <ALT-Pro>” to find what models of endoscope can be automatically leak-tested. (The previous models of the 180 series require a scope ID tag.)

App.

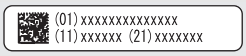
Specification

Environment

Operating environment	**Ambient temperature**	10 – 40°C (50 – 104°F)
	Relative humidity	30 – 85% (without condensation)
	Atmospheric pressure	700 – 1060 hPa (0.7 – 1.1 kgf/cm^2) (10.2 – 15.4 psia)
	Elevation	3000 meters (maximum)
Standard storage environment (e.g. within the hospital)	**Ambient temperature**	5 – 40°C (41 – 104°F)
	Relative humidity	10 – 95%
	Atmospheric pressure	700 – 1060 hPa (0.7 – 1.1 kgf/cm^2) (10.2 – 15.4 psia)
Transportation and storage environment	**Ambient temperature**	–47 to +70°C (–52.6 to +158°F)
	Relative humidity	10 – 95%
	Atmospheric pressure	700 – 1060 hPa (0.7 – 1.1 kgf/cm^2) (10.2 – 15.4 psia)

Specifications

Compatible endoscope		Olympus flexible endoscopes: (Refer to “List of Compatible Endoscopes <ALT-Pro>” in Appendix.)
Number of endoscope tested		1 or 2
ALT time setting		Approximately 2 minutes and 20 seconds
Pump		Diaphragm type pump (Pump life: 700 hours)
Leakage testing functions/ performance	**Automated leakage test**	Detects automatically for a pin hole by feeding air into the endoscope and computing a changed air pressure.
	Manual leakage test	Feeds air into the endoscope to visually inspect for any air bubbles emerging from the immersed endoscope.
	Self-check	Automatically checks if this equipment functions properly.
Control		Microprocessor control
Operation		Operating with switches on control panel
Indicators		The LCD displays the following: (1) Stand-by state, (2) ID data of endoscopes and operators (e.g., Model name, S/N, and Control number), (3) Processes, and (4) Results of automated leakage tests, (5) Error codes
ID data entry		The internal RFID reader of this equipment automatically enters the ID data of endoscopes (Model name and S/N) and operators (name and control number)
Data storage		Approximately 13,000 logs can be stored in this equipment and downloaded to Olympus-designated portable memories.
Print mode options		Either Manual Printing or Auto Printing can be selected. In the Manual Printing, the printed data can be selected from “Last”, “Today”, and “Select Date”.
Endoscope connection		The specified tube (the ALT-Pro leak test air tube) can be connected airtightly by rotating it about 90 degrees with a single motion.
Size	**Dimensions**	280 (W) × 201 (H) × 184 (D) mm
	Dimensions (maximum)	280 (W) × 201 (H) × 202 (D) mm
	Weight	5.5 kg
Power supply	**Rated voltage**	100 – 120 V AC
	Voltage fluctuation	Within ±10%
	Rated frequency	50/60 Hz
	Frequency change	Within ±1 Hz
	Rated input	45 VA
	Fuse rating	2 A, 250 V (Time-lag Type)
	Fuse size	ø 5 × 20 mm
EMC	**Applied standard: IEC 61326-1: 2005**	This equipment complies with the standards listed in the left column. CISPR 11 of emission: Group1,Class B

Electrical safety	**Applicable standard; IEC 61010-1: 2001**	This equipment complies with the standard listed in the left column. Installation category: II Pollution degree: 2
Radio transmitter	**Compliance**	ISO/IEC 18000-3 (Mode1)
	Center frequency	13.56 MHz
	Modulation	ASK (Amplitude shift keying)
	Effective radiated Power	100 mW ±20%
Memory backup	**Lithium battery life**	4 years
UDI label  		The UDI label is required by some countries' regulations regarding the identification of a medical device also known as Unique Device Identification (UDI). The following information is being coded in the 2-dimensional barcode (GS1 Data Matrix): • (01) 14-digit GS1 Global Trade Item Number; • (11) 6-digit date of manufacture; • (21) 7-digit serial number.

App.

EMC information

This model is intended for use in the electromagnetic environments specified below. The user and the medical staff should ensure that it is used only in these environments.

❍ Magnetic emission compliance information and recommended electromagnetic environments

Emission standard	Compliance	Guidance
RF emissions CISPR 11	Group 1	This instrument uses RF (Radio Frequency) energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
Radiated emissions CISPR 11	Class B	This instrument’s RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
Main terminal conducted emissions CISPR 11		
Harmonic emissions IEC 61000-3-2	Not applicable	Power supply specification of this instrument is less than 220 VAC, and this instrument is exempt from requirements of IEC 61000-3-2.
Voltage fluctuations/flicker emissions IEC 61000-3-3	Not applicable	Power supply specification of this instrument is less than 220 VAC, and this instrument is exempt from requirements of IEC 61000-3-3.

Electromagnetic immunity compliance information and recommended electromagnetic environments

Immunity test	IEC 61326-1 test level	Compliance level	Guidance
Electrostatic discharge (ESD) IEC 61000-4-2	Contact: ±2, ±4 kV Air: ±2, ±4, ±8 kV	Same as left	Floors should be made of wood, concrete, or ceramic tile that hardly produces static. If floors are covered with synthetic material that tends to produce static, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 61000-4-4	±2 kV for power supply lines ±1 kV for input/output lines	Same as left	Mains power quality should be that of a typical commercial (original condition feeding the facilities) or hospital environment.
Surge IEC 61000-4-5	Differential mode: ±0.5, ±1 kV Common mode: ±0.5, ±1, ±2 kV	Same as left	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions, and voltage variations on power supply input lines IEC 61000-4-11	< 5% U_T (> 95% dip in U_T) for 0.5 cycle 40% U_T (60% dip in U_T) for 5 cycle 70% U_T (30% dip in U_T) for 25 cycle < 5% U_T (> 95% dip in U_T) for 5 seconds	Same as left	Mains power quality should be that of a typical commercial or hospital environment. If the user of this instrument requires continued operation during power mains interruptions, it is recommended that this instrument be powered from an uninterruptible power supply or a battery.
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	30 A/m	Same as left	It is recommended to use this instrument by maintaining enough distance from any equipment that operates with high current.

NOTE

U_T is the AC mains power supply prior to application of the test level.

Cautions and recommended electromagnetic environment regarding portable and mobile RF communications equipment, such as cellular phones

Immunity test	IEC 61326-1 test level	Compliance level	Guidance
Conducted RF IEC 61000-4-6	3 Vrms (150 kHz – 80 MHz)	3 V (V_1)	Formula for recommended separation distance (V_1=3 according to the compliance level) $d = \left[\frac{3.5}{V_1}\right]\sqrt{P}$
Radiated RF IEC 61000-4-3	10 V/m (80 MHz – 1 GHz) 3 V/m (1.4 GHz – 2 GHz) 1 V/m (2 GHz – 2.7 GHz)	10 V/m (E_1) 3 V/m (E_2) 1 V/m (E_3)	Formula for recommended separation distance (E_1=10, E_2=3, E_3=1 according to the compliance level) $d = \left[\frac{3.5}{E_1}\right]\sqrt{P}$ 80 MHz – 800 MHz $d = \left[\frac{7}{E_1}\right]\sqrt{P}$ $d = \left[\frac{7}{E_2}\right]\sqrt{P}$ $d = \left[\frac{7}{E_3}\right]\sqrt{P}$

NOTE

- Where “P” is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer, and “d” is the recommended separation distance in meters (m).
- This instrument complies with the requirements of IEC 61326-1: 2005. However, if used in an electromagnetic environment that exceeds the required maximum noise level, electromagnetic interference may occur on this equipment.

❍ Recommended separation distance between portable and mobile RF communication equipment and this equipment

Rated maximum output power of transmitter P (W)	Separation distance according to frequency of transmitter (m) (calculated as V_1=3 and E_1=10)		
	150 kHz – 80 MHz $d = 1.2\sqrt{P}$	80 MHz – 800 MHz $d = 0.35\sqrt{P}$	800 MHz – 1 GHz $d = 0.7\sqrt{P}$
0.01	0.12	0.04	0.07
0.1	0.38	0.12	0.23
1	1.2	0.35	0.70
10	3.8	1.2	2.3
100	12	3.5	7

NOTE

The guidance may not apply in some situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people. Portable and mobile RF communications equipment such as cellular phones should be used no closer to any part of this instrument, including cables than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter.

FCC and IC information	This equipment complies with part15 of the FCC rules and the IC RSS210.	**FCC ID: S8Q-RU8354** **IC: 4763B-RU8354**
FCC WARNING	Change or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.	

App.

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■ newlib

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